

Classical and Quantum Query Complexity of Scalar Newton Quantities

Abstract

When H is the Hessian and b the gradient of a barrier function, $b^*H^{-1}b$ is the squared Newton decrement. The cost of estimating this number can differ from that of preparing or writing out the Newton direction. We study the query complexity of estimating it to relative error ϵ when $\kappa^{-1}I \preceq H \preceq I$, given sparse access to H , with at most d nonzeros per row, and sampling-and-query access to b . A classical estimator that explores neighborhoods of sampled coordinates uses $\kappa\epsilon^{-2}(d+1)^{O(\sqrt{\kappa}\log(2/\epsilon))}$ queries at constant confidence. Separate sparse families show that the statistical factor $\kappa\epsilon^{-2}$ and the exponential factor are each necessary, even with full sampling access to the matrix. An explicit composition gives single instances that require both factors at once in some parameter ranges, but it does not force the product of the two full factors. In the quantum model where H/α is a block of a unitary, a lower bound for 3×3 matrices matches, up to logarithmic factors, the known variable-time upper bound $\tilde{O}(\alpha\kappa/\epsilon)$. We construct sparse LPs and SOCPs whose optimal values, optimizer coordinates, or Newton decrements are classically hard to estimate, and every query to the full sampling-and-query interface of the formulation can be simulated with $O(1)$ queries to the hidden input. We also show that this cost adds up when the requested outputs (several path scalars or one endpoint coordinate) encode independent instances, and we use examples and reuse bounds to explain why one-shot hardness cannot simply be multiplied by an iteration count. For Newton matrices N formed by a well-conditioned sparse matrix plus a sampled low-rank correction, including Lorentz and generalized-power cone systems under stated assumptions, we give classical algorithms that estimate $b^*N^{-1}b$ and sample from an approximate solution of $Nx = b$. All results are query and arithmetic bounds; they do not establish a quantum speedup for interior-point methods.

1 Introduction

An interior-point method may need only one number before it decides how to move: the squared Newton decrement $b^*H^{-1}b$, where H and b are the Hessian and gradient of the barrier subproblem at the current point. It measures local centrality and Newton progress, but by itself it does not certify global optimality. A linear-system solver may instead return the Newton direction as a vector, as samples from its squared coordinates, or as a normalized quantum state. A normalized state loses the scale that a scalar estimate needs, and writing out a whole vector can cost much more than estimating one number derived from it. This matters when assessing quantum speedups of interior-point methods, which must also read the data, choose steps, and report an answer.

We study the complexity of this scalar and when it carries over to outputs of an optimization problem. The central problem is to estimate

$$q(H, b) = b^*H^{-1}b, \quad \kappa^{-1}I \preceq H \preceq I,$$

to relative error ϵ . Classical algorithms get sparse row access to H , with at most d nonzeros per row, and sampling-and-query access $\text{SQ}(b)$: coordinate values, the norm $\|b\|$, and samples of an

index i with probability proportional to $|b_i|^2$. Our classical lower bounds hold even with full matrix SQ access, which adds the row and column norms of H and the associated samples. Quantum algorithms get either coherent access to the entries of H or block access, a unitary whose top-left block is H/α . These two models differ even for diagonal 3×3 matrices: our block-access lower bound uses such matrices, and one exact entry query solves the same instances.

1.1 Results and contribution

Table 1 summarizes the bounds for a single scalar.

Upper and lower bounds for the scalar. Theorem 3.4 estimates q with $O(\kappa\epsilon^{-2}\log(1/\zeta)(d+1)^{O(\sqrt{\kappa}\log(2/\epsilon))})$ queries and arithmetic operations at failure probability ζ . It combines known ingredients: a Chebyshev residual polynomial, local sparse evaluation, and importance sampling. Because the form is positive, the Kantorovich inequality bounds the relative second moment of the estimator by $O(\kappa)$, so the statistical factor is $\kappa\epsilon^{-2}$, not quadratic in κ . Theorem 4.1 carries the matrix-function lower bound of Montanaro–Shao over to positive definite forms of row sparsity $s \geq 5$: estimating two such forms with public right-hand sides (known without queries) requires $((s-1)/2)^{\Omega(\sqrt{\kappa}\log(1/\epsilon))}$ classical queries, up to a polynomial denominator; the family is realized at the analytic center of a sparse box LP. Theorem 4.5 gives a two-sparse family of dimension $2M$ with the sharp lower bound $\Omega(\min\{M, \kappa\epsilon^{-2}\})$, or $\Omega(\kappa\epsilon^{-2}\log(1/\zeta))$ at failure probability ζ when M is large enough.

Theorem 5.2 shows when both kinds of cost fall on one form: suppose a promise problem g with randomized query complexity $R(g)$ is encoded, under the access and promise conditions of the theorem, as a form equal to a public positive baseline plus $C\Phi$, with signal-to-baseline ratio χ (C divided by the baseline) at most $5/3$. Then, for $0 < \epsilon \leq \chi/200$, estimating the average of such forms over many independent blocks to relative error ϵ needs $\Omega((\chi/\epsilon)^2 R(g))$ classical full-SQ queries. Relative error is measured against the whole value, so we compute the baseline exactly for an explicit witness clock (Proposition 5.3), a sparse matrix that applies circuit gates along a path of time steps. This gives, for example, a single-form lower bound $\epsilon^{-2}s^{\Omega(\sqrt{\kappa})}$, up to polynomial denominators, and a bound with a factor κ^2 in a restricted high-accuracy regime (Theorem 5.4), but not the product of the two full lower bounds. We also prove limits for specific constructions. In the cyclic construction, the contrast (the ratio of the two possible values of the form) is at most $(1 + K/L)^2$, where K^2 bounds the condition number, so contrast of the order of the condition number requires a bounded clock length L . The two-cluster construction gives only constant contrast. In a Schur-complement construction, a large form value raises the condition number.

For plain block access with normalization α , Theorem 6.1 matches the known variable-time negative-power norm algorithm with an $\Omega(\alpha\kappa/\epsilon)$ lower bound on block queries; its explicit unitary completion shows why this bound does not transfer to exact entry queries. For coherent sparse access, the proved bounds are $\Omega(\sqrt{\kappa}/\epsilon)$ and $\tilde{O}(d\kappa/\epsilon)$.

Optimization outputs and repeated queries. Section 7 constructs sparse LPs and SOCPs whose optimizer coordinate, optimal value, or Newton decrement encodes the amplitude Φ of a Forrelation circuit. The underlying cyclic clock matrix runs the circuit forward, holds, and undoes it. This matrix and its Newton Hessian have public row and column norms, so full SQ access to the formulation can be simulated; its inverse applied to the start vector has an exactly known norm, and the clock parameters trade off conditioning, sparsity, and accuracy. For the optimizer coordinate, classical full-SQ algorithms need $\Omega(2^{n/2}/n)$ queries, where n is the number of work qubits of the Forrelation circuit, while a quantum algorithm with the corresponding coherent input oracle needs

$O(1)$ queries to the hidden Forrelation sign tables (Theorem 7.2). Scaling an objective tilt by the exact public norm of an inverse-transpose vector, rather than by an upper bound on it, enlarges the hidden signal in the optimal value (Theorem 7.5). For an LP on a one-dimensional affine slice, whose barrier has parameter one, the classical upper and lower bounds for the optimal value have the same exponential dependence in the accuracy regime of Theorem 7.6; the reduced objective is not assumed given, since eliminating the equality constraints requires hidden information.

Section 8 encodes B independent instances in the changes of one coordinate between checkpoints on an LP central path, in the predictor decrements of an ordinary SOCP, and, through their parity, in one bounded optimizer coordinate. Because these outputs contain information about all B instances, returning them requires $\Omega(BL_g)$ classical full-SQ queries, where $L_g = R(g)$ (Theorems 8.2, 8.3, and 8.6). Conversely, in Section 9 a sparse KKT system has the same hard normalized solution along the whole central path, although reaching a small objective gap takes many bounded local steps; learning the hidden input once suffices. Reusing the span of stored solutions, and solving the system anew only when the smallest current relative residual over that span fails a test, requires at most r such refresh solves if the solution directions span a space of dimension r and computations are exact (Proposition 9.3), and at most $2r - 1$ if they lie close enough to one and the stated error bounds hold (Theorem 9.4); residual tests and explicit outputs are counted separately.

Structured Newton systems. Theorem 10.1 handles $N = S + CXLX^TC^T$, where $S = CC^T$ is a well-conditioned sparse full-rank base and $CXLX^TC^T$ is a low-rank correction whose symmetric core L may be singular or indefinite. Each column of X is a known sparse contraction of a unit source vector with SQ access, and no nearly zero column of CX is normalized. The algorithm builds one fixed approximate solution with a reusable setup, and each draw returns a sample or a failure flag at bounded cost; Corollary 10.2 gives relative estimation of the inverse form. Lorentz and generalized-power barriers have this structure under explicit bounds on eccentricity and base conditioning, when certain scalars of each cone block are supplied; for the power cone, we prove a query lower bound for computing one of them, a weighted geometric mean, from vector access alone. For a full classical Newton direction, we give separate solve bounds using eccentricity profiles (how many cone blocks have large eccentricity) and latent sparsity (a supplied tree decomposition of small width).

To our knowledge, the following are new: the full-SQ lower bounds for positive forms, the single-form composition and baseline calculations, the parameterized scalar realizations in optimization problems and along paths with the examples that limit reuse, and the theorem for a sparse full-rank base with low-rank corrections under source-vector access. We claim no priority for inverse-form estimation, sparse polynomial dequantization, Forrelation matrix constructions, hardness of scalar optimization outputs, low-rank cone algebra, or subspace recycling.

1.2 Relationship to prior work

Li–Sra–Jegelka develop Gaussian quadrature for matrix inverse forms [30], and Bizas–Mitrouli–Turek give further analytic and heuristic estimates [10]; these methods count matrix–vector products. Our classical model instead explores neighborhoods of sampled coordinates, trading dependence on dimension for exponential dependence on the polynomial degree.

Gharibian–Le Gall already give sparse polynomial overlap estimation from sampling access [22, Theorem 4.1]. Montanaro–Shao relate the query complexity of matrix functions to approximation degree and construct the weighted Forrelation clocks used here [32, Theorem 1.8]. Cifuentes et al. classify matrix-function tasks under several access and measurement models [19], and Edenhofer–Hasegawa–Le Gall apply sparse polynomial evaluation to spectral sums, including reciprocal traces

Access and problem	Bound	Regime and location
Classical sparse H , SQ(b); relative scalar	$O(\kappa\epsilon^{-2} \log(1/\zeta)(d+1)^{O(\sqrt{\kappa} \log(2/\epsilon))})$ upper	$\kappa \geq 1$, $0 < \epsilon \leq 1/2$; Theorem 3.4
Classical full SQ; two-sparse family	$\Omega(\min\{M, \kappa\epsilon^{-2}\})$ lower in dimension $2M$; $\Omega(\kappa\epsilon^{-2} \log(1/\zeta))$ for sufficiently large M	$\kappa \geq 4$, $\epsilon \leq 1/128$; Theorem 4.5
Classical full SQ; s -sparse clock family	$\Omega\left(\frac{((s-1)/2)^{\Omega(\sqrt{\kappa} \log(1/\epsilon))}}{\log(s)\sqrt{\kappa} \log(1/\epsilon)}\right)$ lower	$\kappa \geq 4$, $s \geq 5$, small ϵ ; Theorem 4.1
Classical full SQ; one-form composition	$\Omega((\chi/\epsilon)^2 R(g))$ lower, where χ is the signal-to-baseline ratio and $R(g)$ the inner randomized query cost	$0 < \chi \leq 5/3$, $\epsilon \leq \chi/200$; Theorem 5.2
Quantum plain block encoding of normalization α	$\Omega(\alpha\kappa/\epsilon)$ lower and $\tilde{O}(\alpha\kappa/\epsilon)$ upper in block calls	$\alpha \geq 1$, $\kappa \geq 4$, $\epsilon \leq 1/16$; Theorem 6.1
Quantum coherent sparse access	$\Omega(\sqrt{\kappa}/\epsilon)$ lower and $\tilde{O}(d\kappa/\epsilon)$ upper; the canonical two-sparse sign-block family attains the lower rate	$d \geq 2$, $\kappa \geq 4$, $\epsilon \leq 1/128$; Corollary 4.6, (33)

Table 1: Selected query bounds for relative estimation of inverse forms. Lower-bound constructions choose their dimensions; the other conditions of the cited theorems apply. Quantum rows count input-state preparation separately and use constant failure probability. The clock lower bound is proved for estimating two public forms, and hence also bounds a routine that estimates the form for any given b . The composition row requires the theorem’s oracle-preserving inner family. The rows are separate statements: their lower bounds must not be multiplied.

[20, Theorem 3.1 and Section 3.3]. Our target is a positive inverse form with an arbitrary vector and relative error, which requires controlling the baseline and the variance; the polynomial method and its origin in approximation theory come from this prior work. Le Gall’s robustness results already cover approximate length-squared sampling for sparse and low-rank dequantization [29]; our finite-error analysis, for approximate samples and estimator values, is only a sufficient condition for our estimator, not a general robustness framework.

Block encodings and polynomial transformations are established quantum tools [23], and our block-access upper bound specializes the norm estimator of Chakraborty–Gilyén–Jeffery [15, Theorem 33 in the full version]. Alase et al. study scalar observables of linear-system solutions [2]; their tight bound counts queries to a block of the observable for additive error in an expectation value, while ours counts queries to the system matrix for relative estimation of $b^*H^{-1}b$. Orsucci–Dunjko study preparation of solution states for positive definite systems, and stronger access structures that can improve the dependence on conditioning [36]; Section 6 compares these different access models with ours.

Grønlund–Larsen prove an exponential separation between quantum algorithms and SQ access for outputting linear-system solutions [25], and Apers–Gribling already prove full-SQ hardness for the optimal value of an ordinary LP [5, Lemma 8.5]. Our optimization constructions instead add explicit control of condition number, sparsity, and accuracy, a normalization of the hard scalar, realizations as actual Newton quantities, and an account of the cost of acquiring the input. Zhao et al. study normalized solution observables and Forrelation embeddings under restrictions on streaming space and data samples [40, Task F.1 and Sections E.5–F]; here the input is static and classical storage is unrestricted. General direct-sum, composition, and XOR results are also prior work [8, 11, 13, 16]; we realize them in the stated optimization outputs and prove that every access

operation can be simulated.

Low-rank SQ matrix arithmetic and solution sampling are established [18]; our structured theorem instead has a sparse full-rank base with a low-rank correction, and does not require normalized transformed columns as input. Hessian identities for cones and their sparse implementations are likewise established [3, 17, 24, 26, 38]; we prove the spectral and sampling conditions needed to use it with local queries. Krylov recycling and greedy reduced bases [9, 37], and sparse and low-treewidth elimination [21, 39], are the classical background for our results on reuse and on full output.

1.3 Scope and organization

Section 2 states all oracle and ideal-arithmetic assumptions. Sections 3–5 give classical algorithms, lower-bound families, and composition; Section 6 treats coherent access; Sections 7–9 treat optimization outputs, composition over a path, and reuse; and Section 10 treats structured cone systems.

The results are query and arithmetic bounds; they do not establish a quantum speedup for general interior-point methods, a uniform bound on bit complexity, or that sampling oracles, which change with the iterate, can be maintained at no cost. The general classical upper and lower bounds do not match in their joint dependence on all parameters, nor do the bounds for coherent sparse access. None of the optimization or structured results depends on closing these gaps.

2 Problems and access models

All vector norms are Euclidean and all matrix norms are spectral unless specified otherwise. The adjoint is denoted by $*$. A Hermitian matrix $H \succ 0$ is supplied with public bounds

$$\kappa^{-1}I \preceq H \preceq I, \quad \kappa \geq 1. \quad (1)$$

Thus κ is a known upper bound on the condition number of H ; statements that require equality say so. For a nonzero $b \in \mathbb{C}^N$, our main target is

$$q(H, b) = b^* H^{-1} b. \quad (2)$$

An estimate has relative error ϵ if $|\hat{q} - q| \leq \epsilon q$. If $H = \nabla^2 F(x)$ and $b = \nabla F(x)$, then q is the squared Newton decrement at x . More generally, b may be the barrier gradient perturbed by a linear objective. A theorem about an arbitrary pair (H, b) becomes a theorem about optimization only once an objective, an interior point, and oracles for them are specified. We give such realizations separately.

In the optimization sections, a barrier ϕ is a C^3 convex function on the relative interior of its feasible affine slice, written in independent coordinates, whose Hessian restricted to the slice is positive definite. Its local norm is $\|h\|_x^2 = h^T \nabla^2 \phi(x) h$, with dual norm $\|g\|_{x,*}^2 = g^T \nabla^2 \phi(x)^{-1} g$. A self-concordant barrier satisfies $|D^3 \phi(x)[h, h, h]| \leq 2\|h\|_x^3$ and diverges at the boundary; a parameter ν additionally bounds $|D\phi(x)[h]| \leq \sqrt{\nu}\|h\|_x$. Stated barrier parameters refer to the given barrier and coordinates, not to every formulation of the same feasible set. For maximizing $c^T x$, the barrier subproblem is $F_\eta(x) = \phi(x) - \eta c^T x$, $\eta > 0$. A central point minimizes F_η ; an analytic center minimizes ϕ . The Newton direction is $-\nabla^2 \phi(x)^{-1} \nabla F_\eta(x)$ and its squared decrement is $\|\nabla F_\eta(x)\|_{x,*}^2$. At a central point, changing the multiplier by $\Delta\eta$ gives a predictor squared decrement $(\Delta\eta)^2 \|c\|_{x,*}^2$. With equality constraints, we use the Hessian and gradient restricted to the tangent space, or explicitly stated KKT equations. The restricted quantities need not have the condition number of an augmented KKT matrix.

Definition 2.1 (Sparse access and vector sampling). Row-sparse access to a matrix supplies the nonzero count in each requested row, the location of its j th nonzero, and its value, at unit query cost. The ordering of the nonzeros is fixed for each input. A known bound d limits the count. Column-sparse access has the analogous meaning. For Hermitian matrices, column access follows from row access by conjugation.

For $v \neq 0$, $\text{SQ}(v)$ supplies coordinate queries, the norm $\|v\|$, and independent samples from $p_v(i) = |v_i|^2 / \|v\|^2$. Zero coordinates have zero sampling probability. These are exact oracles in our main model. Full matrix SQ additionally supplies row and column norms, the Frobenius norm, within-row and within-column squared-magnitude samples, and samples of rows or columns proportional to their squared norms. A zero row or column is recognized by its zero norm, and no sample is requested from it. A lower bound that explicitly grants both full matrix SQ and row-sparse access also provides the locations of nonzero entries.

These oracles must be specified because building a sampling data structure for an arbitrary explicit vector generally requires a pass over that vector. We do not count this preprocessing for an algorithm whose input model already provides $\text{SQ}(v)$. Conversely, a lower-bound reduction must simulate every granted oracle; it cannot ignore hidden information in norms or sampling probabilities. Data are *public* if the algorithm knows them in advance, so using them costs no oracle query; in a lower bound, public data do not depend on the hidden input. Matrix SQ is a classical interface and does not by itself provide a coherent state-preparation unitary.

Our quantum statements specify a unitary U whose top-left block is H/α , controlled calls to U and U^* , and a specified preparation of $b/\|b\|$. The normalization α and the allowed unitary completions are part of this access model. An exact sparse-value oracle, which returns an entry as a number, is a different interface, and a lower bound for black-box block encodings need not hold for it.

We count oracle invocations and ideal real or complex arithmetic operations. Independent uniform random draws from $[0, 1]$ and exact comparisons with computed real probabilities are unit-cost primitives in this model. In particular, the exact rejection samplers below use ideal randomness; their arithmetic bounds do not count a finite-bit implementation of these random choices. An index is one word; its bit length is $\lceil \log_2 N \rceil$. Thus “dimension-independent” means that these counts do not depend on N , not that the bit cost or the cost of loading the input is independent of N . The notation $\tilde{O}$ suppresses logarithmic factors only in the displayed parameters. Our direct classical upper bounds below use ordinary O notation, with unspecified absolute constants in polynomial degrees and local-walk exponents. Unless stated otherwise, $0 < \epsilon \leq 1/2$ and $0 < \zeta < 1/2$ denote accuracy and failure probability, and every classical or quantum lower bound assumes success probability at least $2/3$ on every promised input and counts worst-case queries. Explicit confidence parameters and distributional or expected-cost requirements take precedence over this convention.

Besides a single scalar, we consider a solution-sampling output for a linear system with matrix A and right-hand side b : the algorithm provides coordinate access to one fixed vector $\tilde{x}$ and exact samples from $p_{\tilde{x}}$, where $\|\tilde{x} - A^{-1}b\| \leq \epsilon \|A^{-1}b\|$. This is weaker than writing all coordinates, and stronger than an unspecified approximate output distribution. It does not include an exact norm oracle for $\tilde{x}$. A lower bound that uses a rare event must also specify whether unflagged failures of the algorithm may affect that event. A bound for one of these outputs need not hold for another.

Finally, a public interval $\mu I \preceq H \preceq LI$ reduces to (1) by using H/L with $\kappa = L/\mu$. The inverse quadratic form of H/L must then be divided by L .

3 Classical local algorithms

This section gives classical algorithms that estimate $q(H, b)$ in the access model of Section 2. The ingredients are standard: Chebyshev residual polynomials, local-walk evaluation of sparse matrix polynomials, and importance sampling. Gharibian–Le Gall [22] already estimate sparse-polynomial overlaps; Montanaro–Shao [32] and Cifuentes et al. [19] study matrix-function entry estimation. Our version (Theorem 3.4) guarantees relative error for the positive inverse quadratic form and uses positivity to reduce the statistical factor from κ^2 to κ . We then give two variants for comparison with the optimization constructions: a solution sampler (a stronger output) and an estimator for nonsymmetric matrices.

3.1 A relative inverse polynomial

Lemma 3.1 (Chebyshev residual). *For $K > 1$ and $0 < \eta < 1$, there is a real polynomial P of degree at most $m - 1$, where*

$$m = \left\lceil \frac{\log(2/\eta)}{\log((\sqrt{K} + 1)/(\sqrt{K} - 1))} \right\rceil = O(\sqrt{K} \log(2/\eta)), \quad (3)$$

such that $|1 - \lambda P(\lambda)| \leq \eta$ for $\lambda \in [K^{-1}, 1]$. Consequently, for $K^{-1}I \preceq H \preceq I$,

$$(1 - \eta)H^{-1} \preceq P(H) \preceq (1 + \eta)H^{-1}, \quad \|I - HP(H)\| \leq \eta. \quad (4)$$

For $K = 1$, one may take $P = 1$.

Proof. Put $a = K^{-1}$, $c = (1 + a)/(1 - a)$, and $x(\lambda) = (1 + a - 2\lambda)/(1 - a)$. If T_m is the degree- m Chebyshev polynomial, set

$$r_m(\lambda) = \frac{T_m(x(\lambda))}{T_m(c)}, \quad P(\lambda) = \frac{1 - r_m(\lambda)}{\lambda}.$$

Since $x(0) = c$, the numerator vanishes at zero, so the quotient is a polynomial. On $[a, 1]$, $|T_m(x(\lambda))| \leq 1$. With $\theta = (\sqrt{K} - 1)/(\sqrt{K} + 1)$, the identity $T_m(c) = (\theta^{-m} + \theta^m)/2$ gives $|r_m(\lambda)| \leq 2\theta^m$. The displayed choice of m makes this at most η . Finally $\log((1 + u)/(1 - u)) \geq 2u$ for $u = 1/\sqrt{K}$ proves the degree estimate; the matrix inequalities follow by diagonalizing H . $\square$

Lemma 3.2 (Local evaluation). *If H is d -row-sparse, one coordinate of $P(H)v$, for a specified degree- k polynomial P , can be evaluated with $(d + 1)^{O(k+1)}$ sparse and coordinate queries and arithmetic operations. If row and column sparsity are both bounded by d , a full row or column of $P(H)$ can be materialized in the same bound. More generally, a polynomial $q(A^*A)A^*$ of degree k in A^*A has this property under row and column access to A , with walk length at most $2k + 1$.*

Proof. Expand a requested coordinate of $H^j v$ into matrix walks of length j starting from that coordinate. Each step has at most d choices. Summing for $0 \leq j \leq k$ costs at most $O((k + 1)^2(d + 1)^{k+1})$ operations, including multiplying path weights and aggregating equal endpoints. The same expansion without the terminal coordinate of v materializes the corresponding row; reverse walks materialize a column. One can avoid sorting by retaining the walk list and aggregating equal endpoints by comparisons in quadratic time, still within the stated bound. For $q(A^*A)A^*$ the walks alternate between column and row queries to A . No length- N vector is formed. $\square$

3.2 Positive quadratic forms

Lemma 3.3 (Positive-form second moment and the Kantorovich inequality). *If $P \succ 0$ has spectrum in $[\alpha, \beta]$ and $b \neq 0$, sample $I \sim p_b$ and define*

$$X = \|b\|^2 \frac{(Pb)_I}{b_I}. \quad (5)$$

Then

$$\mathbb{E}X = b^*Pb, \quad \frac{\mathbb{E}|X|^2}{(b^*Pb)^2} \leq \frac{\|b\|^2 \|Pb\|^2}{(b^*Pb)^2} \leq \frac{(\alpha + \beta)^2}{4\alpha\beta}. \quad (6)$$

The last inequality is the classical Kantorovich inequality in an equivalent form; see Lin [31, Eq. (1.1)]. It is sharp.

Proof. The sampling identity follows from $|b_i|^2/b_i = \bar{b}_i$ on the support of b . The second moment is $\|b\|^2 \sum_{i:b_i \neq 0} |(Pb)_i|^2$, which is bounded by the expression shown. Under the spectral weights of $b/\|b\|$, let Y be an eigenvalue of P and $t = \mathbb{E}Y$. Since $(Y - \alpha)(Y - \beta) \leq 0$,

$$\frac{\mathbb{E}Y^2}{(\mathbb{E}Y)^2} \leq \frac{(\alpha + \beta)t - \alpha\beta}{t^2} \leq \frac{(\alpha + \beta)^2}{4\alpha\beta}.$$

The last maximum occurs at $t = 2\alpha\beta/(\alpha + \beta)$. Equality is attained by $P = \text{diag}(\alpha, \beta)$ with squared coordinates of normalized b equal to $\beta/(\alpha + \beta)$ and $\alpha/(\alpha + \beta)$. This proves sharpness for this estimator, without asserting an algorithm-independent lower bound. $\square$

Theorem 3.4 (Relative scalar estimation). *Suppose H satisfies (1) and is d -row-sparse, and that row-sparse access to H and $\text{SQ}(b)$ are available. There is a randomized algorithm returning a relative- ϵ estimate of $q(H, b)$ with failure probability at most ζ , using*

$$O\left(\kappa\epsilon^{-2} \log(1/\zeta)(d+1)^{O(\sqrt{\kappa} \log(2/\epsilon))}\right) \quad (7)$$

queries and ideal arithmetic operations. Full matrix SQ is not needed.

Proof. For $\kappa = 1$, $H = I$ and the norm oracle returns $q = \|b\|^2$ exactly. Otherwise choose the polynomial in Lemma 3.1 with $\eta = \epsilon/4$ and put $Q = b^*P(H)b$. Then $|Q - q| \leq \eta q$. The eigenvalues of $P(H)$ lie in $[1 - \eta, \kappa(1 + \eta)]$, so their ratio is at most $9\kappa/7$, because $\eta \leq 1/8$. Lemma 3.3 gives $\text{Var}(\text{Re } X) \leq C\kappa Q^2$ for an absolute constant C .

Divide independent samples into an odd number $g = O(\log(1/\zeta))$ of groups, each of size $h = \lceil 64C\kappa/\epsilon^2 \rceil$. Chebyshev's inequality bounds by $1/4$ the probability that a group mean differs from Q by more than $\epsilon Q/4$. The median of the group means fails only if at least half the groups fail; the binomial Chernoff bound makes this probability at most ζ for a suitable absolute constant in g . On success,

$$|\hat{q} - q| \leq (\epsilon/4)Q + (\epsilon/4)q \leq (\epsilon/4)(2 + \epsilon/4)q < \epsilon q.$$

Each sample requires one local polynomial evaluation, covered by Lemma 3.2; multiplying its cost by gh proves the result. $\square$

Standard inverse-form quadrature uses full matrix-vector products [30]; the algorithm above explores only neighborhoods of sampled coordinates. The price is cost exponential in the degree: the local factor is $\exp(O(\sqrt{\kappa} \log(d+1) \log(2/\epsilon)))$, so the bound is not polynomial in κ .

Corollary 3.5 (Bilinear forms). *With the same matrix assumptions, $\text{SQ}(a)$ and coordinate access to b suffice to estimate $a^*H^{-1}b$ to additive error*

$$\epsilon \sqrt{(a^*H^{-1}a)(b^*H^{-1}b)}$$

*with the complexity in (7). If the public promise $|a^*H^{-1}b| \geq \gamma \sqrt{(a^*H^{-1}a)(b^*H^{-1}b)}$ holds for $0 < \gamma \leq 1$, replacing ϵ by $\gamma\epsilon$ gives relative error ϵ .*

Proof. Write $A = a^*H^{-1}a$ and $B = b^*H^{-1}b$. The zero-vector cases return zero. The residual representation gives

$$|a^*(P(H) - H^{-1})b| \leq \eta \sqrt{AB}.$$

Indeed, $P(H) - H^{-1} = H^{-1/2}EH^{-1/2}$ for a Hermitian E with $\|E\| \leq \eta$. Sampling from p_a , the random variable $X = \|a\|^2 (P(H)b)_I / a_I$ has mean $a^*P(H)b$ and second moment at most $\|a\|^2 \|P(H)b\|^2$. The pointwise spectral inequality $P(H)^2 \preceq \kappa(1+\eta)^2 H^{-1}$ and $\|a\|^2 \leq A$ bound this by $\kappa(1+\eta)^2 AB$. Apply the group-mean argument separately to real and imaginary parts, with tolerances $\epsilon\sqrt{AB}/4$ and failure probabilities $\zeta/2$. The sample count depends on the known upper bound for the ratio of variance to tolerance squared, so it does not require knowing A or B . Taking $\eta = \epsilon/4$ proves the claim. $\square$

3.3 Sampling a sparse inverse solution

Scalar estimation does not require a solution sampler. For comparison, the following elementary rejection scheme gives this stronger output.

Lemma 3.6 (Sparse-transform rejection sampler). *Let P be invertible, with at most R nonzeros per row and column and singular values in $[a_P, b_P]$. Given $\text{SQ}(v)$ and explicit materialization of the sparse rows and columns of P , one can sample exactly from p_{Pv} using at most $R(b_P/a_P)^2$ trials in expectation. Each trial materializes at most one row and one column and uses $O(R)$ further coordinate operations.*

Proof. Let $c_j^2 = \sum_i |P_{ij}|^2$ and $S_i = \sum_j |P_{ij}v_j|^2$. Sample $j \sim p_v$, accept with probability c_j^2/b_P^2 , and, if accepted, sample i from column j with probability $|P_{ij}|^2/c_j^2$. Compute $y_i = (Pv)_i$ and S_i from row i , and accept i with probability $|y_i|^2/(RS_i)$. The event $S_i = 0$ has zero probability at this stage and can be defined to reject. Column norm bounds and Cauchy–Schwarz show both acceptance probabilities lie in $[0, 1]$. Summing over j , the probability of outputting i in one raw trial is

$$\sum_j \frac{|v_j|^2}{\|v\|^2} \frac{c_j^2}{b_P^2} \frac{|P_{ij}|^2}{c_j^2} \frac{|y_i|^2}{RS_i} = \frac{|y_i|^2}{Rb_P^2 \|v\|^2}.$$

Conditioning on acceptance gives p_{Pv} . Its total probability is at least $a_P^2/(Rb_P^2)$, proving the expected trial count. $\square$

Theorem 3.7 (Solution sampling). *Let $A \in \mathbb{C}^{N \times N}$ be invertible with $\|A\| \leq 1$, $\sigma_{\min}(A) \geq K^{-1}$, and at most d nonzeros per row and column. Given row- and column-sparse access to A and $\text{SQ}(b)$, an algorithm can provide coordinate access to a fixed vector $\tilde{x}$ satisfying*

$$\|\tilde{x} - A^{-1}b\| \leq \epsilon \|A^{-1}b\|, \quad (8)$$

together with exact samples from $p_{\tilde{x}}$, at expected cost

$$K^2(d+1)^{O(K \log(2/\epsilon))} \quad (9)$$

per sample. For positive definite A , this improves to $K^2(d+1)^{O(\sqrt{K} \log(2/\epsilon))}$.

Proof. Apply Lemma 3.1 to $B = A^*A$, whose spectrum is contained in $[K^{-2}, 1]$, and let q be its inverse polynomial with residual $r = 1 - \lambda q(\lambda)$ bounded by ϵ . Put $P = q(A^*A)A^*$ and $\tilde{x} = Pb$. If $x = A^{-1}b$, then

$$Pb - x = -r(A^*A)x,$$

which proves (8). The singular value decomposition of A shows that the singular values of P are $|1 - r(\sigma^2)|/\sigma$; thus they lie in $[1 - \epsilon, K(1 + \epsilon)]$ and $\kappa(P) \leq 3K$. Lemma 3.2 materializes each needed row or column in cost $W = (d + 1)^{O(K \log(2/\epsilon))}$, and also bounds its nonzero count by W . Lemma 3.6 therefore costs $O(K^2 W^2)$ per sample, which has the stated form. For positive definite A , use $P = q(A)$ directly with degree $O(\sqrt{K} \log(2/\epsilon))$. When $K = 1$, the general matrix is unitary and $P = A^*$ is exact; the same rejection proof applies. $\square$

For fixed sparsity and accuracy, this upper bound implies that an N^c classical lower bound with the same access and output requirement needs $K = \Omega(\log N)$ in general and $K = \Omega(\log^2 N)$ in the positive definite case. These are necessary conditions; we do not claim hardness for every matrix at those scales. Andoni–Krauthgamer–Pogrow explicitly ask how to construct an ℓ_2 sampler for a solution from an ℓ_2 sampler for its right-hand side [4, Section 1.4]; their main results instead give coordinate estimates under different hypotheses and error guarantees. Theorem 3.7 states the sampling guarantee and rejection cost explicitly.

3.4 Nonsymmetric inverse forms and optimization values

Theorem 3.8 (Norm-sensitive inverse overlap). *Let M satisfy the general hypotheses of Theorem 3.7. Given $\text{SQ}(a)$ and coordinate access to e , the scalar $a^*M^{-1}e$ can be estimated to additive error $\epsilon \|a\| \|e\|$, with failure probability ζ , at cost*

$$O\left(R_e^2 \epsilon^{-2} \log(1/\zeta) (d + 1)^{O(K \log(2R_e/\epsilon))}\right), \quad (10)$$

where $1 \leq R_e \leq K$ is any supplied upper bound on $\|M^{-1}e\| / \|e\|$. Without a sharper bound, take $R_e = K$. Here $0 < \epsilon \leq R_e/2$, which extends the default range $\epsilon \leq 1/2$ of Section 2 when $R_e > 1$. The modulus $|a^*M^{-1}e|$ admits the same additive accuracy.

Proof. Take the normal-equation residual polynomial from the preceding proof with $\eta = \epsilon/(4R_e) \leq 1/8$, and put $P = q(M^*M)M^*$. Its error satisfies

$$\|Pe - M^{-1}e\| \leq \eta R_e \|e\|, \quad \|Pe\| \leq (1 + \eta) R_e \|e\|.$$

The estimator $X = \|a\|^2 (Pe)_I / a_I$, with $I \sim p_a$, has mean a^*Pe and second moment at most $(1 + \eta)^2 R_e^2 \|a\|^2 \|e\|^2$. The polynomial bias is at most $\epsilon \|a\| \|e\| / 4$. Separate real and imaginary median-of-means estimates give the claimed statistical error using $O(R_e^2 \epsilon^{-2} \log(1/\zeta))$ samples. Local evaluation costs $(d + 1)^{O(K \log(2R_e/\epsilon))}$. The inequality $\|z\| - \|\hat{z}\| \leq |z - \hat{z}|$ proves the modulus claim. $\square$

Corollary 3.5 normalizes the error by the H^{-1} -norms of a and b ; Theorem 3.8 normalizes it by the Euclidean norms of a and e . Neither gives a relative estimate of an overlap that can be small through cancellation. The parameter R_e is useful when an optimization value equals $|a^*M^{-1}e|$ and the construction makes the solution norm public.

3.5 What finite precision changes

Le Gall [29] studies robust dequantization under approximate length-squared sampling. The oracle bounds of this section assume exact sampling and ideal arithmetic; they do not claim that approximate sampling is free, and they give no bound on the bit precision an implementation needs. The following proposition gives a sufficient condition under which the estimator of Theorem 3.4, with approximate samples and inexact arithmetic, keeps relative error ϵ at failure probability 2ζ .

Proposition 3.9 (Robustness to approximate samples and arithmetic). *Run the polynomial estimator of Theorem 3.4 with $L = O(\kappa\epsilon^{-2}\log(1/\zeta))$ samples. Suppose the distribution of each requested sample, conditional on all previous computation, is within total-variation distance δ of p_b . Extend the estimator by the value zero on indices with $b_i = 0$, and have the approximate implementation return zero on those indices. On the support of b , suppose every reported real estimator value $\hat{X}_i$ differs from its exact value $\text{Re } X_i$ by at most $\tau \|b\|^2$. There is a coupling with an exact-oracle execution under which the final medians differ by at most $\tau \|b\|^2$, except on an event of probability at most $L\delta$. In particular, $\tau \leq \epsilon/4$ and $\delta \leq \zeta/L$ retain relative error ϵ and failure probability at most 2ζ for the constants used above.*

Proof. Successive maximal couplings make every sample agree with the exact-oracle sample except with probability at most $L\delta$, by a conditional union bound. An off-support approximate sample necessarily belongs to the disagreement event. On agreement, every group mean changes by at most $\tau \|b\|^2$; the median is Lipschitz with constant one in the maximum coordinate norm. Since $q \geq \|b\|^2$, this is at most τq . The proof of Theorem 3.4 leaves error at most $17\epsilon q/32$, so the additional $\epsilon q/4$ fits within the claimed tolerance. $\square$

If the norm and the sampled coordinate $b_i \neq 0$ are exact, the error condition of Proposition 3.9 holds when the computed polynomial coordinate satisfies

$$|(\widehat{P(H)b})_i - (P(H)b)_i| \leq \tau |b_i|.$$

Alternatively, interval arithmetic can bound the estimator error directly, including its division and norm multiplication. For finite rational input data and a rational spectral bound, the residual polynomial has rational coefficients. A finite local-walk sum and its sampled nonzero denominator can therefore be evaluated to any such tolerance, or exactly in rational arithmetic. This shows implementability, not a dimension-independent bit bound; such a bound must account for small sampled coordinates, input bit lengths, coefficient growth, and the cost of approximate sampling. In particular, boundedness of the polynomial on the spectral interval does not by itself justify a uniform working precision for divisions by arbitrarily small b_i . Proposition 3.9 is the finite-precision claim used here; the other oracle bounds assume ideal arithmetic.

4 Classical lower bounds with full sampling access

This section proves classical lower bounds for relative estimation of $q(H, b)$. Each bound counts queries to a hidden Boolean input; all other input data are public. We grant every operation in Definition 2.1 and verify its simulation, rather than infer sampling hardness from entry-query hardness. The parameters determine the dimensions of the hard instances; we claim no bound exceeding the cost of reading the whole finite input.

4.1 The exponential dependence

Write $a = 1/\kappa$, $m = (1 + a)/2$, $h = (1 - a)/2$, and

$$f_\kappa(x) = \frac{a}{m + hx}, \quad \rho = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}. \quad (11)$$

For a continuous real function f on $[-1, 1]$, write $\deg_\tau(f)$ for the least degree of a polynomial within uniform error τ of f . The classical minimax formula for the reciprocal gives

$$E_n(f_\kappa) = \frac{\kappa - 1}{2\kappa} \rho^n, \quad \deg_\tau(f_\kappa) = \Theta(\sqrt{\kappa} \log(1/\tau)) \quad (\kappa \geq 4, 0 < \tau \leq \tau_0). \quad (12)$$

Here E_n is the best uniform error of degree- n polynomials, including $n = 0$, and τ_0 is any fixed absolute constant in $(0, 3/8)$; see Kraus et al. [28, Corollary 2.2]. Indeed their reciprocal error, after multiplying by a , is $(\sqrt{\kappa} - 1)^2 \rho^{n-1}/(2\kappa)$ for $n \geq 1$, which equals the displayed expression. For $n = 0$ half the range of f_κ gives the same answer. The degree assertion follows because $\log(1/\rho) = \Theta(\kappa^{-1/2})$ and the prefactor lies in $[3/8, 1/2]$. This is a known bound, not a new result.

Theorem 4.1 (Relative inverse-form lower bound). *There is an absolute $\epsilon_0 > 0$ such that, for every $\kappa \geq 4$, integer $s \geq 5$, and $0 < \epsilon \leq \epsilon_0$, a family of real SPD matrices satisfying $aI \preceq H \preceq I$, $\kappa(H) = \kappa$, and row sparsity at most s has the following property. Returning relative- ϵ estimates of both $q(H, b_+)$ and $q(H, b_-)$ for two public unit vectors $b_\pm$ of support two, with success probability $2/3$, requires*

$$\Omega \left(\frac{((s-1)/2)^{\Omega(\sqrt{\kappa} \log(1/\epsilon))}}{\log(s) \sqrt{\kappa} \log(1/\epsilon)} \right) \quad (13)$$

classical queries, even with both full matrix SQ and sparse query access. Consequently a routine that estimates one form for every supplied b has the same worst-case lower bound, up to an absolute factor. The family has a sparse box-LP realization with public analytic center (Proposition 4.2) and a full-SQ constraint oracle using $O(1)$ hidden-input queries per request.

Proof. We use the matrix-function lower bound of Montanaro and Shao [32, Theorem 1.8, Corollary 5.1 and Section 5.1] (the theorem numbering refers to arXiv:2311.06999). For continuous $f : [-1, 1] \rightarrow [-1, 1]$, their construction gives a d -sparse real symmetric contraction A_y and distinct public indices i, j for which additive error $\tau/4$ in $(f(A_y))_{ij}$ costs

$$\Omega \left(\frac{(d/2)^{(\deg_{2\tau}(f)-1)/6}}{\log(d) \deg_\tau(f)} \right). \quad (14)$$

Their construction takes $d = 2^{r+1}$, with Hadamard gates of size $2^r = d/2$. For an arbitrary integer sparsity bound s , let 2^r be the largest power of two at most $(s-1)/2$; the diagonal shift below adds one nonzero per row. For $s \geq 5$, $r \geq 1$ and $2^{2r} \geq 2^{r+1} > (s-1)/2$, so $\log(2^r) \geq \frac{1}{2} \log((s-1)/2)$. The exponential rate thus drops by at most a factor 2, which the Ω in the exponent of (13) absorbs.

To extend their conclusion to full SQ access, we need the structure of their input. Their matrix is a weighted Feynman path

$$A_y = \sum_{t=1}^T \beta_t (|t\rangle\langle t-1| \otimes U_t + |t-1\rangle\langle t| \otimes U_t^T). \quad (15)$$

The positive weights β_t are public. Each real orthogonal U_t is a public block Walsh–Hadamard gate, a public identity, or a diagonal hidden-sign gate. The latter implements one query to either of two Boolean functions on n bits. The source problem distinguishes

$$\Phi(y) = \langle 0 | W D_1 W D_2 W | 0 \rangle \geq 3/5 \quad \text{from} \quad |\Phi(y)| \leq 1/100, \quad W = H_2^{\otimes n}, \quad (16)$$

where H_2 is the normalized two-dimensional Hadamard matrix. Its randomized query lower bound is $\Omega(2^{n/2}/n)$ [1, Theorem 1]. Factoring each W into local Hadamard layers produces the sparsity dependence in (14). The block-diagonal orthogonal change of basis (the gauge) with blocks $V_0 = I$ and $V_t = U_t \cdots U_1$ conjugates A_y to $J \otimes I$, where J is the public scalar Jacobi matrix with weights β_t . Thus its spectrum is public, and the relevant entry has the form $c\Phi(y)$ with a public $c \geq \tau$ (a public endpoint sign can be absorbed). In the even-function branch of the construction, the queried clock positions are the second and penultimate vertices; identity gates at both ends give the same formula. The gauge also shows that the diagonal entries at both queried positions are independent of y .

Set $H_y = mI + hA_y$. Then $aI \preceq H_y \preceq I$ and $f_\kappa(A_y) = aH_y^{-1}$. Adjoin the public scalar blocks $a, 1$ to make the condition number exactly κ , with right-hand sides zero on these blocks. The shift adds one nonzero per row. For $b_\pm = (e_i \pm e_j)/\sqrt{2}$, polarization gives

$$(f_\kappa(A_y))_{ij} = \frac{a}{2}(q(H_y, b_+) - q(H_y, b_-)).$$

Each form lies in $[1, \kappa]$, so relative ϵ errors propagate to additive error at most ϵ . Set $\tau = 4\epsilon$ in (14), use (12), and amplify the two scalar calls by a constant number of repetitions. Choosing ϵ_0 sufficiently small makes the degree numerator positive uniformly in κ . This proves the query bound.

It remains to show that full SQ access can be simulated. In every clock row, the supports of the two adjacent transitions and the diagonal are disjoint. A unitary transition has squared row and column mass one; a Hadamard row is uniform on a public support and a sign-gate row has one public location. Hence the squared row norm of H_y is $m^2 + h^2(\beta_t^2 + \beta_{t+1}^2)$, omitting nonexistent endpoint edges. A within-row sample first chooses one of these three parts in proportion to its public squared mass, then a uniform Hadamard location or the single sign location. Nonzero counts, locations and their ordering are public. Requested values need at most one sign query. Row-norm sampling first chooses a public clock position and then a uniform work index; the Frobenius norm is a public sum. The same argument applies to columns and the scalar pads. The public weight table can be preprocessed freely in this hidden-input query model. Both right-hand sides have public exact SQ. Therefore no granted norm or sampling request reveals any extra hidden information. The LP realization follows from Proposition 4.2 below. $\square$

Proposition 4.2 (Sparse analytic-center realization). *Every clock $H = mI + hA_y$ above has a factor $C_y^T C_y = H$ with $O(s)$ nonzeros per row and column and public full-SQ metadata (sparsity pattern, entry magnitudes, and row, column and Frobenius norms). The box LP $\max_x \eta b^T x$ subject to $-1 \leq C_y x \leq 1$ has at $x = 0$ the barrier Hessian $2H$ and squared Newton decrement $\eta^2 q(H, b)/2$.*

Proof. Put $d_0 = \sqrt{m}$, $e_t = h\beta_t/d_{t-1}$, and $d_t = \sqrt{m - e_t^2}$. These are positive: their squares are the pivots of the scalar positive definite tridiagonal matrix $mI + hJ$. The lower bidiagonal block matrix B with $B_{tt} = d_t I$ and $B_{t,t-1} = e_t U_t$ obeys $BB^T = H$; its diagonal blocks are $(d_t^2 + e_t^2)I = mI$ and its off-diagonal blocks are $e_t d_{t-1} U_t = h\beta_t U_t$. Take $C_y = B^T$, and factor the public scalar pads separately. Only a diagonal block and one transition meet any row or column. Their disjoint supports and public magnitudes prove both sparsity and the full-SQ simulation, exactly as above; no product $U_t \cdots U_1$ appears in this factor. The paired logarithmic barrier $\phi(x) = -\sum_j \log(1 - (C_y x)_j^2)$

has gradient zero and Hessian $2C_y^T C_y$ at zero. The factor has full column rank, so zero is the unique analytic center. For $\phi(x) - \eta b^T x$, the Newton gradient there is $-\eta b$, giving the asserted decrement. Choosing $\eta = 1/(4\sqrt{\kappa})$ makes its square at most $1/32$ for unit b . The barrier parameter grows with the number of interval constraints; this is a lower bound for estimating the decrement at one point, not an iteration lower bound. $\square$

4.2 Finite rational instances and public preprocessing

Theorem 4.1 only asserts that suitable real public clock weights exist. They need not carry infinite-precision information about the hidden input: the next proposition shows that short dyadic weights suffice, so the public data have a finite description and are computable, though not necessarily efficiently.

Proposition 4.3 (Rational realization). *For rational κ, ϵ and $s \geq 9$, the clock weights may be chosen dyadic with $O(\log(\kappa/\epsilon))$ fractional bits each, and (13) still holds after the base replacement $(s-1)/2 \mapsto \max\{4, (s-1)/8\}$. The matrix and two unit right-hand sides can be rational. An exactly rational $O(s)$ -sparse constraint factor also exists. A single terminating algorithm generates the public tables for all parameters. We claim no polynomial preprocessing time and no bound on the factor's bit length.*

Proof. Let $c \geq \tau$ be the public coefficient of the unrounded clock, where $\tau = \Theta(\epsilon)$ with a sufficiently large absolute constant, fixed below. Set $\theta = \tau/(128\kappa)$ and choose L with $2^{-L} \leq \tau/(256\kappa)$. Round each edge by

$$\hat{\beta}_t = 2^{-L} \lfloor 2^L(1 - \theta)\beta_t \rfloor.$$

The Jacobi matrix is nonnegative and bipartite, so its norm is its Perron root. Entrywise monotonicity gives $\|\hat{J}\| \leq 1 - \theta$, and the tridiagonal row-sum bound gives $\|J - \hat{J}\| \leq 2(\theta + 2^{-L})$. For symmetric contractions X, Y , the resolvent identity yields

$$\|f_\kappa(X) - f_\kappa(Y)\| \leq ah \|(mI + hX)^{-1}\| \|(mI + hY)^{-1}\| \|X - Y\| \leq \frac{\kappa}{2} \|X - Y\|. \quad (17)$$

Thus the coefficient changes by at most $3\tau/256$, and $|\hat{c}| \geq 253\tau/256$. An additive $\tau/4$ estimator of $\hat{c}\Phi$ estimates Φ with error at most $64/253$, strictly below half the promise gap 0.59 . In particular, rounding cannot disconnect the two queried clock positions. The gauge identity makes this perturbation bound uniform over every hidden input.

Choose a power-of-four Hadamard block size, so every normalized gate entry is rational; the largest allowed size is at least both 4 and $(s-1)/8$ for $s \geq 9$. Use the rational unit vectors $(3e_i \pm 4e_j)/5$. Their difference gives the entry through

$$a(H^{-1})_{ij} = \frac{25a}{48}(q_+ - q_-),$$

with propagated error at most $25\epsilon/24$. Taking, for example, $\tau = 5\epsilon$ suffices. The scalar pads are rational. Squared norms and sampling probabilities are rational and public; their square roots are public algebraic numbers, without hidden information.

For completeness, an exact rational constraint factor avoids the irrational square roots of a Cholesky factorization. Write the rounded off-diagonal coefficient as $w_t = h\hat{\beta}_t$. The block LDL^T factorization uses rational pivots

$$D_0 = m, \quad \ell_t = w_t/D_{t-1}, \quad D_t = m - w_t^2/D_{t-1} > 0, \quad L_{t,t-1} = \ell_t U_t,$$

with identity diagonal blocks. For $D_t = p_t/q_t$ in lowest positive terms, the four-square theorem gives integers $z_{t,r}$ such that $p_t q_t = \sum_{r=1}^4 z_{t,r}^2$. Replace the block row $\sqrt{D_t}(L^T)_{t,*}$ by the four rational block rows $(z_{t,r}/q_t)(L^T)_{t,*}$. Their Gram matrix is exactly LDL^T . Each row has one identity block and one gate block; column sparsity increases by at most four. Zero rows can be deleted publicly. All metadata remain public. The scalar pads are treated in the same way.

The finite table has $O(T \log(\kappa/\epsilon))$ fractional bits, where $T = \Theta(\sqrt{\kappa} \log(1/\epsilon))$. A terminating generator can enumerate dyadic tables of this clock length and test, using exact rational arithmetic, the strict spectral margin and the normalized coefficient condition

$$a \left| [(mI + h\hat{J})^{-1}]_{i,j} \right| > 9\tau/10.$$

Here i, j are the queried clock vertices: the endpoints in the endpoint branch, and the second and penultimate vertices in the even branch. Equivalently, after absorbing the known public endpoint sign, the test is $a[(mI + h\hat{J})^{-1}]_{i,j} > 9\tau/10$. Positivity of rational matrices is decidable by exact principal-minor or elimination tests; the tested quantity is the corresponding normalized entry of $f_\kappa(\hat{J})$. The rounded weights constructed above pass the test, so the search terminates. A four-square search then produces the constraint factor. This proves computability, but gives neither a polynomial bound for this search nor a stable, efficient inverse-spectral construction. Preparing the public data is free only because the theorem counts hidden-input queries. $\square$

The clock family also illustrates a caveat for coherent quantum access to the hidden signs. Its forms have the public representation $q_\pm = D \pm C\Phi$, with $|C| \leq D \leq \kappa$ by Cauchy–Schwarz for the positive definite inverse of the scalar clock. Put $q_* := \min_{\Phi \text{ in the source promise}} \pm(D \pm C\Phi)$; positivity gives $q_* \geq 1$ for the realizable inputs. Coherent amplitude estimation of the Forrelation circuit to additive error $O(\epsilon q_*/|C|)$ costs $O(1 + |C|/(\epsilon q_*))$ hidden-sign queries, and suffices for relative estimates of the two forms. If $|C| \leq \epsilon q_*$, outputting the public baseline D already suffices with no hidden query. On clocks with public coefficient $c = \Theta(\epsilon)$ we have $|C| = \Theta(\kappa\epsilon)$, so the cost in the remaining case is $O(\kappa/q_*)$. It is $O(1/\epsilon)$ when $\kappa\epsilon = O(1)$, or under an additional noncancellation bound $q_* = \Omega(|C|)$. A constant-query quantum algorithm for the Forrelation promise does not by itself estimate the forms numerically at that cost. We assume no uniform noncancellation bound for all minimax clocks.

4.3 The statistical factor, confidence, and finite dimension

Lemma 4.4 (Two Hamming spheres). *For integers $1 \leq t \leq M/4$ and $1 \leq \Delta \leq t/4$, distinguishing an M -bit string of Hamming weight t from one of Hamming weight $t + \Delta$ with success probability $2/3$ requires $\Omega(\min\{M, tM/\Delta^2\})$ randomized bit queries.*

Proof. Use the uniform measures on the two spheres. Conditional on any transcript with j distinct queries and z observed ones, the next fresh bit has probabilities $(t - z)/(M - j)$ and $(t + \Delta - z)/(M - j)$. Adaptively choosing its address does not change these probabilities. Stop a T -query transcript if $z \geq t/2$, where $T \leq cM$ for a small absolute c . The chance of this stopping event is at most $2T(t + \Delta)/(Mt) \leq (5/2)c$ under either measure, by the expected number of ones in sampling without replacement. Before stopping, both probabilities are at least $t/(2M)$, and both are bounded away from one. Their difference is $\Delta/(M - j)$. The elementary bound $D_{\text{KL}}(\text{Ber}(p) \parallel \text{Ber}(q)) \leq (p - q)^2/(q(1 - q))$ therefore bounds the stopped transcript's divergence by $CT\Delta^2/(tM)$, by the chain rule. Pinsker's inequality bounds its total variation by the square root of half this quantity. Restoring the stopped suffix changes total variation by at most the sum of the two stopping probabilities. For T a sufficiently small constant times $\min\{M, tM/\Delta^2\}$ the resulting

total variation is below $1/3$. No such transcript distinguishes the hypotheses with average success $2/3$. Internal randomness can be included in the transcript; equivalently apply the distributional minimax principle. $\square$

Theorem 4.5 (Statistical lower bound). *For $\kappa \geq 4$, $0 < \epsilon \leq 1/128$, and every integer $M \geq 1$, there are two-sparse real SPD matrices of dimension $2M$ and one public unit vector for which relative inverse-form estimation costs*

$$\Omega(\min\{M, \kappa/\epsilon^2\}) \quad (18)$$

classical full-SQ queries. If $0 < \zeta \leq 1/10$ and $M \geq C\kappa\epsilon^{-2}\log(1/\zeta)$, failure at most ζ requires $\Omega(\kappa\epsilon^{-2}\log(1/\zeta))$ queries. Both conclusions also hold for the squared Newton decrement of a box LP with a two-sparse constraint factor, at its public analytic center.

Proof. For $z \in \{0, 1\}$ set

$$B_z = \begin{pmatrix} m & (-1)^z h \\ (-1)^z h & m \end{pmatrix}, \quad H_x = \bigoplus_{j=1}^M B_{x_j}, \quad b = M^{-1/2} \bigoplus_{j=1}^M (1, 1)^T / \sqrt{2}.$$

Each block has eigenvalues $a, 1$. Its supports, entry magnitudes, row and column norms and Frobenius norm are independent of the sign. Thus every matrix or vector SQ operation is public, except a queried entry sign, which uses one input bit. The vector $(1, 1)^T / \sqrt{2}$ is an eigenvector of B_z with eigenvalue one for $z = 0$ and a for $z = 1$, giving the exact identity

$$q(H_x, b) = 1 + (\kappa - 1)|x|/M. \quad (19)$$

For $M \geq \kappa/(8\epsilon)$ put $u = M/\kappa$, $t = \lfloor u \rfloor$ and $\Delta = \lceil 8\epsilon u \rceil$. Then $u \geq 1/(8\epsilon)$, $t \leq M/4$, and $\Delta \leq 8\epsilon u + 1 \leq (u - 1)/4 \leq t/4$. The last inequality follows by taking the worst endpoint $\epsilon = 1/128$, $u = 16$, or directly by rearranging. The value at weight t is below two, and the increment is at least $8\epsilon(\kappa - 1)/\kappa \geq 6\epsilon$. Therefore the two relative-error intervals are disjoint. Moreover $t = \Theta(u)$ and $\Delta = \Theta(\epsilon u)$, so Lemma 4.4 proves (18). For smaller M , weights zero and one give an increment exceeding 6ϵ and again disjoint intervals; locating a uniformly hidden one costs $\Omega(M)$ queries, as required.

For confidence, use independent Bernoulli bits with parameters $p_0 = 1/\kappa$ and $p_1 = (1 + 16\epsilon)/\kappa$. Under the assumption $M \geq C\kappa\epsilon^{-2}\log(1/\zeta)$ of the theorem, Chernoff bounds give $|x_j| - Mp_j \leq 2\epsilon M/\kappa$ with probability at least $1 - \zeta/10$ for either j , by increasing C . On these events the scalar bands are separated by at least $12\epsilon(\kappa - 1)/\kappa \geq 9\epsilon$ and are below three; their relative- ϵ bands are disjoint. Thresholding the estimator thus tests the two product distributions with each error at most 1.1ζ . Each fresh addressed bit has the same Bernoulli law regardless of prior queries, and its KL divergence is at most $C'\epsilon^2/\kappa$. The transcript divergence for T queries is at most $C'T\epsilon^2/\kappa$. Data processing to the binary test bounds it below by $\Omega(\log(1/\zeta))$ uniformly for $\zeta \leq 1/10$. This proves the high-confidence assertion, including randomized algorithms by placing their random seed in the transcript. Public SQ requests contribute no divergence.

Finally take $u_0 = (1 + \sqrt{a})/2$, $v_0 = (1 - \sqrt{a})/2$ and $C_z = \begin{pmatrix} u_0 & (-1)^z v_0 \\ (-1)^z v_0 & u_0 \end{pmatrix}$. Then $C_z^2 = B_z$. Its full-SQ metadata are sign-independent and it is two-sparse. The box LP with factor $\bigoplus_j C_{x_j}$ has Hessian $2H_x$ and decrement $\eta^2 q(H_x, b)/2$ at zero, as before. $\square$

Corollary 4.6 (Coherent counting comparison). *For $\kappa \geq 4$, $0 < \epsilon \leq 1/128$, and $M \geq C\kappa/\epsilon$, the canonical coherent versions of the sign-block oracles of Theorem 4.5 (specified in the proof) require $\Omega(\sqrt{\kappa}/\epsilon)$ quantum queries, and $O(\sqrt{\kappa}/\epsilon)$ suffice on this family.*

Proof. The sparse-value oracle is reversibly simulable with a constant number of bit queries; all other unitaries are fixed public circuits. This fixes their unitary completions and excludes unspecified side information. The weight-decision consequence of Nayak and Wu [35, Theorem 1.1] is $\Omega(\sqrt{M/\Delta} + \sqrt{t(M-t)/\Delta})$ for the two weights used above. In the stated range $M \geq C\kappa/\epsilon$, its second term is $\Omega(\sqrt{\kappa}/\epsilon)$. For the upper bound prepare a uniform block index, query its bit and rotate an ancilla with success probability $1/\kappa$ or one according to that bit. The average success probability is $p = q(H_x, b)/\kappa \geq 1/\kappa$. Amplitude estimation [12, Theorem 12] estimates p relatively with $O(1/(\epsilon\sqrt{p}))$ calls and constant success probability. This simple upper bound relies on the public common eigenbasis of the blocks; the corollary does not give that complexity for general sparse inverse forms. $\square$

Theorems 4.1 and 4.5 give, on different families, lower bounds for the exponential and statistical factors of (7); separate lower bounds do not multiply. The next section proves product lower bounds for estimating a single inverse form and states which stronger products they do not establish.

5 When the statistical and exponential factors multiply

This section asks when the exponential and statistical factors of the upper bound apply to one family at once. Direct sums preserve spectra and sparsity, but do not automatically multiply query lower bounds. What matters is the relative variation of a *single positive form*, not an off-diagonal entry obtained by subtracting two forms. We prove a distributional product lower bound that does not require each block to take one of two exact values. We then show that three specific constructions, the cyclic clock $A = I - \gamma U$, the two-cluster construction, and a Schur-complement extension, cannot give the stronger product with a condition-dependent statistical factor. These results do not rule out other constructions.

5.1 Averaging and contrast

Suppose G_y satisfies $aI \preceq G_y \preceq I$ with a public unit vector v . For M inputs define

$$H_y = \bigoplus_{j=1}^M G_{y^{(j)}}, \quad b = M^{-1/2} \bigoplus_{j=1}^M v, \quad q(H_y, b) = M^{-1} \sum_j q(G_{y^{(j)}}, v). \quad (20)$$

If all inner instances have identical public norm tables and magnitude patterns, global SQ first selects a uniform block and then uses its inner SQ interface; norms, supports and sparsity within a block are unchanged. The same holds for sparse constraint factors: their direct sum realizes (20) as one analytic-center squared Newton decrement, multiplied by $\eta^2/2$.

Suppose each inner form takes one of two exact values, q_0 or $q_1 = q_0(1 + D)$, labelled by a partial Boolean function g of the inner input; we call $1 + D$ the contrast. An input with label-one fraction p has scalar $q_0(1 + Dp)$. In the high-contrast range $D \geq 1$, choosing $p = \Theta(1/D)$ keeps the scalar of order q_0 , while relative error ϵ resolves a change $\Theta(\epsilon/D)$ in p . Lemma 4.4 then gives an outer cost $\Omega(\min\{M, D/\epsilon^2\})$, and sampling blocks gives a matching upper bound up to confidence factors. Thus a statistical factor κ/ϵ^2 from this averaging construction requires contrast $1 + D = \Theta(\kappa)$, not merely a large absolute difference between two forms of order κ .

A *realization* of g is a family of SPD matrices G_y , indexed by the inputs y of g , with a public unit vector v . We call a realization *oracle-preserving* if every granted sparse count, location and value request, vector coordinate request, norm request, and sampling request can be simulated with $O(1)$ queries to the hidden input, with all remaining data public. This requirement also covers every

granted constraint-factor request. The constant is uniform over the problem parameters and the requested operations.

Proposition 5.1 (Conditional high-contrast reduction). *Let g be a nonconstant partial Boolean problem with bounded-error randomized query complexity $R(g) = L$. Suppose g has an oracle-preserving sparse SPD realization, with identical public full-SQ metadata, whose forms take the exact levels described above. For $D \geq 1$ and sufficiently small ϵ , one relative inverse-form estimate on a direct sum of $M = \Theta(D/\epsilon^2)$ blocks requires $\Omega(DL/\epsilon^2)$ queries.*

Proof. Choose a small fixed $\gamma > 0$, a weight $t = \lfloor \gamma M/D \rfloor$, and $\Delta = \lceil C_1 \epsilon M/D \rceil$, with C_1 an absolute constant large enough to separate relative output intervals. Choose $M = C_2 D/\epsilon^2$ rounded to an integer, and then choose ϵ sufficiently small and C_2 sufficiently large. These choices ensure $1 \leq \Delta \leq t/4$, $t \leq M/4$, $q_t = q_0(1 + Dt/M) = \Theta(q_0)$, and $q_{t+\Delta} - q_t \geq C_1 \epsilon q_0$. The outer partial function F distinguishes weights $t, t+\Delta$. Lemma 4.4 gives $R(F) = \Theta(M)$. Theorem 2 of Chakraborty et al. [16] explicitly allows a partial outer function and proves $R(F \circ g^M) = \Omega(R(F)R(g))$ whenever $R(F) = \Theta(M)$. Equation (20) and the disjoint relative output intervals reduce this composed decision problem to the scalar estimate with constant query overhead. $\square$

If the forms with a given label only lie within relative half-width η of their level, the weighted average can move by $\eta q_0(1 + Dp)$ even with all labels fixed. The reduction therefore remains valid for $\eta \leq c\epsilon$ with a sufficiently small absolute c . With one absolute half-width δq_1 for both levels, it requires $\delta \leq c\epsilon/(1 + D)$ instead. A constant-gap Forrelation promise satisfies neither narrow-level condition. Moreover, the unrestricted statement $R(F \circ g) = \Omega(R(F)R(g))$ is false for partial functions [8]; Proposition 5.1 relies on the linear outer complexity.

5.2 A distributional product for a single positive form

We first state the consequence of known composition theorems that we use. A noisy bit query of bias γ returns the correct bit with probability $(1 + \gamma)/2$ and costs γ^2 . Write $\text{noisyR}(F)$ for the minimum worst-case expected cost of a bounded-error algorithm in this model. Theorems 24 and 35, Definitions 33–34 and the minimax observation following Definition 34 of Ben-David and Blais [8] imply the following: for every nonconstant partial Boolean inner function g , there are distributions μ_0, μ_1 on its two fibers such that, for every outer F , there is a distribution ν on $\text{Dom}(F)$ for which computing F under

$$z \sim \nu, \quad Y_j \sim \mu_{z_j} \text{ independently conditional on } z \quad (21)$$

requires $\Omega(\text{noisyR}(F)R(g))$ expected queries. Here the inner distributions can be chosen to witness the Shaltiel-free randomized query complexity of g (their Definition 23: the largest Hellinger distinguishing cost, their Definition 22, of a pair of distributions on the two fibers of g), which their Theorem 24 lower-bounds by $\Omega(R(g))$. The order of quantifiers matters: one pair μ_0, μ_1 works for every outer F , and ν is chosen afterwards. The proof of their Theorem 35 establishes the statement for that fixed pair; this is stronger than a lower bound under some unstructured composed distribution. An expected-cost lower bound also covers estimators with a worst-case query bound. We use the theorem numbering of arXiv:2002.10809v2.

Theorem 5.2 (Distributional scalar product). *Let g be a nonconstant partial Boolean problem with an oracle-preserving SPD realization G_y and a public unit vector v such that*

$$q(G_y, v) = d + C\Phi(y), \quad C > 0, \quad d > 0, \quad \chi = C/d \leq 5/3, \quad (22)$$

where $|\Phi| \leq 1/100$ on zero inputs and $3/5 \leq \Phi \leq 1$ on one inputs. Assume the identical public full-SQ metadata needed for (20). If $0 < \epsilon \leq \chi/200$, there is a family of $M = 4\lceil \chi/\epsilon \rceil^2$ blocks for which relative ϵ estimation of the single averaged form requires

$$\Omega((\chi/\epsilon)^2 R(g)) \quad (23)$$

classical full-SQ queries. With sparse constraint factors for the blocks, the same bound holds for the squared Newton decrement of one box LP.

Proof. Set $n = \lceil \chi/\epsilon \rceil \geq 200$, $M = 4n^2$, and let F distinguish the weights $M/2 \pm 24\sqrt{M} = 2n^2 \pm 48n$. They are integers in $[0, M]$. The Hamming-sphere argument proving Lemma 4.4 applies equally for $t = \Theta(M)$ bounded away from both endpoints: before a small fraction of M queries, the two Bernoulli probabilities are bounded away from zero and one, so the per-query divergence is $O(1/M)$. Thus $R(F) = \Theta(M)$. Observation 23 of Chakraborty et al. [16] gives $\text{noisyR}(F) = \Omega(R(F)^2/M) = \Omega(M)$. The reverse bound follows by reading all bits.

Choose the distributions in (21). Let $\Delta = \mathbb{E}_{\mu_1} \Phi - \mathbb{E}_{\mu_0} \Phi \geq 59/100$. For each fixed outer input of one of the two weights, the conditional mean of $M^{-1} \sum_j \Phi(Y_j)$ depends only on that weight. Their means differ by $48\Delta/\sqrt{M} = 24\Delta/n$. Hoeffding's inequality, since every Φ lies in $[-1, 1]$, gives

$$\mathbb{P} \left[\left| M^{-1} \sum_j \Phi(Y_j) - \mathbb{E}[M^{-1} \sum_j \Phi(Y_j) \mid z] \right| > \frac{24\Delta}{4\sqrt{M}} \mid z \right] \leq 2e^{-24^2 \Delta^2 / 32} < 0.004.$$

Hence the two mean scalar values are separated by $S = 24\Delta C/n$, and each realized scalar is within $S/8$ of its mean except on the displayed event. Because $n \leq (201/200)\chi/\epsilon$, we have $S > 14\epsilon d$. Every scalar is at most $(1 + \chi)d < 3d$. Amplify the estimator's success to $11/12$ by a constant number of independent repetitions and a median. Its absolute error is then at most $3\epsilon d$, and $S/8 + 3\epsilon d < S/2$. Thresholding at the midpoint of the two public means computes F with distributional error below $1/3$. This threshold may depend on the fixed hard distributions; such nonuniform dependence is allowed in the distributional lower bound. The composition theorem therefore requires $\Omega(MR(g))$ queries. All scalar-input queries have constant-cost hidden-input simulations, proving the result. $\square$

We now apply Theorem 5.2 to the clock of Theorem 4.1. By gauge equivalence, the average of the two queried diagonal entries of the inverse is a public number d . The positive polarization form is exactly $q_+(y) = d + \kappa c \Phi(y)$, where c is the public normalized endpoint coefficient. Thus $C = \kappa c$ and $\chi_c = \kappa c/d$. We have $d \leq \kappa$, so $\chi_c \geq c$. Positive definiteness bounds the absolute cross entry by the average of its two diagonals; applying this at an input with $\Phi \geq 3/5$ gives $\chi_c \leq 5/3$. For a sufficiently small coefficient scale c , the source problem of the Montanaro–Shao clock has randomized query complexity

$$L_c = \Omega \left(\frac{((s-1)/2)^{\Omega(\sqrt{\kappa} \log(1/c))}}{\log(s) \sqrt{\kappa} \log(1/c)} \right). \quad (24)$$

Since clock sizes are discrete, we use an attainable coefficient within an absolute factor of the desired scale; this changes no bounds, and the explicit construction below shows that such coefficients exist. Theorem 5.2 gives $\Omega((\chi_c/\epsilon)^2 L_c)$ for a single form whenever $\chi_c \geq 200\epsilon$. In particular a fixed small coefficient gives $\epsilon^{-2} s^{\Omega(\sqrt{\kappa})}$ up to the displayed denominator, on one instance. For coefficient scales $c = c_0 \epsilon^\alpha$, $0 \leq \alpha < 1$, the conservative $\chi_c \geq c$ gives

$$\epsilon^{-2(1-\alpha)} s^{\Omega(\sqrt{\kappa}[1+\alpha \log(1/\epsilon)])},$$

again with the denominator of (24) and sufficiently small ϵ . Both factors of each bound hold simultaneously on one family, but these bounds do not establish the product of both full factors of (7).

5.3 A witness clock with an exact diagonal

We now compute the signal-to-baseline ratio χ for an explicit clock. The construction uses best approximation on a restricted interval to produce a dual witness on all of $[-1, 1]$; we do not claim that the restricted approximant is minimax on all of $[-1, 1]$.

Proposition 5.3 (Explicit witness diagonal). *For every $\kappa \geq 4$ and integer $r \geq 1$, there is an odd-order zero-diagonal Jacobi contraction J of order $2r + 5$ with positive edges, whose second and penultimate coordinates satisfy*

$$(f_\kappa(J))_{2,2r+4} = c_r, \quad (f_\kappa(J))_{2,2} = (f_\kappa(J))_{2r+4,2r+4} = d', \quad (25)$$

where

$$c_r = \frac{C\delta^r}{A^2 - 1}, \quad d' = \frac{CA}{A^2 - 1} = \frac{(\kappa + 1)(\kappa^2 + 14\kappa + 1)}{4\kappa(\kappa^2 + 6\kappa + 1)},$$

with $A = 4(m/h)^2 - 3$, $C = 4am/h^2$, and $\delta = A - \sqrt{A^2 - 1}$. In particular $1/4 < d' < 3/5$, $\chi_{c_r} = c_r/d' = \Theta(c_r)$, and $r = \Theta(\sqrt{\kappa} \log(1/c_r))$ at sufficiently small scales.

Proof. We give a full derivation because statistical composition depends on the diagonal, not only on the approximation degree. The even part of f_κ is $am/(m^2 - h^2x^2)$. On $x^2 \in [1/2, 1]$, set $t = 4x^2 - 3$; it becomes $q(t) = C/(A - t)$. Put $\omega = \sqrt{A^2 - 1}$. The degree- r reciprocal minimax error is $E_r = C\delta^r/\omega^2$. Write T_j and U_j for the first- and second-kind Chebyshev polynomials, respectively: $T_j(\cos \varphi) = \cos(j\varphi)$ and $U_j(\cos \varphi) = \sin((j+1)\varphi)/\sin \varphi$, with continuous extension at the endpoints. The scalar polynomial U_j is distinguished from the matrix-valued clock gate U_t by its argument and context. The normalized error is

$$V_r(t) = \frac{N_r(t)}{A - t}, \quad N_r(t) = (At - 1)T_r(t) + \omega(t^2 - 1)U_{r-1}(t).$$

To verify this expression, write $t = \cos \varphi$. The half-angle identity

$$\tan(\psi/2) = \sqrt{(A+1)/(A-1)} \tan(\varphi/2)$$

gives $V_r(t) = \cos(r\varphi + \psi)$. Also $N_r(A) = \omega^2(A + \omega)^r$, so $C - E_r N_r(t)$ vanishes at $t = A$; dividing by $A - t$ yields a degree- r polynomial. The phase increases strictly from zero to $(r+1)\pi$, producing $r+2$ alternating extremal points. The alternation theorem therefore verifies the minimax claim.

Let these points be $t_0 < \dots < t_{r+1}$, and define

$$R_r(t) = (At - 1)U_{r-1}(t) + \omega T_r(t), \quad L_r(t) = (t^2 - 1)R_r(t).$$

The interior extrema are exactly the r roots of R_r (use $\sin(r\varphi + \psi) = 0$). Thus the t_j are the simple roots of L_r , whose leading coefficient is positive, and $V_r(t_j) = \text{sgn } L'_r(t_j) =: \sigma_j$. Set $\mu_j = |L'_r(t_j)|^{-1} / \sum_k |L'_r(t_k)|^{-1}$. Partial fractions at infinity show

$$\sum_j \frac{P(t_j)}{L'_r(t_j)} = 0 \quad (\deg P \leq r). \quad (26)$$

Hence the signed weights $\sigma_j \mu_j$ annihilate every polynomial of degree at most r and pair with q to E_r . At $x_j = \sqrt{(t_j + 3)/4}$, put signed mass $\sigma_j \mu_j/2$ on each of $\pm x_j$. Odd moments cancel, and even moments through degree $2r$ vanish by (26). This is a full-domain dual witness supported inside $[-1, 1]$.

We now realize its cross entry explicitly, in the form underlying Montanaro and Shao [32, Lemma 2.5]. Put $B = (2 \sum_j \mu_j/x_j^2)^{-1}$ and choose the positive symmetric probability measure with mass $1/2$ at zero and $B\mu_j/(2x_j^2)$ at each $\pm x_j$. Gram–Schmidt on $1, x, \dots, x^{2r+4}$ gives its Jacobi matrix J . The symmetry makes its diagonal zero, positivity of the distinct-node measure makes all edges positive, and its eigenvalues are the support points, so it is a contraction. Since $\int x^2 dw = B$, the second orthonormal polynomial is $x/\sqrt{B}$; its spectral masses at $\pm x_j$ are $\mu_j/2$ and at zero are zero. Interpolate the values σ_j by a degree- $(r+1)$ polynomial $S(t)$. Equation (26) implies that $xS(4x^2 - 3)/\sqrt{B}$ is orthogonal to every lower odd polynomial and, by symmetry, every even polynomial. Its norm is one, since $S(t_j)^2 = 1$. Its leading coefficient is positive: the divided difference of the ordered alternating values is a sum of positive terms. Thus it is exactly the penultimate orthonormal polynomial, of degree $2r+3$. Its spectral masses agree with those of the second coordinate, and their signed cross masses are $\sigma_j \mu_j/2$. The cross entry is consequently $E_r = c_r$, and both diagonals are $d' = \sum_j \mu_j q(t_j)$.

To evaluate that average, partial fractions of N_r/L_r give

$$S_0 := \sum_j |L'_r(t_j)|^{-1} = \frac{N_r(A)}{L_r(A)}, \quad S_1 := \sum_j \frac{q(t_j)}{|L'_r(t_j)|} = -C \left(\frac{N_r}{L_r} \right)' (A).$$

Indeed $N_r(t_j)/(A - t_j) = \sigma_j$, and differentiating the partial-fraction expansion gives the second identity. If $\mathcal{S} = T_r(A) + \omega U_{r-1}(A) = (A + \omega)^r$, then $N_r(A) = \omega^2 \mathcal{S}$ and $L_r(A) = \omega^3 \mathcal{S}$. Using $T'_r = rU_{r-1}$ and $\omega^2 U'_{r-1}(A) = rT_r(A) - AU_{r-1}(A)$ gives

$$\frac{N'_r(A)}{N_r(A)} = \frac{r}{\omega} + \frac{A}{\omega^2}, \quad \frac{L'_r(A)}{L_r(A)} = \frac{r}{\omega} + \frac{2A}{\omega^2}.$$

It follows that $d' = S_1/S_0 = CA/(A^2 - 1)$, as claimed. The rational expression and $1/4 < d' < 3/5$ follow by substitution and multiplication of positive denominators. Finally $\log(1/\delta) = \text{arcosh}(A) = \Theta(\kappa^{-1/2})$ and $C/(A^2 - 1) = \Theta(1)$ for $\kappa \geq 4$. $\square$

In the lifted clock, place public identity gates before and after the hard circuit, so that its second and penultimate positions carry the Forrelation amplitude. Fitting an unpadding depth- $\Theta(r)$ circuit between these positions gives the exponent in (24); a bounded number of extra identity gates pads the length. Successive coefficients c_r have ratio δ , bounded below by an absolute constant for $\kappa \geq 4$, so there is always an attainable coefficient within a constant factor of any sufficiently small requested scale. The inverse baseline is $d = \kappa d'$. Consequently, this explicit family has statistical factor $\Theta((c_r/\epsilon)^2)$, with no additional factor growing with κ .

5.4 A constant-weight path at high accuracy

A clock on a path with constant edge weights has a smaller baseline and gives a different product.

Theorem 5.4 (High-accuracy path product). *There are absolute constants $c_* > 0$ and κ_0 such that for $\kappa \geq \kappa_0$, $0 < \epsilon \leq c_* \kappa^{-3/2}$, and $q = 2^r$ with integer $r \geq 1$ (the dimension of one r -qubit gate block, not the form $q(H, b)$), a full-SQ family with matrix sparsity at most $2q + 1$, condition number*

exactly κ , and one public right-hand side requires

$$\Omega\left(\frac{\kappa^2 q^{\ell/2}}{r\ell}\right), \quad \ell = \Theta\left(\sqrt{\kappa}\left[1 + \log \frac{1}{\kappa^{3/2}\epsilon}\right]\right) \quad (27)$$

queries to estimate its inverse form to relative error ϵ . It has an $O(q)$ -sparse box-LP realization with the same query simulation.

Proof. Take an n -vertex scalar path J_n with all off-diagonals $1/2$. Lift its transitions to the Forrelation circuit as in (15). Let $H = mI + hJ_y$, and append the scalar pads $a, 1$. Set $z_0 = m/h = (\kappa+1)/(\kappa-1)$, $R = z_0 + \sqrt{z_0^2 - 1} = (\sqrt{\kappa} + 1)/(\sqrt{\kappa} - 1)$, and $\theta = \log R$. The determinant recurrence, with $D_0 = 1, D_1 = m$, gives $D_n = (h/2)^n U_n(z_0)$. Cofactors give the normalized endpoint coefficient and diagonal

$$c_n = a|(H^{-1})_{1n}| = \frac{2a}{hU_n(z_0)}, \quad d'_n = a(H^{-1})_{11} = \frac{2aU_{n-1}(z_0)}{hU_n(z_0)}, \quad \chi_n = \frac{c_n}{d'_n} = \frac{1}{U_{n-1}(z_0)}. \quad (28)$$

The opposite endpoint has the same diagonal. Choose the sign of the public two-sparse vector to cancel the alternating cofactor sign; its form is $\kappa d'_n + \kappa c_n \Phi$. Since $U_j(z_0) = \sinh((j+1)\theta)/\sinh\theta$, we obtain

$$c_n = \frac{16\sqrt{\kappa}}{(\kappa-1)^2} \frac{1}{R^{n+1} - R^{-(n+1)}}, \quad \chi_n = \frac{4\sqrt{\kappa}}{\kappa-1} \frac{1}{R^n - R^{-n}}, \quad \frac{\chi_n}{c_n} = \frac{\kappa-1}{4} \frac{\sinh((n+1)\theta)}{\sinh(n\theta)}.$$

Uniformly for $n\theta \geq 1$ and $\kappa \geq 16$ this implies $c_n = \Theta(\kappa^{-3/2}e^{-n\theta})$, $d'_n = \Theta(\kappa^{-1})$ and $\chi_n = \Theta(\kappa c_n)$.

To make longer paths encode harder source problems, take a work register of $r\ell$ qubits, decompose each of the three full Hadamards into ℓ disjoint r -qubit layers, and use the two hidden sign queries. There are exactly $3\ell + 2$ transitions and $n = 3(\ell + 1)$ vertices. Choose the largest such n with $c_n \geq \epsilon$. For sufficiently small c_* this length has $n\theta \geq 1$. Consecutive admissible lengths differ by three and the displayed formula bounds their coefficient ratio by an absolute constant. Hence $c_n = \Theta(\epsilon)$, and solving the exponential formula gives the stated ℓ . Further, $\chi_n \geq (\kappa - 1)\epsilon/4$. Choose, for example, $\kappa_0 = 1024$ so $\chi_n \geq 200\epsilon$. The source Forrelation problem has randomized complexity $\Omega(2^{r\ell/2}/(r\ell))$. Apply Theorem 5.2 to $M = 4\lceil\chi_n/\epsilon\rceil^2 = \Theta(\kappa^2)$ blocks. This proves (27).

The unlifted path has norm less than one, so the affine shift obeys the promised spectral bounds, and the public pads make the condition exact. Every transition has public squared row and column mass $1/4$. Interior squared row norms of H are $m^2 + h^2/2$ and boundary norms are $m^2 + h^2/4$. All conditional and global samples are public mixtures of uniform Hadamard supports, a single sign location, and the diagonal; values cost one hidden-input query. Sparsity is at most $2q + 1$. Proposition 4.2 supplies the sparse factor with the same simulation. Direct sums preserve these properties. $\square$

The value of κ_0 only simplifies constants in applying Theorem 5.2. Theorem 5.4 is restricted to high accuracy: compared with the witness clock of Section 5.3 at the same coefficient scale, the endpoint-coefficient prefactor $\Theta(\kappa^{-3/2})$ of the constant path shortens the admissible hard circuit by order $\sqrt{\kappa} \log \kappa$. Its explicit outer factor κ^2 therefore does not improve the leading exponential term.

5.5 Proved limitations of particular contrast constructions

Proposition 5.5 (Contrast bound for cyclic clocks). *Let U be unitary on an L -position clock, and let e satisfy $U^L e = \sigma e$ for $\sigma \in \{1, -1\}$, with $e, Ue, \dots, U^{L-1}e$ orthonormal. For $0 < \gamma < 1$ set $A = I - \gamma U$, $G = AA^*/(1 + \gamma)^2$, and $K = (1 + \gamma)/(1 - \gamma)$. Then $\kappa(G) \leq K^2$ and*

$$\frac{q_+}{q_-} = \left(\frac{1 + \gamma^L}{1 - \gamma^L}\right)^2 \leq (1 + K/L)^2, \quad q_\sigma = e^* G^{-1} e. \quad (29)$$

Proof. The singular values of $I - \gamma U$ lie between $1 - \gamma$ and $1 + \gamma$. The finite geometric identity gives $A^{-1}e = (1 - \sigma\gamma^L)^{-1} \sum_{t=0}^{L-1} \gamma^t U^t e$. Since A is normal, $q_\sigma = (1 + \gamma)^2 \|A^{-1}e\|^2$. Orthogonality makes the numerator $\sum_t \gamma^{2t}$ common to both signs, proving the equality. Write the ratio as $\coth^2((L/2) \log((K+1)/(K-1)))$. The inequalities $\log((K+1)/(K-1)) \geq 2/K$ and $\coth x \leq 1 + 1/x$ prove the upper bound. $\square$

For a real circuit this is an ordinary real SPD construction. If each transition is q -sparse in rows and columns, G is $(2q + 1)$ -sparse. Public scalar pads can make its condition number exactly K^2 . Equation (29) bounds contrast by $O(1 + K^2/L^2)$ even for an *exact* sign relation $U^L e = \sigma e$. Since the final condition number is K^2 , contrast of that order requires bounded L . This limits the stated construction $A = I - \gamma U$, not all sparse linearizations.

Proposition 5.6 (Two-cluster and Schur-complement limitations). *The following two assertions hold.*

1. If $J_y = J_y^T$, $J_y^2 = I$, and orthogonal public unit vectors x, z satisfy $x^T J_y x = z^T J_y z = 0$, put $u_y = x^T J_y z$ and $G_y = mI + hJ_y$. For $v = (x + z)/\sqrt{2}$,

$$q(G_y, v) = 1 + \frac{\kappa - 1}{2}(1 - u_y). \quad (30)$$

Hence the promise $u_y \in [3/5, 1]$ does not keep the scalar bounded by $O(1)$ uniformly as κ grows. Such a uniform bound requires $1 - u_y = O(1/\kappa)$.

2. Let $G \succ 0$ have largest eigenvalue one and condition number K_0 . For an SPD extension $\mathcal{H} = \begin{pmatrix} \alpha & w^T \\ w & G \end{pmatrix}$ with Schur complement $S = \alpha - w^T G^{-1} w > 0$,

$$e_0^T \mathcal{H}^{-1} e_0 = S^{-1}, \quad \kappa(\mathcal{H}) \geq \frac{1 + \|G^{-1}w\|^2}{S}. \quad (31)$$

If $w = tv$ with $\|v\| = 1$, $v^T G^{-1} v \geq cK_0$, $w^T G^{-1} w \geq c_0 > 0$, and $S = \Theta(1/R)$, then $\kappa(\mathcal{H}) = \Omega(K_0 R)$.

Proof. Since $m^2 - h^2 = a$, $(mI + hJ)^{-1} = (mI - hJ)/a$. Substitution gives (30). At $u = 3/5$ its value is $\kappa/5 + 4/5$, while for $|u| \leq 1/100$ it is of order κ ; so the contrast between the two intervals is only an absolute constant, uniformly in κ . For the second assertion, block inversion gives the inverse entry. The vector $(1, -G^{-1}w)$ has Rayleigh quotient $S/(1 + \|G^{-1}w\|^2)$, and interlacing gives $\lambda_{\max}(\mathcal{H}) \geq 1$, proving (31). Finally Cauchy–Schwarz for the spectral measure of v gives $v^T G^{-2} v \geq (v^T G^{-1} v)^2$. Consequently

$$\|G^{-1}w\|^2 = t^2 v^T G^{-2} v \geq (w^T G^{-1} w)(v^T G^{-1} v) \geq c_0 c K_0.$$

Combine this with (31) and $S = \Theta(1/R)$. $\square$

Under the hypotheses of the Schur-complement construction, a bound $\overline{K}$ on the final condition number forces $R = O(\overline{K}/K_0)$. If the clock contributes a factor $\exp(A\sqrt{K_0})$ for fixed $A > 0$, maximizing over K_0 the condition-dependent part of the intended product gives

$$\max_{1 \leq K_0 \leq \overline{K}} \frac{\overline{K}}{K_0} e^{A\sqrt{K_0}} = \max\{\overline{K}e^A, e^{A\sqrt{\overline{K}}}\}.$$

Indeed the logarithm, in $t = \sqrt{K_0}$, is $\log \overline{K} - 2 \log t + At$, which is convex and hence maximized at an endpoint. This shows how the condition-number bound limits this specific construction. It is

not a general upper bound on sparse SPD query complexity, and it does not affect the factor ϵ^{-2} already proved in Theorem 5.2.

A similar limit applies to amplification inside the clock. If a coherent repetition or polynomial amplification procedure uses r_{amp} invocations of a depth- D source circuit, placing that procedure in a clock of length T requires $r_{\text{amp}}D \leq O(T)$. Standard repetition to reduce a constant decision error to $O(\epsilon/\kappa)$ uses $r_{\text{amp}} = O(\log(\kappa/\epsilon))$ invocations. Thus one cannot use the exponent of the original depth- T Forrelation circuit and also amplify its promise without lengthening the clock. We do not claim that every structure-specific amplification has this cost. Nor does a narrow acceptance-probability interval by itself give what applying Proposition 5.1 with contrast of order κ would require: an oracle-preserving sparse SPD realization with the narrow levels of that proposition whose contrast is of the order of its condition number.

In summary, we prove separate lower bounds on explicit families: (13), (18), (23), and (27). For worst-case complexity over the corresponding dimensions, their maximum is a valid lower bound. We do not claim the full product $\kappa\epsilon^{-2}S^{\Omega(\sqrt{\kappa}\log(1/\epsilon))}$. It would require more, for example a hard single-form family with contrast of order κ and the narrow levels of Proposition 5.1. The proved products and construction limitations do not depend on that open question.

6 Coherent access: a tight bound for block encodings

Quantum lower bounds for $b^*H^{-1}b$ depend on whether an algorithm can query matrix entries or only a unitary with H as a normalized block. For the latter we prove a lower bound (Theorem 6.1) that matches, up to logarithmic factors, the known upper bound of Chakraborty, Gilyén, and Jeffery, and we show that the hard instances become easy with exact entry queries.

An $(\alpha, a, \delta_{\text{in}})$ block encoding is a unitary U_H such that

$$\|H - \alpha(|0^a\rangle\langle 0^a| \otimes I)U_H(|0^a\rangle\langle 0^a| \otimes I)\| \leq \delta_{\text{in}}.$$

The algorithm may call U_H , $U_H^\dagger$, and their controlled versions. A second controlled unitary prepares $|b\rangle = b/\|b\|$, and $\|b\|$ is known. In the *plain block-encoding model*, these are the only oracles. The normalization $\alpha \geq 1$ is part of the input; it is not the condition number.

Theorem 6.1 (Relative inverse forms under block access). *Let $\kappa^{-1}I \preceq H \preceq I$, $\kappa \geq 4$, and $0 < \epsilon \leq 1/16$. At constant failure probability, the worst-case number of block queries needed to estimate $Q = b^*H^{-1}b$ to relative error ϵ is*

$$\Omega(\alpha\kappa/\epsilon) \quad \text{and} \quad \tilde{O}(\alpha\kappa/\epsilon).$$

The upper bound is a special case of the variable-time algorithm of Chakraborty, Gilyén, and Jeffery for negative-power norms [15, Theorem 33 in the full version]. With implementation costs T_H, T_b for the two controlled unitaries, its cost at failure probability ζ is

$$O\left(\frac{\log^3 \kappa}{\epsilon} \left[\alpha\kappa(T_H + a) \log^2 \frac{\kappa}{\epsilon} + \sqrt{\kappa} T_b \right] \log \frac{\log \kappa}{\zeta} \right). \quad (32)$$

Their theorem applies here when $\delta_{\text{in}} = o(\epsilon/[\kappa \log^3(\kappa/\epsilon)])$. The lower bound holds even for exact encodings in dimension three with public b .

Proof. Set $R_b = \|H^{-1/2}|b\rangle\|$. The cited theorem with negative-power parameter $c = 1/2$ estimates R_b multiplicatively. If $|\hat{R}_b/R_b - 1| \leq \epsilon/3$, then $|\hat{R}_b^2/R_b^2 - 1| \leq \epsilon$ for $\epsilon \leq 1$. Thus $\|b\|^2 \hat{R}_b^2$ estimates

Q , with the displayed cost. This invocation estimates a norm directly; preparing the normalized inverse state alone would not provide Q .

For the lower bound, let

$$H_z = \text{diag}(\kappa^{-1}, 2(1 + 4\epsilon z)/\kappa, 1), \quad b = e_2, \quad z \in \{0, 1\}.$$

Both matrices have condition number exactly κ , and $Q_0 = \kappa/2$, $Q_1 = \kappa/[2(1 + 4\epsilon)]$. Their relative-error intervals are disjoint, since $(1 - \epsilon)(1 + 4\epsilon) > 1 + \epsilon$. For a Hermitian contraction A , define the exact unitary completion

$$W(A) = \begin{pmatrix} A & \sqrt{I - A^2} \\ \sqrt{I - A^2} & -A \end{pmatrix}.$$

Take $U_z = W(H_z/\alpha)$. The only changing entry of H_z/α lies in $[0, 5/8]$, where $x \mapsto \sqrt{1 - x^2}$ has bounded derivative. Consequently $\|U_0 - U_1\| = O(\epsilon/(\alpha\kappa))$. Replacing the oracle calls one at a time bounds the Euclidean distance between final purified states of any T -query algorithm by $T\|U_0 - U_1\|$; inverse and controlled calls satisfy the same bound. Measurements and adaptive operations can be deferred or purified. Distinguishing the two disjoint output intervals with constant bias requires constant trace distance, and hence $T = \Omega(\alpha\kappa/\epsilon)$. $\square$

The lower bound fails for this family under exact sparse-value access: one query to the second diagonal entry reveals z . For coherent d -sparse access, the standard block encoding has normalization at most $d\|H\|_{\max} \leq d$ [23]. Together with Corollary 4.6, this bounds the worst-case number Q_{sparse} of coherent d -sparse queries for relative- ϵ estimation of Q over d -sparse H with $\kappa^{-1}I \preceq H \preceq I$, with the preparation of $|b\rangle$ counted separately, for $d \geq 2$, $\kappa \geq 4$, $0 < \epsilon \leq 1/128$, and dimensions large enough for the counting family:

$$\Omega(\sqrt{\kappa}/\epsilon) \leq Q_{\text{sparse}} \leq \tilde{O}(d\kappa/\epsilon). \quad (33)$$

These bounds do not match in general. The lower bound is proved for the two-sparse sign-block family, whose matrices commute and have public eigenvalues; the hidden input changes only how much weight b places on each eigenvalue. Its complexity $\Theta(\sqrt{\kappa}/\epsilon)$ therefore does not contradict Theorem 6.1, whose lower bound comes from an unknown eigenvalue.

6.1 Why a single polynomial transform loses a square-root factor

A simpler, weaker algorithm block-encodes $B = H^{-1/2}/(2\sqrt{\kappa})$ and applies it to $|b\rangle$. Its success probability is

$$p_b = \frac{\langle b|H^{-1}|b\rangle}{4\kappa} \in [1/(4\kappa), 1/4].$$

Amplitude estimation [12] gives additive error $O(\sqrt{p_b}/L + L^{-2})$ after L uses of this circuit. Relative error ϵ therefore needs $O(\sqrt{\kappa}/\epsilon)$ uses in the worst case. An error $\xi = O(\epsilon/\sqrt{\kappa})$ in the block encoding of B suffices, because it changes the success probability by at most $2\xi\sqrt{p_b} + \xi^2$. A negative-power polynomial approximation bounded on the whole signal domain $[-1, 1]$ [23] implements this transform with $\tilde{O}(\alpha\kappa)$ block queries, for a total cost of $\tilde{O}(\alpha\kappa^{3/2}/\epsilon)$. The variable-time algorithm behind (32) avoids multiplying these two worst-case costs.

Proposition 6.2 (Bounded-transform obstruction). *Suppose $\kappa \geq 8$, $\alpha \geq 1$, and a degree- n complex polynomial P satisfies $|P(x)| \leq 1$ for every $x \in [-1, 1]$. If P approximates $1/(2\sqrt{\alpha\kappa x})$ to absolute error $1/32$ at both $x_1 = 1/(\alpha\kappa)$ and $x_2 = 4/(\alpha\kappa)$, then $n = \Omega(\alpha\kappa)$. The same conclusion holds for a bounded Laurent transform of the signal phase with bandwidth n .*

Proof. The target values are $1/2$ and $1/4$, so the real part of P changes by at least $3/16$ on an interval of length $3/(\alpha\kappa)$. At some $\xi \in [x_1, x_2] \subset [0, 1/2]$, its derivative has magnitude at least $\alpha\kappa/16$. Bernstein’s polynomial inequality gives $|\operatorname{Re} P'(\xi)| \leq n/\sqrt{1-\xi^2} \leq 2n/\sqrt{3}$. For a Laurent transform $F(e^{i\theta})$ bounded by one on the unit circle, the two phases $\arccos x_1, \arccos x_2$ are separated by $\Theta(1/(\alpha\kappa))$. The trigonometric Bernstein inequality $\|dF/d\theta\|_\infty \leq n \|F\|_\infty$ gives the same bound. $\square$

The proof uses the elementary Bernstein inequalities only for bounded polynomials and trigonometric polynomials. The obstruction is interior: the relevant eigenvalues of H/α occupy a small part of the domain on which the transform must stay bounded. Removing the parity restriction of QSVT, for example by generalized signal processing [34], keeps this boundedness and so does not remove the obstruction. The proposition gives no lower bound for variable-time, rational, or postselected algorithms.

Orsucci and Dunjko also show that the access model matters [36, Proposition 6 and Sections 4.2–4.3]. They prove a worst-case $\Omega(\kappa)$ query lower bound for preparing solution states of positive definite systems under their matrix oracles. With a normalized encoding of $I - \eta A$ instead, their inverse approximation has degree $\tilde{O}(\sqrt{\kappa})$ for fixed $\eta > 0$, because its steep part lies near an endpoint of the signal domain. This shifted encoding is an additional access assumption; obtaining it from an arbitrary encoding of A is not free. Their output is a solution state and their construction uses shifted access, whereas we estimate a scalar to relative error with an unshifted bounded transform. The square-root polynomial degree also omits their state-preparation cost.

6.2 Relation to earlier scalar quantum algorithms

Scalar readout of a linear-system solution is prior work. Alase et al. [2] study additive estimation of x^*Cx for $Ax = b$. For expectation values in a given state they prove the tight complexity $\Theta(\alpha_C/\epsilon)$ in queries to an encoding of the *observable* C , and use it to lower-bound the linear-system scalar task. They also explicitly distinguish block and entry access for a fixed computational-basis state: their block-query lower bound still holds, whereas one exact-entry query suffices [2, discussion after Theorem IV.13 and Appendix A]. Our lower bound counts queries to the inverted matrix H , concerns relative error, and gains an additional factor κ from an eigenvalue that depends on the hidden bit. We claim neither the first scalar-output quantum algorithm nor the negative-power estimator. Theorem 6.1 contributes the matching condition-number dependence for relative estimation of the inverse form, and an explicit example of the block–entry access difference for this task. Likewise, a constant-query quantum decision of an underlying Forrelation promise is not automatically a constant-query estimator of every associated inverse form; see the discussion of q_* after Proposition 4.3.

7 Scalar outputs of sparse conic and linear programs

The matrix lower bounds of the preceding sections need not disappear when the scalar must be an ordinary optimization output: we construct sparse SOCPs and LPs in which a designated optimizer coordinate, the optimal value, or a Newton decrement encodes the same hidden Forrelation amplitude. We also determine how the bounds change when the output is a sample. The clock construction builds on the Forrelation matrix-function reductions of Montanaro and Shao [32] and the inverse-sampling separation of Grønlund and Larsen [25]; neither is new here.

7.1 A cyclic inverse with public norms

Let $q = 2^r$, $r \geq 1$, and let $n = r\ell$ be the number of work qubits. A two-function Forrelation circuit V has three global Walsh–Hadamard transforms and two hidden diagonal sign gates. Its real amplitude $\Phi = \langle 0^n | V | 0^n \rangle$ obeys the promise

$$|\Phi| \leq 1/100 \quad \text{or} \quad \Phi \geq 3/5.$$

The randomized query lower bound is $\Omega(2^{n/2}/n)$ [1]. Split each global transform into ℓ transforms on disjoint r -qubit blocks. Padding with at most one identity layer makes the number T of forward layers even, where $T = 3\ell + 2$ or $3\ell + 3$. Each layer is q -sparse.

Use a clock of length $L = 3T$. The transitions run V forward, hold its final state for T steps, then run its gates in reverse transposed order. Let U be this real orthogonal clock transition. Let W_t be the product of the transitions up to clock time t , with $W_0 = I$. The orthogonal gauge $W = \sum_{t=0}^{L-1} |t\rangle\langle t| \otimes W_t$ satisfies

$$W^T U W = S \otimes I, \quad U^L = I, \quad (34)$$

where $S|t\rangle = |t+1 \bmod L\rangle$. In particular $W_t = V$ for $T \leq t < 2T$. For $K \geq 3$, define

$$\gamma = \frac{K-1}{K+1}, \quad \alpha_K = \log \frac{K+1}{K-1}, \quad M = \frac{I - \gamma U}{1 + \gamma}, \quad e = |0, 0^n\rangle. \quad (35)$$

Here M need not be symmetric; the associated Newton Hessian will be SPD. Since L is even, the spectrum of S contains 1 and -1 . Thus

$$\|M\| = 1, \quad \sigma_{\min}(M) = K^{-1}, \quad \kappa_2(M) = K, \quad \text{nnz}(M_{i,:}), \text{nnz}(M_{:,j}) \leq q + 1. \quad (36)$$

Lemma 7.1 (Inverse history and full sampling access). *Write $z_0 = \gamma^{3T}$, $S_0 = \sum_{t=0}^{3T-1} \gamma^{2t}$, $S_I = \sum_{t=T}^{2T-1} \gamma^{2t}$, and $z = \gamma^{2T}$. Then*

$$x_* := M^{-1}e = \frac{1+\gamma}{1-z_0} \sum_{t=0}^{3T-1} \gamma^t U^t e, \quad R^2 := \|x_*\|^2 = K \frac{1+z_0}{1-z_0}, \quad (37)$$

$$p := S_I/S_0 = \frac{z}{1+z+z^2}, \quad K_{\text{eff}} := K/R = \sqrt{K \frac{1-z_0}{1+z_0}}. \quad (38)$$

Each call to a full SQ oracle for M , for

$$H_0 = 2M^T M = \frac{2}{(1+\gamma)^2} [(1+\gamma^2)I - \gamma(U + U^T)], \quad (39)$$

or for $M^T e$ can be simulated with at most a constant number of hidden sign queries. The Hessian has condition number K^2 and sparsity at most $2q + 1$.

Proof. Multiplying the finite geometric series by $I - \gamma U$ proves the inverse formula. Different clock positions are orthogonal and each $U^t e$ is a unit vector. Hence $R^2 = (1+\gamma)^2 S_0 / (1-z_0)^2$; summing the geometric progression and using $(1+\gamma)/(1-\gamma) = K$ gives (37). Partitioning S_0 into three equal clock segments gives (38). Normality of U gives (39) and the spectrum assertion.

For the access claim, the diagonal, forward-clock, and reverse-clock supports are disjoint. A transition is either a public Hadamard block, a hidden diagonal sign, or an identity. Each of its rows and columns has squared norm one, and its entry magnitudes and locations are public. Therefore

every row and column of M has squared norm $(1 + \gamma^2)/(1 + \gamma)^2$; every row and column of H_0 has the same public squared norm $4[(1 + \gamma^2)^2 + 2\gamma^2]/(1 + \gamma)^4$. Their row-norm sampling distributions are uniform. A within-row sample chooses a clock block with public probabilities, then samples a Hadamard row uniformly or returns the single location of a sign entry. The corresponding column rules are identical. Locations and norms need no hidden-input query; a value uses at most one. The vector $M^T e$ is a public-norm mixture of the clock-zero basis vector and a row of the last transition. The same simulation supplies its SQ interface. $\square$

Fix a sufficiently small constant $\delta_0 > 0$. For $0 < \delta \leq \delta_0$, choose an allowed T nearest to $\log(1/\delta)/\alpha_K$. Allowed lengths have bounded gaps and $\gamma \geq 1/2$, so, with universal constants,

$$p = \Theta(\delta^2), \quad R = \Theta(\sqrt{K}), \quad \ell = \frac{\log(1/\delta)}{3\alpha_K} + O(1), \quad N = 3Tq^\ell. \quad (40)$$

Taking δ_0 small ensures $\ell \geq 1$. All lower bounds below hold for the dimensions N selected in this way, not for every prescribed N .

7.2 An optimizer coordinate and approximately optimal feasible points

Let $Q_{N+1} = \{(t, u) : t \geq \|u\|\}$ be the Lorentz cone. Consider

$$\max_{x, u} 2e^T u \quad \text{subject to} \quad u = Mx, \quad (1, u) \in Q_{N+1}. \quad (41)$$

Its unique optimizer is (x_*, e) , and its optimal value is two for every hidden input, so estimating this value is trivial. Substituting $u = Mx$ gives the barrier $\phi(x) = -\log(1 - \|Mx\|^2)$. This barrier has parameter one, whereas the standard barrier of the full Lorentz cone has parameter two. It is self-concordant as an affine restriction. In the coordinates $u = Mx$, the squared dual local norm of its gradient is $2\|u\|^2/(1 + \|u\|^2) \leq 1$, proving the parameter bound. The analytic center is zero for every hidden input. At this center, for $F_\eta(x) = \phi(x) - 2\eta e^T Mx$,

$$\nabla F_\eta(0) = -2\eta M^T e, \quad \nabla^2 F_\eta(0) = H_0, \quad d_\eta = \eta M^{-1} e, \quad \Lambda_\eta^2 = 2\eta^2. \quad (42)$$

For $\eta < 1$, this full Newton step is strictly feasible; for $\eta \leq 1/(4\sqrt{2})$ its decrement is at most $1/4$. The central path is $x(\eta) = \rho(\eta)x_*$, where $\rho/(1 - \rho^2) = \eta$ and $0 < \rho < 1$. These identities concern the reduced Hessian, not an augmented KKT matrix with redundant free variables.

For t in the hold segment, or plateau, $I := \{T, \dots, 2T - 1\}$, where $W_t = V$, set $i_t = (t, 0^n)$ and

$$a = \sum_{t \in I} \frac{\gamma^t}{R\sqrt{S_I}} e_{i_t}, \quad h = \sqrt{p}. \quad (43)$$

Then $\|a\| = 1/R$ and $a^T x_* = h\Phi$. To make this linear observable an optimizer coordinate, we append free variables satisfying

$$y_T = a_{i_T} x_{i_T}, \quad y_{t+1} = y_t + a_{i_{t+1}} x_{i_{t+1}}, \quad w = y_{2T-1}. \quad (44)$$

The chain has $O(T)$ rows of at most three entries. Each selected column of x gains one nonzero entry. These equalities and their norms are public, so full SQ access to the augmented formulation mixes public data with the interface of Lemma 7.1. The maximum row and column sparsity remains $s = \Theta(q)$. The free accumulator variables add no cone and can be eliminated when computing barrier derivatives.

Theorem 7.2 (Sparse scalar coordinate separation). *For the parameters in (40), estimating the optimizer coordinate $w_* = h\Phi$ to additive error $c\delta$, for a sufficiently small universal c , requires*

$$Q_{\text{classical}} = \Omega\left(\frac{q^{\ell/2}}{r\ell}\right) \quad (45)$$

full-SQ input queries at bounded error. The same bound applies to returning this coordinate of any exactly feasible point with objective gap $\tau \leq c_0\delta^2/K$, for a sufficiently small constant c_0 . A coherent input oracle is source-equivalent if each of its calls can be simulated with $O(1)$ hidden Forrelation sign queries and vice versa. With such an oracle, estimating the exact optimizer coordinate to the stated accuracy takes $O(1)$ hidden queries and $\text{polylog } N$ public gates.

Proof. The coordinate identity follows by taking the overlap of (37) with (43). Its low and high intervals are separated by $(3/5 - 1/100)h = \Theta(\delta)$. Each formulation query costs a constant number of hidden sign queries, so the Forrelation lower bound proves (45). If $(\hat{x}, \hat{u})$ is exactly feasible and has gap τ , then

$$\|\hat{u} - e\|^2 \leq 2 - 2e^T \hat{u} \leq \tau, \quad |a^T \hat{x} - h\Phi| \leq \frac{K}{R} \sqrt{\tau} \leq \sqrt{K\tau}.$$

Exact accumulator feasibility equates the returned coordinate with $a^T \hat{x}$. The specified τ therefore preserves the promise gap. Points that violate the equalities or the cone constraint by a small residual would need a separate error analysis.

A coherent query to a designated transition entry recovers any desired hidden sign, and conversely each coherent matrix-oracle call uses a constant number of sign queries. Run the source circuit and estimate its real amplitude to a sufficiently small constant error, then multiply by public h . $\square$

The feasible-point lower bound uses such a point only as a source of one accurate coordinate, which the quantum algorithm of Theorem 7.2 estimates without outputting the vector.

Since $\alpha_K = \Theta(1/K)$, the logarithm of (45) is $\Omega(K \log s \log(1/\delta))$, even with the denominator $r\ell$ and the $O(1)$ rounding term in ℓ kept. This exponent grows with $\log s$, not with the ambient dimension. With k -Forrelation, which uses a fixed number k of hidden sign tables, the lower bound becomes $\tilde{\Omega}_k(N^{1-1/k})$, a larger power of N for $k \geq 3$ (Corollary 7.3).

Corollary 7.3 (Fixed-order Forrelation). *Fix $k \geq 2$. Replace V by the k -Forrelation circuit and use the primary promise*

$$|\Phi| \leq \alpha_k := 2^{-5k-1} \quad \text{or} \quad \Phi \geq \beta_k := 2^{-5k}.$$

Then $T = (k+1)\ell + O(k)$ and

$$\ell = \frac{\log(1/\delta)}{(k+1)\alpha_K} + O_k(1).$$

At coordinate accuracy $c(\beta_k - \alpha_k)\delta$, the lower bound is

$$\Omega_k\left(\frac{q^\ell}{r\ell}\right)^{1-1/k} = \tilde{\Omega}_k(N^{1-1/k}). \quad (46)$$

For exactly feasible points, an objective gap of $c_k\delta^2/K$, with $c_k = 2^{-O(k)}$, suffices for the same decision. The coherent scalar algorithm uses $2^{O(k)}$ hidden-input queries.

Proof. Theorem 1.3 and Corollary 1.4 of Bansal and Sinha [6] give, for fixed k , the stated randomized lower bound and a $2^{O(k)}$ -query bounded-error algorithm. A problem whose answer depends only on $|\Phi|$ may also allow the high case $\Phi \leq -\beta_k$; this only enlarges the promise domain, so the lower bound still applies. The $k+1$ global Hadamards give the claimed clock length. All remaining identities and oracle simulations are unchanged. Estimating the amplitude to error $O(\beta_k - \alpha_k)$ proves the numerical upper bound, and the feasible-point error $\sqrt{K\tau}$ proves the last accuracy claim. $\square$

7.3 A hard optimal value from a tilt with public norm

The optimal value of (41) is public, but adding a public tilt to the objective makes the value depend on Φ . Define

$$g = M^{-T}a, \quad G = \|g\|.$$

Although g depends on the hidden matrix M , the plateau I makes G exactly public (Lemma 7.4), so the quadratic tilt term in the optimal value is known rather than an unknown error.

Lemma 7.4 (Public tilt norm and plateau optimality). *Let $m = (I - \gamma S)/(1 + \gamma)$ and $\bar{a} = \sum_{t \in I} \gamma^t |t\rangle / (R\sqrt{S_I})$. Then*

$$G = \|m^{-T}\bar{a}\| = \Theta(\sqrt{K}), \quad h \leq G \leq K/R \leq \sqrt{K}. \quad (47)$$

For any clock vector $\bar{c}$ supported on I , with $b_c = m^{-T}\bar{c} \neq 0$,

$$\frac{|(b_c)_0|}{\|b_c\|} \leq \left(\sum_{t=0}^{T-1} \gamma^{-2t} \right)^{-1/2} = \Theta(\delta/\sqrt{K}). \quad (48)$$

Thus, up to constant factors, the readout (43) maximizes the normalized inverse-transpose overlap (48) among clock readouts supported on the plateau I .

Proof. On the plateau $W_t = V$. Applying (34) gives

$$W^T a = \bar{a} \otimes V^T |0^n\rangle, \quad W^T g = (m^{-T}\bar{a}) \otimes V^T |0^n\rangle.$$

Hence G does not depend on the hidden input. Its value follows from the finite geometric inverse of m^T , requiring no hidden-input queries. If $b = m^{-T}\bar{a}$, then $b_0 = \bar{a}^T m^{-1} |0\rangle = \sqrt{S_I/S_0} = h$. Since $\bar{a}_t = 0$ for $0 \leq t < T$, the equation $b_t - \gamma b_{t+1} = 0$ gives $b_t = h\gamma^{-t}$ there. Hence

$$G^2 \geq h^2 \sum_{t=0}^{T-1} \gamma^{-2t} = \frac{1-z}{1+z+z^2} \frac{\gamma^2}{1-\gamma^2} = \frac{1-z}{1+z+z^2} \frac{(K-1)^2}{4K}. \quad (49)$$

For the chosen $z = \Theta(\delta^2)$ this is $\Omega(K)$. The upper bound is $\|m^{-1}\| \|\bar{a}\| = K/R$. The same homogeneous recurrence for b_c proves (48); evaluating its geometric sum gives the last order estimate. For any public readout on $(t, 0^n)$ over this plateau, its hidden overlap is $(b_c)_0 \Phi$ and its direction norm is $\|b_c\|$. Our choice has ratio $h/G = \Theta(\delta/\sqrt{K})$. $\square$

Theorem 7.5 (Optimal value and direct Newton decrement). *Keep the accumulator equalities (44) and replace the objective in (41) by $e^T u + \lambda w$, where $\lambda = 1/(2G)$. Then*

$$\text{OPT}^2 = \frac{5}{4} + \frac{h}{G} \Phi, \quad \frac{1}{2} \leq \text{OPT} \leq \frac{3}{2}. \quad (50)$$

Estimating the optimal value to additive error $c(\beta - \alpha)\delta/\sqrt{K}$ has lower bound (45) for the two-function family and (46) for fixed-order Forrelation, where α and β are the corresponding source thresholds. With the source-equivalent coherent oracle, an algorithm attains that accuracy with $O(1)$ or $2^{O(k)}$ hidden-input queries, respectively.

At the public analytic center, with objective multiplier $\eta > 0$, the squared Newton decrement is

$$\Lambda_\eta^2 = \frac{\eta^2}{2} \left(\frac{5}{4} + \frac{h}{G} \Phi \right). \quad (51)$$

The same lower bounds hold for estimating Λ_η^2 to additive error $c(\beta - \alpha)\eta^2\delta/\sqrt{K}$, even when the algorithm is given full SQ access to the reduced Hessian and its right-hand side.

Proof. Eliminating the equalities gives $w = g^T u$. Maximizing $(e + \lambda g)^T u$ on the unit ball gives $\text{OPT} = \|e + \lambda g\|$. Since $e^T g = h\Phi$ and $\lambda^2 G^2 = 1/4$, this proves (50). The bound $h/G \leq 1$ proves its value range. The low and high cases have respectively absolute squared deviations from $5/4$ at most $\alpha h/G$ and at least $\beta h/G$. On $[1/2, 3/2]$ squaring and taking square roots have constant Lipschitz bounds, proving the accuracy transfer. The objective has only the public e and w coefficients, so its access adds no hidden-input query. The coherent algorithm outputs $\sqrt{5/4 + (h/G)\Phi}$ after estimating Φ to a small multiple of $\beta - \alpha$; clipping $\hat{\Phi}$ to $[-1, 1]$ is harmless.

The reduced gradient is $-\eta c$, where $c = M^T e + \lambda a = M^T(e + \lambda g)$. Multiplying by $(2M^T M)^{-1}$ gives (51). The supports of $M^T e$ and a occupy disjoint public clock regions, and both component norms are public. Thus $\text{SQ}(c)$ is a known mixture, with at most one hidden-input query for a value. Lemma 7.1 supplies full SQ access to the Hessian. This proves the last claim. $\square$

The eliminated tilt coefficients stay bounded: every singular value of m^{-T} is at least one, so $G \geq \|\bar{a}\| = 1/R$ and hence $\|\lambda a\| \leq 1/2$. The optimality statement in Lemma 7.4 covers only readouts on this plateau with a normalized objective direction of constant scale, and does not rule out other scalar reductions. For comparison, if only the bound $G^2 \leq K$ were used, one could take the smaller public tilt $\lambda = \theta h/K$. Then

$$\text{OPT}_\lambda^2 = 1 + 2\theta h^2 \Phi / K + \theta^2 h^2 G^2 / K^2.$$

Taking $\theta \leq c(\beta - \alpha)$ separates the cases at accuracy $\Theta(\theta(\beta - \alpha)\delta^2/K)$, even without computing G . Using the exact public G therefore increases the gap between the low- and high-case optimal values from order δ^2/K to $\delta/\sqrt{K}$.

For requested value accuracy ε in the two-function case, choose $\delta = \Theta(\varepsilon\sqrt{K})$ within the small- δ regime. The lower bound then grows as

$$\exp \left(\Omega \left(K \log s \log \frac{1}{\varepsilon\sqrt{K}} \right) \right),$$

and (45) gives the exact denominator. For the reduced-Hessian condition number $\kappa_H = K^2$, substitute $K = \sqrt{\kappa_H}$. This realizes the lower bound as an optimal value in a high-accuracy regime; it does not replace Theorem 4.1, which holds over a wider parameter range.

7.4 Two LP realizations and their access requirements

The decrement (51) also arises in an LP with box constraints. Set $c = M^T e + \lambda a$ as above and consider

$$\max_x c^T x \quad \text{subject to} \quad -\mathbf{1} \leq Mx \leq \mathbf{1}.$$

The logarithmic barrier of the paired constraints, $\phi_{\text{box}}(x) = -\sum_i \log(1 - (Mx)_i^2)$, has analytic center zero and Hessian H_0 there. Its decrement is exactly (51). Full SQ access to $[M; -M]$ and c is simulated as before, so the decrement lower bound under direct full SQ access to the Newton system already holds for an LP with scalar inequality constraints. We do not claim that the optimal value of the box LP equals that of the SOCP.

For the unperturbed objective $2e^T Mx$, the box barrier subproblem in $u = Mx$ is

$$F_\eta(u) = -\sum_i \log(1 - u_i^2) - 2\eta e^T u.$$

Separability gives $u(\eta) = \rho(\eta)e$, with $\rho/(1 - \rho^2) = \eta$. Therefore the central path lies on the same ray as the Newton step in (42), and its accumulator coordinate is $\rho h\Phi$. Its Hessian is at least $2I$ in u coordinates. Consequently, if $F_\eta(\hat{u})$ exceeds its minimum by at most τ , then $\|\hat{u} - \rho e\| \leq \sqrt{\tau}$ and the accumulator error is at most $\sqrt{K}\tau$. For any fixed public $\eta > 0$, the coordinate lower bound therefore still holds for points with barrier-subproblem gap at most $c\rho(\eta)^2(\beta - \alpha)^2\delta^2/K$, for a sufficiently small absolute constant $c > 0$. Again, the accumulator equalities must hold exactly, or the reported scalar must be computed as $a^T \hat{x}$. The box barrier has parameter N , since each paired interval barrier has parameter one; this realization gives no small iteration bound.

A second LP replaces the full-dimensional box by a one-dimensional affine slice. Let $\tilde{a} = Ra$, so $\|\tilde{a}\| = 1$, and set

$$\max_{x,t} \tilde{a}^T x \quad \text{subject to} \quad Mx = te, \quad -1 \leq t \leq 1. \quad (52)$$

Replace a by $\tilde{a} = Ra$ in the accumulator equalities (44) and use w as the objective. This rescales only the readout coefficients of x , makes the objective one-sparse, and leaves the projection of the feasible set onto (x, t) unchanged.

Theorem 7.6 (Affine-slice value and matched upper exponent). *The LP (52) has a unit-norm objective vector and full-SQ input access with a constant number of hidden-input queries per call. Its exact value is*

$$\text{OPT} = Rh|\Phi| = \Theta(\sqrt{K}\delta)|\Phi|. \quad (53)$$

Its equality-constraint matrix without the accumulator rows, $A_{\text{eq}} = [M \ -e]$, has $K \leq \kappa_2(A_{\text{eq}}) \leq \sqrt{2}K$. Estimating the value to additive error $c(\beta - \alpha)\sqrt{K}\delta$ has the classical lower bounds (45) and (46), and coherent algorithms achieve it with the corresponding constant or $2^{O(k)}$ number of hidden-input queries. In particular, for fixed k and constant additive value accuracy, one can take $\delta = \Theta_k(K^{-1/2})$, which gives a lower bound with exponent $\Omega_k(K \log s \log K)$.

For the two-function family, at additive accuracy ε with $\varepsilon/\sqrt{K}$ sufficiently small, the logarithms of the classical lower and upper bounds agree up to constant factors:

$$\log Q = \Theta\left(K \log s \log \frac{\sqrt{K}}{\varepsilon}\right) \quad (54)$$

when the displayed exponent dominates the polynomial and rounding factors. The exact lower bound remains (45), with $\ell = \lceil \log(\sqrt{K}/\varepsilon) + O(1) \rceil / (3\alpha_K) + O(1)$. The bracketed $O(1)$ term comes from constant accuracy factors and changes ℓ by $O(K)$; the final $O(1)$ term accounts for rounding to an allowed length.

Proof. Invertibility forces $x = tM^{-1}e$, so maximizing over $|t| \leq 1$ gives (53). The added column $-e$ changes only one public row norm of M ; its location, norm, and all conditional mixture weights are public. The objective $\tilde{a}$ is also public. Moreover,

$$K^{-2}I \preceq A_{\text{eq}}A_{\text{eq}}^T = MM^T + ee^T \preceq 2I.$$

The K^{-2} eigenspace of MM^T has multiplicity $q^\ell \geq 2$. A vector in it orthogonal to e preserves the smallest eigenvalue. The largest singular value is at least $\|M\| = 1$, proving the condition bounds. No such bound is asserted for the accumulator-augmented KKT matrix. The value identity and the source gap prove the lower and coherent upper bounds exactly as before.

For the classical upper bound, apply Theorem 3.8 to $(M, \tilde{a}, e)$, using the exact public $R_e = R = \Theta(\sqrt{K})$. The theorem permits $0 < \varepsilon \leq R/2$, which contains the chosen small- δ value-accuracy regime even when $\varepsilon > 1/2$. The absolute-value function is 1-Lipschitz, so the theorem estimates (53) at cost

$$O\left(R^2 \varepsilon^{-2} \log(1/\zeta) (q+2)^{O(K \log(2R/\varepsilon))}\right).$$

This yields the upper exponent in (54) without constructing a nullspace basis, an optimizer, or an SQ interface to $M^{-1}e$. Substituting $\delta = \Theta(\varepsilon/\sqrt{K})$ in the lower family yields the matching exponent. $\square$

In its intrinsic coordinate t , the affine slice has the barrier $\varphi(t) = -\log(1-t^2)$ with parameter one. Indeed,

$$\varphi' = \frac{2t}{1-t^2}, \quad \varphi'' = \frac{2(1+t^2)}{(1-t^2)^2}, \quad \varphi''' = \frac{4t(3+t^2)}{(1-t^2)^3}.$$

The gradient bound is $(\varphi')^2/\varphi'' = 2t^2/(1+t^2) \leq 1$. Squaring the self-concordance inequality $|\varphi'''| \leq 2(\varphi'')^{3/2}$ reduces to $(1-t^2)^2(t^2+2) \geq 0$. At the analytic center, the tangent space is $\text{span}\{(M^{-1}e, 1)\}$, the barrier Hessian in t is two, and the projected objective is $Rh\Phi$. Therefore

$$\Lambda_\eta^2 = \frac{\eta^2}{2} R^2 h^2 \Phi^2 = \frac{\eta^2}{2} \text{OPT}^2. \quad (55)$$

Estimating this squared decrement to additive error $c(\beta^2 - \alpha^2)\eta^2 K \delta^2$ has the same source lower bound. Here the scalar projected right-hand side is *not* given, since it already contains the hard quantity. If a nullspace basis and the projected objective were supplied, the program would reduce to an explicit problem on an interval. Thus, in this parameter-one realization, the hardness lies in eliminating the equality constraints and reading out the scalar, not in repeated high-dimensional barrier computations.

7.5 Replacing the large cone by three-dimensional cones

A binary tree of three-dimensional Lorentz cones can replace the single large cone. Let D be the smallest power of two at least N , pad u by zeros to D leaves, and for each internal node v impose $(t_v, z_{v,0}, z_{v,1}) \in Q_3$, where its children are leaf coordinates or lower node variables. Fix the root t to one. Recursive use of $t_v^2 \geq z_{v,0}^2 + z_{v,1}^2$ shows that every feasible u satisfies $\|u\| \leq 1$. Choosing each non-root t_v as the norm of the leaf coordinates below v proves the converse. Public two-sparse equalities that copy shared variables put the constraints in standard product-cone form. The construction uses $D-1$ cones, each new variable appears in a constant number of constraints, and all new SQ data are public, so all optimizer-coordinate and optimal-value results above still hold.

The first Newton system can also be computed exactly. After the copied variables are identified, the product barrier is

$$\Psi(t, u) = - \sum_{v \text{ internal}} \log(t_v^2 - z_{v,0}^2 - z_{v,1}^2).$$

First leave all D leaves free, with only the root fixed to one. Put $J = \log_2 D$. At $u = 0$, the analytic center of this full tree has a common value t_d° at each depth d , where

$$(t_d^\circ)^2 = \frac{D - 2^d}{2^d(D-1)}, \quad 0 \leq d < J. \quad (56)$$

To verify this, set $y_d = 2^d t_d^2$, $y_0 = 1$, and $y_J = 0$. The barrier on depth-symmetric points, up to a constant, is $-\sum_{d=0}^{J-1} 2^d \log(y_d - y_{d+1})$. Its minimizing gaps are $2^d/(D-1)$, giving (56). This point is stationary in all tree variables: automorphisms are transitive at each depth, so vanishing of each depth derivative makes every coordinate derivative vanish. Strict convexity of the sum of Lorentz barriers on the affine slice then proves it is the unique center; each nonzero variable direction affects at least one strictly convex cone term. All leaf derivatives vanish at $u = 0$. This full-tree center lies on the affine slice fixing the padded leaves to zero, so restriction preserves its minimizing property and uniqueness. This argument does not require the padded problem itself to retain the full tree symmetry.

Each leaf enters quadratically only in its bottom cone. Hence at this center

$$\nabla_{ut}^2 \Psi = 0, \quad \nabla_{uu}^2 \Psi = c_D I, \quad c_D = 2(D-1).$$

After $u = Mx$, the x Hessian block is $c_D M^T M$ with zero cross block. For the objective $2e^T u$, the first Newton direction in x is $(2\eta/c_D)M^{-1}e$. For the normalized tilt, the squared decrement is

$$\Lambda_\eta^2 = \frac{\eta^2}{c_D} \left(\frac{5}{4} + \frac{h}{G} \Phi \right). \quad (57)$$

The tree-variable block is public, has constant degree, and can be preprocessed without hidden-input queries. Thus full SQ access to the Newton system is still simulable, but the gap between the low- and high-case squared decrements is $\Theta((\beta - \alpha)\eta^2\delta/(c_D\sqrt{K}))$. The standard product barrier has parameter $2(D-1)$. Neither a small condition number of the full lifted Hessian nor a polylogarithmic iteration bound follows from the reduced clock block. The norm tree is the standard nested Lorentz-cone representation of a Euclidean norm [3], not a new cone lift.

7.6 Sampling outputs and the rare-event cost

Let Π be the projection onto the coordinates $(t, 0^n)$ with t in the hold segment. The normalized inverse history $v = x_*/R$ satisfies

$$\|\Pi v\|^2 = p\Phi^2. \quad (58)$$

If $0 \leq \varepsilon_v < 1$ and $\|\hat{x} - x_*\| \leq \varepsilon_v R$, then

$$\frac{(\sqrt{p}|\Phi| - \varepsilon_v)_+^2}{(1 + \varepsilon_v)^2} \leq \frac{\|\Pi \hat{x}\|^2}{\|\hat{x}\|^2} \leq \frac{(\sqrt{p}|\Phi| + \varepsilon_v)^2}{(1 - \varepsilon_v)^2}.$$

For a fixed source gap $\beta > \alpha$, choosing $\varepsilon_v \leq c(\beta - \alpha)\sqrt{p}$ with a sufficiently small constant makes the low- and high-case probabilities of sampling a hold-segment coordinate differ by $c'p$, with both $O(p)$. By the variance bound for a Bernoulli mean, averaging $O(1/p)$ independent samples therefore decides the source promise with constant error. It follows that generating one sample needs

$$\Omega\left(p \frac{q^{\ell/2}}{r\ell}\right) \quad (59)$$

queries in the two-function case. With the parameters (40), $p = \Theta(\delta^2)$, so this bound is (45) times $\Theta(\delta^2)$. The argument also applies to mixtures of such laws whose components each come from a vector satisfying the same perturbation bound.

Equation (59) assumes that the overall sample law satisfies this guarantee. It also holds if failures are flagged and retrying has bounded expected cost, or if unflagged failures (contamination) have

probability at most $c'p$ for a sufficiently small constant. A constant unflagged failure probability on each fresh sample call is not enough: it can overwhelm the vanishing plateau probability. A different output model allows one randomized setup that produces a fixed good sampler with constant probability, whose repeated draws all have a good sample law. Taking a constant number of independent setups and a majority vote of their source decisions gives the usual bounded-error reduction, provided that setup succeeds with probability at least $1/2 + c_{\text{setup}}$ for an absolute constant $c_{\text{setup}} > 0$, and each decision conditional on a good setup errs with probability at most $c_{\text{setup}}/2$. Each setup then decides correctly with probability bounded above one half, even if bad setups always err, so setup failure need not vanish. With Q_{setup} and Q_{draw} the query costs of one setup and one draw, and Q_{source} the randomized query lower bound for the source problem, the resulting cost bound is $Q_{\text{setup}} + O(1/p)Q_{\text{draw}} = \Omega(Q_{\text{source}})$. It gives no lower bound on a draw when preprocessing is not counted. In Theorem 3.7, the polynomial approximation is deterministic, so its setup cannot fail; the rejection-sampling randomness only draws samples from a fixed approximate vector. If the sampler is promised relative vector error ε_v , choosing instead $p = \Theta(\varepsilon_v)$ gives the weaker but useful form

$$\Omega\left(\frac{\varepsilon_v q^{\ell/2}}{r\ell}\right), \quad \ell = \frac{\log(1/\varepsilon_v)}{6\alpha_K} + O(1),$$

under the same contamination condition, for vector error sufficiently small relative to the fixed source gap $\beta - \alpha$. The sharper perturbation inequality above also allows $p = \Theta(\varepsilon_v^2)$ with a sufficiently large fixed prefactor, at the cost of a factor ε_v^2 from repetition. The scalar estimation in Theorem 7.2 uses ordinary bounded error, with no repetition factor or vanishing failure probability.

At fixed sufficiently small vector error and fixed q , choosing $p = \Theta(1)$ gives $\underline{T} = \Theta(K)$ and $\log N = \Theta(K)$. The fixed- k construction then gives a classical lower bound of $\hat{\Omega}_k(N^{1-1/k})$ for sampling, with $\kappa(M) = \Theta(\log N)$ and $\kappa(H_0) = \Theta(\log^2 N)$. These condition numbers match the scales shown to be necessary after Theorem 3.7. For full SQ access to the first Newton system, use $(H_0, M^T e)$; its solution is $M^{-1}e/2$, and its full-SQ simulation was proved above. An exactly feasible point with SOCP objective gap τ has relative vector error at most $K_{\text{eff}}\sqrt{\tau} \leq \sqrt{K}\tau$. Thus, if $\tau \leq \varepsilon_v^2/K$, samples from such a point satisfy this relative-error guarantee.

The family also has explicit upper bounds. Classically, reading all hidden sign tables costs $O_k(q^\ell)$ queries; computing their circuit prefixes and sampling the public geometric clock gives the exact inverse-state distribution in $q^\ell \text{poly}_k(r, \ell)$ arithmetic operations. A designated matrix value query recovers each sign, so this upper bound uses the same input interface. Consequently the logarithm of the classical query complexity is $\Theta(K \log s \log(1/\varepsilon_v))$ whenever its leading term dominates the displayed repetition and denominator factors. A quantum algorithm prepares the public clock state with geometric amplitudes and applies each gate controlled on the clock having passed that gate's position. Forward and reverse segments use each hidden sign gate twice in total. This prepares the exact normalized history in the ideal circuit model. For any requested Euclidean state error $0 < \eta < 1$, synthesize the public circuit to operator error at most η : allocating error at most η/G_{public} to each of the G_{public} public gates suffices by telescoping. The implementation retains $O(k)$ hidden-input queries and public gate cost polynomial in $n, T, \log(1/\eta)$. Taking $\eta \leq \varepsilon_v$ identifies the synthesized normalized state with a vector $\hat{x}/R$ satisfying the relative-vector sampling tolerance above. This algorithm uses the source-equivalent coherent oracle; a classical SQ data structure alone does not provide its controlled sign queries.

A generic upper bound also holds for *sampling* from an approximately optimal feasible point. The normal-equation polynomial used in Theorem 3.7 can produce $z = Pe$ with $\|Mz - e\| \leq \xi$, at degree $O(K \log(1/\xi))$. This follows directly from the singular-value residual bound for MP , not

just the relative error in z . For $\xi < 1/2$, the normalized point

$$x = z / \|Mz\|, \quad u = Mz / \|Mz\|$$

is exactly feasible for (41) and has objective gap $O(\xi^2)$: its squared angular distance from e is bounded by a constant times $\|Mz - e\|^2$. Scaling z does not change its squared-coordinate sample law. Setting $\xi = \Theta(\varepsilon_v / \sqrt{K})$ therefore gives a sample from a point of gap $O(\varepsilon_v^2 / K)$ with exponential cost $\exp(O(K \log(s+1) \log(\sqrt{K}/\varepsilon_v)))$, up to polynomial factors. This bound covers samples from the x block only; it does not provide the normalization or the explicit coordinates of the feasible point.

7.7 Relation to prior work

Forrelation decision and its near-linear classical lower bounds are due to Aaronson–Ambainis and Bansal–Sinha [1, 6]. The history-state sampling obstruction is already present in Grønlund–Larsen [25]. Ellipsoids, norm trees, and their SOCP formulations are standard. Apers and Gribbling [5, Lemma 8.5] already prove a full-SQ classical lower bound for constant-error LP optimal-value estimation. Therefore neither an SQ lower bound for optimization nor the hardness of scalar optimization outputs is new here. To the best of our knowledge, the following are new: the joint dependence on conditioning, sparsity, and accuracy for these specific formulations; the observation that $G = \|M^{-T}a\|$ is public, which permits the larger gap between optimal values, and the optimality of this readout on the plateau; and exact realizations of the hardness as a Newton decrement under direct full SQ access to the Newton system and as the value of an affine-slice LP with a parameter-one barrier, each under its stated access model.

The coherent upper bounds are explicit algorithms for these structured instances, not a general quantum interior-point method for arbitrary SOCPs or LPs. Quantum SOCP methods that reconstruct classical Newton directions produce a different output [27]. Building an explicit matrix table or outputting an entire optimizer has dimension-dependent cost and lies outside the scalar query models studied here.

8 Scalar composition along a path and at its endpoint

The preceding constructions encode one source instance in one scalar. We compare three ways of using B independent instances in one optimization problem: reporting several scalar diagnostics along the central path, encoding the B answers in one coordinate of the final optimizer, and merely revisiting one fixed hidden input. Only the first two give a direct-sum lower bound. Throughout, the input is the *formulation*, with the full SQ interface of Section 7; an exact oracle for a later iterate is not supplied for free.

8.1 A direct sum for classical formulation queries

Let g be the promised two-function Forrelation predicate, with $|\Phi| \leq \alpha = 1/100$ in case zero and $\Phi \geq \beta = 3/5$ in case one. Write $L_g = R_{1/3}(g)$ for its worst-case bounded-error randomized query complexity. The cyclic instances satisfy

$$L_g = \Omega\left(\frac{q^{\ell/2}}{r\ell}\right), \quad a_i^T M_i^{-1} e = h\Phi_i, \quad \|M_i\| = 1, \quad \sigma_{\min}(M_i) = K^{-1},$$

where $h = \Theta(\delta)$ and the one-block dimension is $N_0 = \Theta(Tq^\ell)$. Here T remains the clock length of Section 7; the positive integer B denotes the number of independent blocks.

Lemma 8.1 (Classical direct sum). *Suppose every query to a formulation built from B independent promised inputs (blocks) can be simulated by $O(1)$ classical queries, each reading one block that is named by the query or sampled with public probabilities. Then an output from which all B promise bits can be recovered, with joint success probability at least $2/3$, requires $\Omega(BL_g)$ formulation queries.*

Proof. Constant-error randomized query complexities are comparable by worst-case amplification. Minimax therefore gives a distribution μ for which any randomized algorithm using fewer than cL_g queries has average success at most $9/16$. Embed a target drawn from μ in a uniform hidden block I , and fill the other blocks independently from μ . The simulator may sample and retain complete filler inputs for free: only queries to the target count. The assembled input has distribution μ^B and is independent of I . Each of the $O(Q)$ simulated queries made by a Q -query formulation algorithm thus touches the target with average probability $1/B$. This also holds for global SQ samples with nonuniform public block weights. The expected target cost is $O(Q/B)$. Truncate after twelve times that expectation. Markov's inequality loses at most $1/12$ success, leaving average target success at least $7/12$. Since $7/12 > 9/16$, the chosen distribution forces $Q/B = \Omega(L_g)$. No amplification of a distributional average advantage is used. $\square$

This elementary argument is a special case of known randomized direct-sum results [11]. It does not apply to a coherent query in superposition over all blocks.

8.2 Successive scalar increments of a box central path

Consider the single LP

$$\max 2 \sum_{i=1}^B \theta_i e^T M_i x_i, \quad -\mathbf{1} \leq M_i x_i \leq \mathbf{1}, \quad \theta_i = \Gamma^{-i}. \quad (60)$$

A public three-sparse partial-sum chain defines the free coordinate $w = \sum_i a_i^T x_i$, as in (44). Define

$$\rho(z) = \frac{2z}{1 + \sqrt{1 + 4z^2}}, \quad \rho(0) = 0.$$

Since $\rho/(1 - \rho^2) = z$, the logarithmic-barrier central point at multiplier η is unique, with blocks $x_i(\eta) = \rho(\eta\theta_i)M_i^{-1}e$. At the checkpoints $\eta_j = \Gamma^{j+1/2}$, $0 \leq j \leq B$, the B increments of w , indexed by $j = 1, \dots, B$, satisfy

$$D_j := w(\eta_j) - w(\eta_{j-1}) = h \sum_{i=1}^B d_{j-i} \Phi_i, \quad d_m = \rho(\Gamma^{m+1/2}) - \rho(\Gamma^{m-1/2}). \quad (61)$$

The kernel d_m is nonnegative and sums to one over all m . The identities $\rho(z) \leq z$ and $1 - \rho(z) = \rho(z)/(z(1 + \rho(z))) \leq 1/(2z)$ imply

$$d_0 \geq 1 - \frac{3}{2\sqrt{\Gamma}}, \quad |D_j/h - d_0 \Phi_j| \leq 1 - d_0 =: \tau_\Gamma. \quad (62)$$

These bounds follow by telescoping; restricting the sum to the B actual blocks only reduces its off-diagonal mass.

Theorem 8.2 (Scalar-increment trajectory). *For $\Gamma = 2^{20}$, returning all B increments in (61) to additive error $0.21h$, jointly with probability $2/3$, requires $\Omega(BL_g)$ classical full-SQ queries. With the source-equivalent coherent oracle, $O(B)$ queries suffice at this accuracy, and $O(B \log(2B))$ at the stronger accuracy $h/100$. The LP has dimension $\Theta(BN_0)$, row and column sparsity $O(q)$, and checkpoint multiplier range $\log(\eta_B/\eta_0) = B \log \Gamma$.*

Proof. The low and high intervals for D_j/h are separated by at least $d_0(\beta - \alpha) - 2\tau_\Gamma > 0.586$. Error 0.21 therefore still identifies the bit. Each matrix query touches one block; the added readout chain and objective coefficients are public. Row and column norms and sampling probabilities follow from public squared magnitudes. The chain has $O(BT)$ rows and constant incidence, so it preserves the claimed dimension and sparsity. Lemma 8.1 proves the lower bound.

For accuracy $h/100$, estimate each Φ_j to error 0.005, with failure probability $1/(3B)$, and return $hd_0\hat{\Phi}_j$. Equation (62) bounds the total error by $0.005h + 0.0015h < h/100$. Constant-accuracy source estimation followed by median amplification costs $O(\log(2B))$ queries per block. For the looser accuracy, use the coherent Boolean recovery theorem of Buhrman et al. [14, Corollary 3]. It recovers all B bits jointly using $O(B)$ controlled calls to their bounded-error constant-query subroutines and their inverses. Return zero for a low bit and $0.8hd_0$ for a high bit. Their errors are at most $(0.01d_0 + \tau_\Gamma)h < 0.012h$ and $(0.2d_0 + \tau_\Gamma)h < 0.202h$, respectively. $\square$

The $O(B)$ bound needs Boolean recovery: constant individual success does not imply constant joint success. Boolean recovery returns bits, not real-valued estimates, whereas the absolute checkpoint values $w(\eta_j)$ involve sums of the actual amplitudes. Estimating each amplitude to error $O(1/B)$ with failure probability $O(1/B)$ gives an elementary quantum upper bound $O(B^2 \log(2B))$ for absolute checkpoint accuracy $O(h)$; we do not claim a matching linear upper bound for this output.

The same lower bound holds for exactly feasible approximate centers with barrier-subproblem gap $\tau_{\text{ctr}} \leq ch^2/(BK)$, for a sufficiently small constant c . Indeed, the interval barrier Hessian is at least $2I$ in $u_i = M_i x_i$ coordinates, so strong convexity gives $\|\hat{u} - u\| \leq \sqrt{\tau_{\text{ctr}}}$. Since $\|M_i^{-T} a_i\| \leq K/R \leq \sqrt{K}$,

$$|\hat{w} - w| \leq \sqrt{BK\tau_{\text{ctr}}}.$$

An increment D_j differences two such values, so for small c its error stays well below the decoding gap of Theorem 8.2. The accumulator equality must hold exactly, or its residual must be included in this error budget. At the analytic center the reduced Hessian has condition number K^2 ; there is no uniform condition-number bound over the whole scaled trajectory. Each paired interval barrier has parameter one (Section 7), so the product barrier has parameter BN_0 .

For the fixed- k promise in Corollary 7.3, choose $\sqrt{\Gamma} \geq 12/(\beta_k - \alpha_k)$ and use error $c_k h$, $c_k > 0$ sufficiently small. The same proof gives

$$\tilde{\Omega}_k(BN_0^{1-1/k}) = \tilde{\Omega}_k(B^{1/k} \mathcal{N}^{1-1/k}), \quad \mathcal{N} = \Theta(BN_0),$$

and a coherent upper bound $2^{O(k)} B \log(2B)$ for the numerical estimates. One representative value per recovered bit need not suffice: the positive high interval $[\beta_k, 1]$ can be much wider than the decoding gap.

8.3 Ordinary SOCP predictor decrements

Let $g_i = M_i^{-T} a_i$, and use the public common norm $G = \|g_i\|$ from Lemma 7.4. Put

$$v_i = e + \frac{g_i}{2G}, \quad C = \frac{5}{4}, \quad \xi = \frac{h}{CG}, \quad \|v_i\|^2 = C(1 + \xi \Phi_i), \quad 0 < \xi \leq \frac{1}{4}.$$

The restriction $\xi \leq 1/4$ holds when the public upper bound on δ is small enough. On the product of B unit balls consider

$$\max \sum_i \theta_i \left(e^T u_i + \frac{a_i^T x_i}{2G} \right), \quad u_i = M_i x_i, \quad \|u_i\| \leq 1, \quad \theta_i = \frac{\Gamma^{-i}}{\sqrt{C}}. \quad (63)$$

The barrier is $-\sum_i \log(1 - \|u_i\|^2)$, with the conventional Lorentz parameter $2B$. At the exact central point for $\eta_j = \Gamma^j$, $j = 1, \dots, B$, consider the relative update $\eta_j \mapsto (1 + \sigma)\eta_j$, where $\sigma = \sigma_0/\sqrt{B}$ and σ_0 is a small constant.

Theorem 8.3 (Predictor-decrement trajectory). *Let $\Gamma \geq 256$, and let Λ_j^2 be the squared predictor decrement at η_j for the update $\eta_j \mapsto (1 + \sigma)\eta_j$. For $j = 1, \dots, B$,*

$$P_j := \frac{\Lambda_j^2}{\sigma^2} = \sum_i f\left(\Gamma^{j-i} \sqrt{1 + \xi \Phi_i}\right), \quad f(z) = 1 - \frac{1}{\sqrt{1 + 4z^2}}. \quad (64)$$

Returning all B predictor decrements Λ_j^2 to error $\sigma^2 \xi / 100$, with joint success $2/3$, requires $\Omega(BL_g)$ classical full-SQ formulation queries. Their scale is $\sigma^2 \xi = \Theta(\delta / (B\sqrt{K}))$, and $\Lambda_j^2 \leq \sigma_0^2$. With the source-equivalent coherent oracle, $O(B \log(2B))$ queries suffice at this accuracy. For $\Gamma = 2^{20}$ and the looser error $0.036\sigma^2 \xi$, $O(B)$ queries suffice, and the same classical lower bound holds.

Proof. Write $s_i = \|v_i\|$. The central point is $u_i = \rho(\eta\theta_i s_i)v_i/s_i$. At radius r_i , the radial barrier-Hessian eigenvalue is $2(1 + r_i^2)/(1 - r_i^2)^2$. The multiplier update leaves residual $2\sigma\eta\theta_i v_i$. Its squared inverse-Hessian norm is $2\sigma^2 r_i^2/(1 + r_i^2) = \sigma^2 f(\eta\theta_i s_i)$. This proves (64) and the bound on its size.

Subtract the public baseline $P_j^0 = \sum_i f(\Gamma^{j-i})$ and set $\psi_z(x) = f(z\sqrt{1 + \xi x}) - f(z)$. Direct differentiation gives

$$\psi'_z(x) = \frac{2\xi z^2}{(1 + 4z^2(1 + \xi x))^{3/2}}, \quad \frac{2\xi}{6^{3/2}} \leq \psi'_1(x) \leq \frac{\xi}{4} \quad (|x| \leq 1). \quad (65)$$

The other blocks contribute absolute total at most

$$T_\Gamma \leq \xi \left(\frac{2}{3^{3/2}(\Gamma - 1)} + \frac{2}{\Gamma^2 - 1} \right) \leq \frac{0.4\xi}{\Gamma - 1}, \quad \Gamma \geq 256. \quad (66)$$

For scales $z = \Gamma^m \geq \Gamma$, use $\psi'_z \leq 2\xi/(3^{3/2}z)$; for $z \leq \Gamma^{-1}$, use $\psi'_z \leq 2\xi z^2$, and sum both geometric series. The low/high gap for $P_j - P_j^0$ is at least

$$\xi \left(\frac{2\beta}{6^{3/2}} - \frac{\alpha}{4} - \frac{0.8}{\Gamma - 1} \right) > 0.075\xi.$$

Lemma 8.1 proves the classical lower bound. The two objective pieces in each block occupy disjoint u and x registers, with public squared norm $4\theta_i^2(1 + \|a_i\|^2/(4G^2))$. Thus objective SQ, as well as equality-matrix SQ, has constant source-query simulation cost.

Estimating each active amplitude to error 0.02 and failure $1/(3B)$, then returning $P_j^0 + \psi_1(\hat{\Phi}_j)$, gives error at most $0.005\xi + T_\Gamma < 0.01\xi$. For $\Gamma = 2^{20}$, recover all promise bits jointly and output the midpoint of $P_j^0 + \psi_1([-\alpha, \alpha])$ or $P_j^0 + \psi_1([\beta, 1])$. On the low interval, $\psi'_1 \leq 2\xi/(5 - 0.01)^{3/2} < 0.180\xi$, so the midpoint error is less than 0.0018ξ . On the positive high interval, $\psi'_1 \leq 2\xi/5^{3/2}$, so its midpoint error is at most $0.4\xi/5^{3/2} < 0.035778\xi$. Adding (66) keeps both below 0.036ξ . The gap is still greater than $0.079\xi > 2(0.036\xi)$. $\square$

For fixed k , the same proof uses $\Delta_k = 2\beta_k/6^{3/2} - \alpha_k/4 > 0$, chooses $\Gamma_k \geq \max\{256, 1 + 1.6/\Delta_k\}$, and takes error $c_k\sigma^2\xi$ with $c_k < \Delta_k/8$. It gives the same classical lower bound $\hat{\Omega}_k(BN_0^{1-1/k})$ and numerical coherent upper bound $2^{O(k)}B \log(2B)$. The formulation has dimension $\Theta(BN_0)$ and sparsity $O(q)$. The reporting points span a range $(B - 1)\log \Gamma$ in the logarithm of the multiplier. An algorithm may take intermediate short steps; the diagnostic at each reporting point concerns the small update σ , not a jump by a factor Γ . We claim no condition-number bound for the Hessian along the full trajectory.

Corollary 8.4 (Cost of a dynamic block-norm interface). *Consider an interface, constructed from the formulation, that returns for each $j = 1, \dots, B$ the norm $\|u_j(\eta_j)\|$ of cone block j at the central point for η_j , to additive error $\xi/50$, with joint success probability at least $2/3$. Its total classical query cost, including setup, is $\Omega(BL_g)$. In particular, if each request after setup costs $O(1)$ queries, the setup cost plus $O(B)$ is $\Omega(BL_g)$.*

Proof. The norm equals $\rho(\sqrt{1 + \xi\Phi_j})$. In this argument $\rho < 2/3$ and $\rho'(z) = (1 - \rho^2)^2/(1 + \rho^2) \geq 25/117$. Consequently its derivative in Φ_j is at least $25\xi/(117\sqrt{5})$. The promise gap in the norm is greater than $0.056\xi > 2\xi/50$, so Lemma 8.1 applies. $\square$

Free exact SQ access to the central points for $\eta_1, \dots, \eta_B$, including the block norms $\|u_j(\eta_j)\|$, would reveal the promise bits. Theorem 8.3 is therefore not a lower bound for algorithms given that stronger input; the corollary bounds the cost of constructing it. It does not apply to a normalized quantum state output without its norm.

8.4 Encoding promised overlaps in one bounded optimizer coordinate

A sum of promised amplitudes need not determine the sum of the promise bits: the intervals for different bit patterns can overlap. The following constant-size LP avoids this problem: its optimizer contains the promise bit itself.

Lemma 8.5 (LP threshold encoding). *Suppose $\varphi \in [-1, 1]$ satisfies $|\varphi| \leq \alpha$ or $\varphi \geq \beta$, with $0 \leq \alpha < \beta \leq 1$. Put $c = (\alpha + \beta)/2$, $\Delta = (\beta - \alpha)/2$, $J = 1/\Delta$, $M_0 = 2J$, and $C_0 = 4J$. The LP*

$$\max C_0 z - M_0 r + t, \quad 0 \leq z \leq 1, \quad 0 \leq r \leq 2, \quad 0 \leq t \leq z, \quad t \leq Jr, \quad r \geq z(\varphi - c) \quad (67)$$

has a unique optimizer with $(z, t) = (1, g)$, where g is the promise bit. Every feasible point has objective gap at least $|t - g|$.

Proof. For a low input the last constraint is redundant. For fixed z , the best t is $\min\{z, Jr\}$ and the penalty coefficient $M_0 = 2J$ uniquely sets $r = t = 0$. Then $z = 1$ is unique. Moreover the gap is $C_0(1 - z) + M_0 r - t \geq t$. For a high input set $d = \varphi - c \geq 1/J$. Feasibility forces $r \geq dz$, so the unique best choices are $r = dz, t = z$. The remaining coefficient $C_0 + 1 - M_0 d$ is positive since $d < 1$. Thus $z = t = 1$ uniquely. For every feasible point, $f \leq (C_0 - M_0 d)z + t$ and $C_0 - M_0 d > 0$, giving $f^* - f \geq 1 - t$. $\square$

For each cyclic block, introduce $M_i y_i = z_i e$ and a public sparse readout chain defining $q_i = a_i^T y_i / h$. These linear equalities enforce $q_i = z_i \Phi_i$. Replace the last constraint of (67) by $r_i \geq q_i - c z_i$. There is no product of two optimization variables: Φ_i is implicit input data, and its product with z_i is implemented by linear equations. Give block i the objective weight $\theta_i > 0$. Set $p_1 = t_1$ and, for $2 \leq i \leq B$, impose

$$\begin{aligned} p_i &\leq p_{i-1} + t_i, & p_i &\geq p_{i-1} - t_i, \\ p_i &\geq t_i - p_{i-1}, & p_i &\leq 2 - p_{i-1} - t_i. \end{aligned} \quad (68)$$

These inequalities define the convex hull of the Boolean XOR triples. Their projection onto (p_{i-1}, t_i) is the whole unit square, and Boolean inputs uniquely force $p_i = p_{i-1} \oplus t_i$.

Theorem 8.6 (A bounded endpoint scalar). *The resulting LP has a unique optimizer and $P^* := p_B^* = g_1 \oplus \dots \oplus g_B$. Its dimension is $\Theta(BN_0)$, row and column sparsity $O(q)$, and the logarithmic barrier of its displayed inequalities has parameter at most $12B - 4$. Estimating P^* to additive error*

less than $1/3$ requires $\Omega(BL_g)$ classical full-SQ queries and has query complexity $\Theta(B)$ with the source-equivalent coherent oracle for two-function Forrelation. For fixed k , the classical lower bound is $\tilde{\Omega}_k(BN_0^{1-1/k})$ and the coherent complexity is $\Theta_k(B)$. Every exactly feasible point with objective gap τ satisfies

$$|p_B - P^*| \leq \frac{\tau}{\min_i \theta_i}. \quad (69)$$

Proof. The XOR constraints leave the projection onto all t_i unchanged. Lemma 8.5 fixes each block's optimizer uniquely; then the XOR variables, inverse histories and accumulators are also unique. There are eight scalar inequalities per threshold block and four per later XOR gate. Strict relative feasibility follows by taking $z_i = 3/4$, $r_i = 3/2$, $t_i = p_i = 1/2$, with the equality variables determined accordingly. The resulting scalar product log barrier has parameter at most the number $12B - 4$ of its logarithms, independent of N_0 .

For $B = 1$, the randomized lower bound is the one-block lower bound. For $B \geq 2$, it follows from the strong XOR theorem of Brody et al. [13, Theorem 1.1]: $\bar{R}_\epsilon(\text{XOR}_B \circ g^B) = \Theta(B\bar{R}_{\epsilon/B}(g))$ for constant outer error, where the bar denotes worst-input expected query cost. Expected constant-error and worst-case constant-error costs are comparable: truncate after a constant multiple of the worst-input expectation and amplify the resulting worst-input success. Hence the theorem implies $\Omega(BR_{1/3}(g))$ also for worst-case queries. For the coherent upper bound, jointly recover all promise bits using Buhrman et al. [14, Corollary 3] and take their XOR. This costs $O(B)$ or $2^{O(k)}B$ source queries. For the quantum lower bound, restrict every block to two fixed inputs having opposite promise bits. One source query on this restriction is simulated by $O(1)$ queries to the selecting outer bit. Thus the algorithm computes the parity of B independent bits, which needs $\Omega(B)$ quantum queries [7].

Finally, Lemma 8.5 gives $\tau \geq \sum_i \theta_i |t_i - g_i|$. The XOR facets imply $|p_i - p_{i-1}| \leq t_i$ and $|p_i - (1 - p_{i-1})| \leq 1 - t_i$. Induction, using the appropriate inequality for $g_i = 0$ or 1 , yields $|p_i - (g_1 \oplus \dots \oplus g_i)| \leq \sum_{j \leq i} |t_j - g_j|$, proving (69). $\square$

All new coefficients and supports are public. An old row or column gains only a constant number of public entries, so its new SQ distribution is a public mixture of the old distribution and those entries. Readout chains keep the incidence constant. Hence each full-SQ formulation query costs $O(1)$ source queries. Conversely, the hidden signs remain at known matrix entries, so the source upper bound also holds with formulation access. The factor $1/h$ is public. We claim no condition-number bound for the entire equality system.

By (69), the same lower bound applies to the coordinate p_B of exactly feasible points with objective gap $\tau < 1/3$ when $\theta_i = 1$, and $\tau < \Gamma^{-B}/3$ when $\theta_i = \Gamma^{-i}$. Without the XOR facets, the threshold blocks are separate, and at multiplier η block i is central at multiplier $\eta\theta_i$; a public summation readout $\sum_i t_i$ does not change this. The XOR facets couple the blocks through the barrier, so we do not claim this rescaling identity for the formulation with output p_B . Endpoint hardness therefore does not show that the cost must be paid at each of the B scales along the path.

The threshold encoding tolerates rounding of the public readouts. If $b_i = a_i/h$ is replaced by $\tilde{b}_i$ with $\|\tilde{b}_i - b_i\| \leq \Delta/(10R)$, then $\|y_i\| \leq R$ makes the readout error at most $\Delta/10$. Use the narrower effective half-gap $9\Delta/10$ when choosing J, M_0, C_0 . The optimizer bits then remain exact: although a rounded amplitude can slightly exceed one, its high-branch value $d = \tilde{\Phi}_i - c$ is still below one because $c \geq \Delta$, so the proof of Lemma 8.5 applies with the narrower gap. With the old threshold constants, saturation at the high boundary need not be preserved. Rounding M_i requires a separate error budget for the resolvent.

8.5 A parity endpoint, and what hardness along the path means

A simpler sparse family, whose blocks have no quantum–classical separation, illustrates the distinction between path diagnostics and an endpoint output. Let $\sigma_{i,k} \in \{-1, 1\}$, $1 \leq i \leq B$, $1 \leq k \leq L$, and $h_i = \prod_k \sigma_{i,k}$. Introduce variables $-1 \leq a_i \leq 1$ and chains

$$x_{i,0} = a_i, \quad x_{i,k} = \sigma_{i,k} x_{i,k-1}.$$

A public partial-sum chain defines $W = \sum_i x_{i,L}$. Order all BL signs as $\tau_1, \dots, \tau_{BL}$, and add the carrier chain $r_0 = a_B$, $r_m = \tau_m r_{m-1}$, $w = r_{BL}$. Maximize $2 \sum_i \Gamma^{-i} a_i$, with $\Gamma = 256$. All equalities have at most three nonzeros per row and constant column incidence. There are $2B$ inequalities and $\Theta(BL)$ variables. The optimizer is unique, and direct elimination gives

$$W(\eta) = \sum_i h_i \rho(\eta \Gamma^{-i}), \quad w(\eta) = H \rho(\eta \Gamma^{-B}), \quad w^* = H := \prod_i h_i. \quad (70)$$

At $\eta_j = \Gamma^{j+1/2}$, the increment kernel (61) gives $h_j(W(\eta_j) - W(\eta_{j-1})) \geq 2d_0 - 1 \geq 13/16$, so the path has B separated transitions of size at least $13/16$. At its final checkpoint, $|w(\eta_B) - H| \leq 1/32$, so additive error $1/4$ in this one bounded coordinate computes the parity of all BL signs. Both its quantum and classical query complexities are $\Theta(BL)$, by reading all signs and by the parity lower bound. All nonzero magnitudes, norms and sampling distributions are public, and each queried matrix entry uses at most one source sign; repeated signs refer to the same fixed input bit. The barrier parameter $2B$ and the public $O(B)$ -bit objective scales do not affect these query bounds.

Every feasible point of objective gap τ has $|w - H| = 1 - a_B \leq \tau / (2\Gamma^{-B})$. Equal objective weights remove this exponentially small gap requirement but also the scale separation. At $\eta_f = 16B\Gamma^B$, $|W(\eta_f) - \sum_i h_i| \leq 1/32$; additive error $1/4$ in this unnormalized sum also computes parity by rounding. Normalizing by B changes the required accuracy to $O(1/B)$. More generally, if a block bit has an exact coherent evaluator of cost q_0 , amplitude estimation gives the mean of the B block bits to error ϵ using $O(q_0 \min\{B, 1/\epsilon\})$ queries [12]. Hence a constant-accuracy mean need not cost a factor B , whereas the parity output w does.

The query lower bounds in this section concern specified final outputs or joint outputs at all checkpoints. The input oracle is static: an algorithm that only has to return a final relation may compute it without generating any intermediate iterate, and moving along the path creates no new oracle input. A lower bound for each step along the path therefore needs an online-output, fresh-input, or limited-memory restriction on the model, or a composition theorem for the required final function. Our parity and threshold constructions use the composition theorem; they do not prove per-iteration lower bounds.

The Forrelation, direct-sum, robust Boolean recovery, and XOR theorems used here are prior work. Scalar optimization encodings of Boolean composition also appear in Apers and Gribling [5]. Our contribution in this section is the explicit sparse realization with its quantitative guarantees: the bound (62) on the contribution of the other blocks to each increment, the sensitivity bound (65), and the threshold LP with its objective-gap bound (69). To our knowledge, these oracle-preserving central-path realizations have not appeared in the cited literature; we claim no priority for the query-complexity results they use.

9 What repeated Newton solves can reuse

The direct sums of Section 8 need independent hidden information in the requested output; many central-path movements alone do not force a large cost. We first give a sparse KKT system whose

normalized predictor solution is the same, possibly hard, state at every central multiplier; reading the input once serves every checkpoint. We then bound the number of refresh solves when solutions lie near a low-dimensional subspace and stored solutions are reused after a residual test. Neither result treats residual tests, dense output, or state preparation as free.

9.1 A fixed sparse KKT state at every central multiplier

Let $M \in \mathbb{R}^{n \times n}$ be invertible, with bidirectional sparse access, row and column sparsity at most s , and $\|M\| \leq 1$. For the one-cone program (41), write $v = M^{-1}e$, $a = \rho(\eta)$, and $q_a = 1 - a^2$. We keep both $u = Mx$ and the equality multiplier λ .

Proposition 9.1 (A common KKT ray). *At every exact central point, the augmented predictor equations are*

$$\begin{pmatrix} D_a & 0 & I \\ 0 & 0 & -M^T \\ I & -M & 0 \end{pmatrix} \begin{pmatrix} \dot{u} \\ \dot{x} \\ \dot{\lambda} \end{pmatrix} = \begin{pmatrix} 2e \\ 0 \\ 0 \end{pmatrix}, \quad D_a = \frac{2}{q_a}I + \frac{4a^2}{q_a^2}ee^T. \quad (71)$$

Their unique solution and normalized state are

$$(\dot{u}, \dot{x}, \dot{\lambda}) = \frac{q_a^2}{1 + a^2}(e, v, 0), \quad \frac{(e, v, 0)}{\sqrt{1 + \|v\|^2}}. \quad (72)$$

The KKT matrix is at most $(s + 2)$ -sparse, and each sparse-access query to it can be simulated with $O(1)$ queries to M and M^T . Thus any one-shot lower bound for preparing $|v\rangle$ to sufficiently small constant state error also holds, up to constants, at each single KKT checkpoint.

Proof. Stationarity in x gives $M^T\lambda = 0$, so $\lambda = 0$. The other stationarity equation is $u/(1 - \|u\|^2) = \eta e$; hence $u = ae$ and $x = av$. Differentiating gives $a' = q_a^2/(1 + a^2)$ and (71). The second block row forces $\dot{\lambda} = 0$, the third forces $\dot{u} = M\dot{x}$, and $D_a e = 2(1 + a^2)e/q_a^2$ proves (72). The remaining entries are public diagonal or identity entries because e is a basis vector. This proves sparsity and oracle simulation. Since $\|v\| \geq 1$, the x register has probability at least $1/2$; measuring that register extracts $|v\rangle$ with constant probability. Sufficiently small constant Euclidean state error perturbs this probability and the conditional state by only constant amounts. $\square$

This concerns state output; it does not claim that every scalar derived from that state is hard. It gives no uniform bound on the condition number of the augmented KKT matrix near the boundary.

Conversely, a scalar output can be hard while the solution state is trivial. A signed equality chain $r_0 = \alpha$, $r_i = \sigma_i r_{i-1}$, with $|\alpha| \leq 1$ and objective $\alpha + 2r_L$, reduces to maximizing $(1 + 2H)\alpha$, where $H = \prod_i \sigma_i$. Its optimum is three for $H = 1$ and one for $H = -1$, so estimating it to two-sided relative error below $1/2$ costs $\Theta(L)$ quantum or classical queries. Yet the reduced-coordinate solution lies in a one-dimensional space, so its state is input independent up to global phase; normalization discards exactly the sign and scale that the scalar readout needs. The terminal optimizer coordinate r_L^* equals one in both cases, so it cannot serve as the parity readout. The paired interval barrier has parameter one, but this does not reveal the hidden reduced objective coefficient without the chain signs.

Although its normalized predictor never changes, the instance of Proposition 9.1 can require arbitrarily many bounded local movements. For $\phi(u) = -\log(1 - \|u\|^2)$, direct inversion of its radial Hessian gives

$$\|\nabla\phi(u)\|_{u,*}^2 = \frac{2\|u\|^2}{1 + \|u\|^2} \leq 1.$$

Objective gap at most ϵ , where $0 < \epsilon \leq 1$, implies $e^T u \geq 1 - \epsilon/2$, so $1 - \|u\|^2 \leq \epsilon$ and $\phi(u) - \phi(0) \geq \log(1/\epsilon)$. The gradient bound makes the barrier-metric length of every interior path from the center to such a point at least $\log(1/\epsilon)$. A chord with starting Dikin norm at most $R < 1$ has barrier-metric length at most $\log(1/(1 - R))$. One can see this directly for the ball: along a line $\phi(u + td)$ is the sum of two negative logarithms of affine factors. If $h^2 = d^T \nabla^2 \phi(u) d$, the absolute logarithmic slopes at zero have squared sum h^2 ; at $t \geq 0$ their denominators are at least $1 - th$. Therefore the metric speed is at most $h/(1 - th)$, whose integral from zero to one is $-\log(1 - h)$. Consequently, if each round moves along at most m such chords, then

$$\# \text{rounds} \geq \frac{\log(1/\epsilon)}{m \log(1/(1 - R))}. \quad (73)$$

Pullback by M preserves this barrier metric on the equality slice. Equation (73) concerns this explicit primal barrier and chords of bounded Dikin norm; it is not an iteration lower bound for arbitrary algorithms.

Lemma 9.2 (An elementary hard ray). *There are four-sparse matrices M_σ , of dimension $\Theta(L)$ and condition number $\Theta(L)$, for which preparing $|M_\sigma^{-1}e\rangle$ to sufficiently small constant state error has quantum query complexity $\Theta(L)$. Classically, reading the input and solving uses $O(L)$ queries and arithmetic. These matrices may be used in Proposition 9.1.*

Proof. Take a path $p_0, \dots, p_L$ with hidden edge signs $\sigma_1, \dots, \sigma_L$. Attach a complete public binary tree with $h = 2^{\lceil \log_2 L \rceil}$ leaves at p_0 , and a disjoint copy at p_L . For every nonroot node z impose $y_z - s_z y_{\pi(z)} = 0$, where s_z is the appropriate hidden path sign or the public value one; impose $y_{p_0} = 1$. These equations define a triangular matrix A_σ with row sparsity two and column sparsity four. Its dimension is $N_L = L + 4h - 3$, and $\|A\| \leq \sqrt{\|A\|_1 \|A\|_\infty} \leq \sqrt{8}$. Set $M = A/\sqrt{8}$. The unscaled solution $A^{-1}e$ is one on the first tree and $H = \prod_i \sigma_i$ on the second, with prefix parities on the path. The solution $M^{-1}e = \sqrt{8}A^{-1}e$ has the same normalized state, since its common public factor cancels. This state has constant probability $2h/N_L \geq 2/5$ in the span of the two public uniform leaf states. Measuring their sum and difference identifies H whenever that subspace is observed. A constant number of sufficiently accurate preparations therefore computes parity, which needs $\Omega(L)$ queries [7]. Reading all signs and forward substitution gives the matching upper bound.

Every entry of A^{-1} is zero or a signed ancestor indicator. Its absolute row and column sums are $O(L)$, so $\|A^{-1}\| = O(L)$. The path submatrix of A^{-1} is, after diagonal sign changes, the cumulative-sum matrix; applying it to the uniform vector shows norm $\Omega(L)$. Since $1 \leq \|A\| = O(1)$, the condition number is $\Theta(L)$. The augmented KKT graph also has a width-three tree decomposition: for each nonroot z use the bag $\{\lambda_z, u_z, x_z, x_{\pi(z)}\}$ and join bags along the dependency tree; omit the absent parent variable at the root. $\square$

For these matrices, one $O(L)$ -query solve gives v , which determines every center and predictor in (72) by a public scalar rescaling, however large the movement count in (73). An implicit classical output then consists of a scalar and the stored vector. Repeated dense writes or quantum state preparations still cost work, but no further input queries. More generally, after reading all $O(ns)$ nonzero coefficients of a fixed sparse matrix, an algorithm needs no further coefficient queries for any output those coefficients determine mathematically, with arbitrary storage and computation. The coefficients do not determine the unspecified behavior of an arbitrary unitary completion, and this observation does not bound arithmetic, gate, memory, or bit complexity.

The same reuse applies to scalar outputs. For any fixed public readout c , one has $c^T x(\eta) = \rho(\eta) c^T M^{-1}e$. Since $0 \leq \rho(\eta) < 1$, one estimate of $c^T M^{-1}e$ to additive error ϵ gives all these scalar

center values to the same error. With the plateau readout of Section 7, this one scalar is classically hard, and its reuse needs no state-output lower bound.

Mori et al. [33, Theorem 1] prove, for a constant-sparsity family under coherent sparse position/value access, the sharper one-shot state-preparation lower bound $\Omega(\kappa \log(1/\varepsilon_{\text{state}}))$, where $0 < \varepsilon_{\text{state}} \leq 1/11$ is measured in Euclidean distance between normalized solution states; right-hand-side preparation cost is treated as negligible. It concerns a different output from our scalar inverse forms and does not justify multiplying one-shot hardness by a movement count. The elementary family above already refutes this general inference without a precision-dependent state-preparation lower bound.

9.2 Reusing the span of stored solutions after a residual test

Consider any sequence of nonsingular systems $H_t z_t = f_t \neq 0$ with unit solution rays $\psi_t = z_t / \|z_t\|$. Our procedure maintains the span of stored checkpoint vectors. At time t it minimizes the *current* relative residual $\|H_t y - f_t\| / \|f_t\|$ over this span. If the minimum is too large, it solves the system once (a refresh) and appends the solution ray. Every use of an old vector requires its image under the current operator; we do not assume that old residuals remain valid.

Proposition 9.3 (Exact span). *If the full sequence of solution rays spans a space of dimension r , then with exact residual minimization and exact checkpoint rays the procedure makes at most r refresh solves, for any acceptance threshold in $(0, 1)$. This bound is sharp.*

Proof. If the current ray belongs to the stored span, that span contains the exact solution and its minimum residual is zero. Each rejection therefore adds a linearly independent ray. For sharpness take $H_t = I$, $f_t = e_t$, $t = 1, \dots, r$; every not-yet-stored vector has relative residual one. $\square$

For a robust version, set $A_t = (\|z_t\| / \|f_t\|) H_t$, so $A_t \psi_t = f_t / \|f_t\|$, and assume $\|A_t\| \leq B_0$. Necessarily $B_0 \geq 1$. These normalized operators are only for analysis; the algorithm can minimize the original relative residual and need not know $\|z_t\|$. Let $0 < \eta \leq 1$ be the desired residual scale. Suppose minimization returns a candidate with residual at most $\eta/8$ above the true minimum, and the test estimates that candidate's residual to additive error $\eta/8$. Accept if the reported value is at most $3\eta/4$. An accepted candidate has true relative residual at most $7\eta/8$, whereas rejection implies minimum residual greater than $\eta/2$.

Theorem 9.4 (Numerical span with normalized checkpoints). *Let D be the system dimension and let $1 \leq r \leq D$ be an integer. Suppose every refreshed stored vector $\hat{\psi}_t$ has norm one and $\|\hat{\psi}_t - \psi_t\| \leq \xi$. Define the Kolmogorov width*

$$d_r = \inf_{\dim S=r} \sup_t \text{dist}(\psi_t, S), \quad \Delta = \frac{\eta}{2B_0} - \xi > 0.$$

If

$$d_r + \xi < \frac{\Delta^{2-1/r}}{2r}, \tag{74}$$

then the residual procedure makes at most $2r - 1$ refreshes, on the event that all stated minimization, test, and checkpoint error bounds hold.

Proof. For a stored span V , its best relative residual is at most $B_0 \text{dist}(\psi_t, V)$. Rejection therefore implies $\text{dist}(\psi_t, V) > \eta/(2B_0)$, and the next approximate stored vector has distance greater than Δ

from V . Suppose $2r$ refreshes occur and form a matrix Y from their normalized columns. Its first QR diagonal has magnitude one and all later diagonals exceed Δ , giving

$$\sqrt{\det(Y^*Y)} > \Delta^{2r-1}.$$

Choose an r -space S with all column distances at most $d_r + \xi + \epsilon$, where $\epsilon > 0$ is arbitrary. The first r singular values of Y are at most $\sqrt{2r}$ since its columns have norm one. The remaining r are at most $\|(I - \Pi_S)Y\| \leq \sqrt{2r}(d_r + \xi + \epsilon)$, since $\Pi_S Y$ has rank at most r . Thus

$$\sqrt{\det(Y^*Y)} \leq (2r)^r (d_r + \xi + \epsilon)^r.$$

Letting $\epsilon \downarrow 0$ contradicts (74). Normalization of every stored column is used in this volume bound. $\square$

There is a stronger bound under a correspondingly stronger stability assumption. Suppose the exact rays have rank r , and whenever a new exact ray is dependent on *the exact versions of the previously selected checkpoint rays*, it has a representation in those rays whose coefficients have sum of absolute values at most Λ . If $B_0 \Lambda \xi \leq \eta/2$, replacing that representation by the stored approximate rays has relative residual at most $\eta/2$. It cannot trigger a rejection. Hence at most r refreshes occur. A coefficient bound in an unspecified final basis does not suffice for this adaptive argument, which represents each new ray in the checkpoints already selected. This condition allows checkpoint error linear in η ; the cost of obtaining such checkpoints by tomography still depends on the particular tomography interface.

These statements specialize the established ideas of subspace recycling and greedy reduced bases [9, 37] to checkpoints accepted after a residual test. They bound refresh counts; they do not give unconditional quantum speedups. With explicit vectors of dimension D and R retained checkpoints, forming current images and solving their least-squares problem costs, for example, $O(R \text{mv}(H_t) + DR^2 + R^3)$ arithmetic per test. Quantum tests have their own input preparation, normalization and coefficient-conditioning costs. If the systems are expressed in a linear nullspace representation enforcing homogeneous feasibility equations, span combinations preserve those equations. For an affine feasibility constraint one must instead retain a particular feasible point and combine homogeneous directions. Neither observation provides access to the cone scaling for free.

9.3 When low numerical rank is justified, and what it does not buy

Holomorphic dependence on the parameter suffices for a small width d_r in Theorem 9.4. Suppose a unit-ray curve on $[-1, 1]$ extends holomorphically to a Bernstein ellipse of parameter $\chi > 1$, continuously to its boundary, with norm at most W . Its vector-valued Chebyshev coefficients satisfy $\|c_j\| \leq 2W\chi^{-j}$ for $j \geq 1$, by the usual contour integral on the ellipse. The degree- $(r-1)$ truncation lies in the span of r coefficient vectors, and its tail is bounded by

$$d_r \leq \sum_{j=r}^{\infty} 2W\chi^{-j} = \frac{2W\chi^{-(r-1)}}{\chi - 1}. \quad (75)$$

Thus, with uniform χ and W , the rank needed for a given width grows logarithmically in the inverse width. These hypotheses do not follow from strict complementarity. For example, the strictly complementary separable QP $\min_{x \geq 0} \frac{1}{2} \|x\|^2 + c^T x$, with distinct $c_i > 0$, has central-path coordinates and predictors

$$x_i(\tau) = \frac{-c_i + \sqrt{c_i^2 + 4\tau}}{2}, \quad x'_i(\tau) = \frac{1}{\sqrt{c_i^2 + 4\tau}}.$$

The coordinate functions of the predictor are linearly independent: analytic continuation of a vanishing linear combination to a neighborhood of $-c_i^2/4$ forces its i th coefficient to vanish, since only that summand has that branch singularity. The normalized predictor rays therefore span the full dimension; common normalization does not change linear relations among coordinate functions on the real interval. Nor is there a uniform complex neighborhood as $\min_i c_i \downarrow 0$. For a normalized-ray example in dimension two, choose $c_2 = 2c_1$. The squared first coordinate of the normalized predictor is $(c_2^2 + 4\tau)/(c_1^2 + c_2^2 + 8\tau)$, with a nonremovable pole tending to zero from below as $c_1 \downarrow 0$. A uniform Bernstein ellipse for normalized rays on $[0, 1]$ is therefore impossible in this family.

Low rank alone also does not make the residual test cheap. With coefficient queries, take $H_0 = H_1 = I_D$, $f_0 = e_0$, and

$$f_1 = e_0 + \sum_{i=1}^{D-1} a_i e_i, \quad a_i \in \{0, 1\}, \quad \sum_i a_i \in \{0, 1\}.$$

The solution span has dimension at most two and $B_0 = 1$. The minimum relative residual from $\text{span}\{e_0\}$ is zero or $1/\sqrt{2}$. Any test separating these cases solves unique search, which requires $\Omega(D)$ randomized or $\Omega(\sqrt{D})$ quantum queries. For completeness, under a uniform marked location a Q -query randomized algorithm hits it with probability at most $Q/(D-1)$ until the transcript changes; the quantum hybrid bound changes the average final-state distance by at most $O(Q/\sqrt{D})$. An exact norm oracle for f_1 would reveal the case immediately, which is why the coefficient-query input is explicit here.

Even a single stored ray does not remove the cost of explicit output. For $H = I$ and $f_b = D^{-1/2}((-1)^{b_1}, \dots, (-1)^{b_D})$, a phase query implemented by the canonical bit oracle prepares the amplitude state. The input for the following lower bound is this bit/value oracle; a phase-only unitary would not expose the overall sign. A classical vector within Euclidean error $1/4$ determines all but at most $D/16$ signs by rounding its coordinates. This requires $\Omega(D)$ classical queries and $\Omega(D)$ writes; its quantum query cost is also $\Omega(D)$. To see the latter without assuming exact recovery, a Q -query final state, as the signs vary, lies in a space of dimension at most $\sum_{j=0}^Q \binom{D}{j}$ by the multilinear query expansion [7]. Any one rounded output is correct for at most $V = \sum_{j \leq D/16} \binom{D}{j} \leq 2^{H_2(1/16)D}$ inputs, where H_2 is binary entropy. For any measurement, its average success on uniform inputs is at most $V \sum_{j \leq Q} \binom{D}{j} / 2^D$: bound each state's density matrix by the projector onto this common space and sum the measurement operators. Constant success consequently requires $Q = \Omega(D)$. The argument separates cheap state preparation from classical vector output, even when the same rank-one target is requested at every checkpoint.

10 Structured cone systems and scalar estimation

A dense cone Hessian need not make its inverse forms hard to estimate. We give two classical algorithms for structured Newton systems. The first assumes a sparse positive definite base plus a few low-rank corrections, with sampling access to the *source* vectors of the corrections. It returns a scalar estimate or a solution sampler without writing out a dense Newton direction. The second uses an explicit factorization and returns the full direction. They need different structural parameters.

The low-rank cone identities and their use in sparse factorizations are established techniques [3, 17, 24], as is matrix arithmetic under sampling-and-query access with dimension-independent cost [18]. The result below combines a full-rank sparse base with possibly indefinite corrections. It needs its own access model and error analysis because sampling from a nearly zero transformed correction column can be arbitrarily expensive; the algorithm never normalizes such a column.

10.1 A sparse base with sampled low-rank corrections

We work over $\mathbb{R}$ in this section. Suppose that $C \in \mathbb{R}^{m \times n}$ has row and column sparsity at most d and that public parameters give the spectral bounds

$$S = CC^T, \quad \kappa^{-1}I \preceq S \preceq I, \quad N = S + CXLX^TC^T, \quad aS \preceq N \preceq \beta S. \quad (76)$$

Here $0 < a \leq 1 \leq \beta$, $X \in \mathbb{R}^{n \times r}$ satisfies $\|X\| \leq 1$, and the explicitly supplied $r \times r$ symmetric matrix L obeys $\|L\| \leq \ell$, where $\ell \geq 1$; L need be neither invertible nor semidefinite. Each column of X is given as $X_j = T_j v_j$, where $\|v_j\| = 1$, $\text{SQ}(v_j)$ is supplied, and T_j is a known contraction with at most one entry per row and column, such as a diagonal map or a coordinate embedding. Each T_j has constant-cost sparse row and column access as in Definition 2.1, including reports of absent entries. The norms of X_j and CX_j are not supplied. For the nonzero right-hand side b , $\text{SQ}(b)$ is supplied; by scaling, we assume $\|b\| = 1$ throughout the proof.

Theorem 10.1 (Sparse base and low-rank corrections). *Assume (76), row and column sparse access to C , and the vector access above. Then a randomized setup has an event of probability at least $1 - \zeta$, the good setup event, on which the coordinate oracle it builds answers queries to one fixed vector $\tilde{y}$ with*

$$\|\tilde{y} - N^{-1}b\| \leq \epsilon \|N^{-1}b\|,$$

and on this event each requested draw returns either a failure flag or a sample with law exactly $p_{\tilde{y}}$ conditional on no flag, with flag probability at most a prescribed $\delta \in (0, 1/2)$. Setup, coordinate queries, and draws have bounded cost even for bad setup outcomes. With $R_* = 2 + (r + 1)\kappa\beta\ell/a$, each is at most

$$(d + 1)^{O(\sqrt{\kappa} \log(R_*/\epsilon))} \text{poly}(R_*, \epsilon^{-1}, \log(1/\zeta), \log(1/\delta)). \quad (77)$$

No oracle for the exact norm of the output is provided. For $r = 0$ it reduces to Theorem 3.7 with a cap on the number of rejection trials.

Proof. Put $D_* = 1 + \ell/a$ and $U_* = D_*\ell$. The matrices

$$K = X^T C^T S^{-1} C X, \quad h = X^T C^T S^{-1} b, \quad J = (I + LK)^{-1}$$

obey $0 \preceq K \preceq I$ and $\|h\| \leq \sqrt{\kappa}$, since $C^T S^{-1} C$ is an orthogonal projection. The identity

$$J = I - LX^T C^T N^{-1} C X, \quad \|J\| \leq D_* \quad (78)$$

follows by multiplying by $I + LK$; the norm estimate uses $N^{-1} \preceq a^{-1}S^{-1}$. Thus Woodbury's identity holds without L^{-1} and gives

$$z = J L h, \quad \|z\| \leq U_* \sqrt{\kappa}, \quad y = N^{-1}b = S^{-1}b - S^{-1}C X z.$$

Use Lemma 3.1 to obtain a polynomial $P = P(S)$ with $\|I - SP\| \leq \eta \leq 1$. Functional calculus gives

$$\|P\| \leq 2\kappa, \quad \|PC\| \leq 2\sqrt{\kappa}, \quad \|C^T P C\| \leq 2. \quad (79)$$

Writing $C = S^{1/2}V$ with $VV^T = I$ proves all three. Let $K_P = X^T C^T P C X$ and $h_P = X^T C^T P b$. Their biases satisfy $\|K_P - K\| \leq \eta$ and $\|h_P - h\| \leq \eta\sqrt{\kappa}$.

Each entry of K_P is estimated by sampling from v_i and evaluating $T_i^T C^T P C T_j v_j$ locally. If $I \sim p_{v_i}$, the variable $(T_i^T C^T P C T_j v_j)_I / (v_i)_I$ has the desired expectation and second moment at most 4; zero sampled denominators have probability zero. For $(h_P)_i$ use $T_i^T C^T P b$ instead; its

second moment is at most 4κ . Median-of-means estimates with entry errors $\rho/(2r)$ and $\nu/(2\sqrt{r})$, respectively, give

$$\|\widehat{K} - K\| \leq \rho, \quad \|\widehat{h} - h\| \leq \nu$$

with joint probability $1 - \zeta$, provided $\eta \leq \rho/2$ and $\eta\sqrt{\kappa} \leq \nu/2$. One may symmetrize $\widehat{K}$ without increasing its error. The total number of local evaluations is

$$O\left[\left(\frac{r^4}{\rho^2} + \frac{r^2\kappa}{\nu^2}\right) \log \frac{r+1}{\zeta}\right]. \quad (80)$$

All estimates sample from the supplied vectors, not from CX_j .

If $U_*\rho \leq 1/2$, perturbing the $r \times r$ inverse gives $\|(I + L\widehat{K})^{-1}\| \leq 2D_*$ and, for $\widehat{z} = (I + L\widehat{K})^{-1}L\widehat{h}$,

$$\|\widehat{z} - z\| \leq 2U_*(\nu + \rho U_*\sqrt{\kappa}).$$

The fixed output $\widetilde{y} = Pb - PCX\widehat{z}$ consequently satisfies

$$\|\widetilde{y} - y\| \leq \eta\kappa(1 + U_*) + 4U_*\sqrt{\kappa}\nu + 4U_*^2\kappa\rho. \quad (81)$$

Here the polynomial bias in $S^{-1}b$ is at most $\eta\kappa$, that in $S^{-1}CXz$ is at most $\eta\sqrt{\kappa}\|z\|$, and the remaining term is bounded using (79). The explicit choices

$$\begin{aligned} \rho &= \frac{\epsilon}{48\beta U_*^2\kappa}, & \nu &= \frac{\epsilon}{48\beta U_*\sqrt{\kappa}}, \\ \eta &= \min\left\{\frac{\epsilon}{12\beta\kappa(1 + U_*)}, \frac{\rho}{2}, \frac{\nu}{2\sqrt{\kappa}}\right\} \end{aligned}$$

make (81) at most $\epsilon/(4\beta)$. Since $N \preceq \beta I$, $\|y\| \geq 1/\beta$, which proves the stated relative error with room to spare. On the good event, $\|\widehat{z}\| \leq 4U_*\sqrt{\kappa}$.

It remains to sample without dividing by small component norms. The raw trial in Lemma 3.6 also applies to a rectangular or singular transform T : if its rows have at most W entries and $B_T \geq \|T\|$, its output probability at coordinate i in one trial is

$$\frac{|(Tv)_i|^2}{WB_T^2\|v\|^2}. \quad (82)$$

Its proof uses only column-norm upper bounds and the Cauchy–Schwarz inequality on rows, so no lower singular-value bound is needed. Choose one $W = (d+1)^{O(\sqrt{\kappa}\log(R_*/\epsilon))}$ bounding the row and column support sizes of all transforms used and the cost of computing their entries; sparse walks through CC^T and the one-entry maps T_j give such a bound. Put $w_0 = Pb$, $w_j = PCT_jv_j$, $c_0 = 1$ and $c_j = -\widehat{z}_j$. Take the raw-trial denominators $D_0 = 4W\kappa^2$ and $D_j = 4W\kappa$ for $j \geq 1$, which equal $WB_T^2\|v\|^2$ for the bounds B_T in (79), and let $Z = \sum_{j=0}^r c_j^2 D_j$. Select j with probability $c_j^2 D_j/Z$ and perform one raw trial for w_j . If it returns i , accept again with probability

$$\frac{|\sum_{j=0}^r c_j w_{j,i}|^2}{(r+1) \sum_{j=0}^r |c_j w_{j,i}|^2}.$$

The denominator is positive whenever this step is reached. Summing over j shows that one combined trial outputs i with probability $|\widetilde{y}_i|^2/((r+1)Z)$. On the good setup event,

$$Z \leq 4W\kappa^2(1 + 16U_*^2), \quad \|\widetilde{y}\| \geq \frac{1}{2\beta}.$$

Thus the total acceptance probability is at least $\pi_* = [16(r+1)W\beta^2\kappa^2(1+16U_*^2)]^{-1}$.

To bound the runtime on every outcome, setup returns a failure flag if $I + L\widehat{K}$ is singular, if the Frobenius norm of its inverse exceeds $2\sqrt{r}D_*$, or if $\|\widehat{z}\| > 4U_*\sqrt{\kappa}$. These checks never reject a good setup and need only dense $r \times r$ arithmetic. Every draw is capped at $\lceil \pi_*^{-1} \log(1/\delta) \rceil$ trials and returns a flag if all fail. Since the trials are i.i.d., truncation does not change the law conditional on acceptance. On a bad setup the loop still terminates, even if $\widetilde{y} = 0$. Each trial costs $O((r+1)W)$; (80), the $r \times r$ solve, and these caps prove (77). For nonunit b , scale the fixed vector back by the known norm of b ; its sampling law is unchanged. $\square$

Corollary 10.2 (Structured positive inverse forms). *Under the same hypotheses, $q = b^T N^{-1} b$ can be estimated to relative error ϵ with failure probability at most ζ , at a cost of the form (77), using only setup and coordinate queries and hence no per-draw failure parameter δ .*

Proof. Normalize b to unit norm. Then $q \geq 1/\beta$ and $\|N^{-1}b\| \leq \kappa/a$. Run the preceding setup with vector error $\epsilon a/(4\beta\kappa)$ and failure budget $\zeta/2$. On the good setup event, $|b^T \widetilde{y} - q| \leq \epsilon/(4\beta)$. Independently sample $I \sim p_b$ and average $\widetilde{y}_I/b_I$ by median of means. Conditionally on a good setup, its second moment is at most $\|\widetilde{y}\|^2 \leq 4\kappa^2/a^2$. Thus $O(\beta^2\kappa^2a^{-2}\epsilon^{-2}\log(1/\zeta))$ coordinate evaluations achieve additional error $\epsilon/(2\beta)$. A union bound proves the claim. The argument uses neither positivity of the approximate linear map nor a promise that its individual correction terms do not cancel. $\square$

10.2 Lorentz and generalized-power barriers

At an interior Lorentz point (t, z) with $t > \|z\|$, let $H = \nabla^2 F_L(t, z)$ for $F_L(t, z) = -\log(t^2 - \|z\|^2)$. Put $r_z = \|z\|$, $s = t^2 - r_z^2 > 0$, $d_x = s/2$, and $\chi = (t + r_z)/(t - r_z)$. If $r_z > 0$, put $u_{\pm} = 2^{-1/2}(1, \pm z/r_z)$. Differentiation and inversion give

$$H^{-1} = d_x I + r_z(t + r_z)u_+ u_+^T - r_z(t - r_z)u_- u_-^T, \quad \chi^{-1}d_x I \preceq H^{-1} \preceq \chi d_x I. \quad (83)$$

Indeed, the eigenvalues are $(t + r_z)^2/2$, $(t - r_z)^2/2$ and $s/2$ on the two radial eigendirections and their orthogonal complement. At $z = 0$ the correction vanishes. For L_c blocks, let $\mathcal{D} = \bigoplus d_x I$, $C = A\mathcal{D}^{1/2}$ and $S = CC^T$. Form X from the orthonormal columns $(1, 0)$ and $(0, z/r_z)$ in each block. The matrix L in (76), defined by $\mathcal{D}^{-1/2}H^{-1}\mathcal{D}^{-1/2} = I + X L X^T$, is explicit, block diagonal, and has norm at most $\chi_{\max} - 1$, where $\chi_{\max}$ is the largest χ over the blocks. Thus $r \leq 2L_c$, $a = \chi_{\max}^{-1}$ and $\beta = \chi_{\max}$ suffice in (76). Supplied block norms and $\text{SQ}(z)$ give all required source access. A known positive scaling of S normalizes its spectral upper bound to one and scales N and the desired output accordingly. Hence Theorem 10.1 and Corollary 10.2 apply to Lorentz cones.

For generalized-power cones, let $x_i > 0$, $\alpha_i > 0$, $\sum_{i=1}^{m_x} \alpha_i = 1$, and

$$p = \prod_i x_i^{2\alpha_i}, \quad s = p - \|z\|^2 > 0, \quad \lambda = p/s, \quad 1 \leq \lambda \leq \Lambda.$$

The barrier

$$F(x, z) = -\log s - \sum_i (1 - \alpha_i) \log x_i \quad (84)$$

has parameter $m_x + 1$ [38]. Its use and low-rank Hessian structure are prior work [17]; closed-form inverse-Hessian operators are given by Kapelevich et al. [26]. Set $k_i = 1 + (2\lambda - 1)\alpha_i$ and $D_0 = \text{diag}(k_i/x_i^2, (2/s)I)$. With $c = \|\alpha\|^2$, define two columns with disjoint supports:

$$X_1 = (\alpha_i/(\sqrt{c}\sqrt{k_i}))_i \oplus 0, \quad X_2 = 0 \oplus z/\|z\|.$$

At $z = 0$, omit the correction: $\lambda = 1$ and $H = D_0$. Otherwise direct differentiation of (84) gives

$$H = D_0^{1/2}(I + XBX^T)D_0^{1/2}, \quad B = \begin{pmatrix} 4\lambda(\lambda - 1)c & -4\lambda\sqrt{c(\lambda - 1)/2} \\ -4\lambda\sqrt{c(\lambda - 1)/2} & 2(\lambda - 1) \end{pmatrix}. \quad (85)$$

Before normalization, the correction is $4\lambda(\lambda - 1)aa^T - (4\lambda/s)(az^T + za^T) + (4/s^2)zz^T$, where $a_i = \alpha_i/x_i$ and the x and z groups are embedded separately; this confirms the signs in B and the diagonal base D_0 .

Lemma 10.3 (Power-cone spectral comparison and estimated coefficient matrix). *The inverse Hessian satisfies*

$$(4\lambda)^{-1}D_0^{-1} \preceq H^{-1} \preceq 4\lambda D_0^{-1}. \quad (86)$$

In $H^{-1} = D_0^{-1/2}(I + XL(t)X^T)D_0^{-1/2}$, the coefficient matrix is symmetric and given by

$$t = \sum_i \frac{\alpha_i^2}{ck_i}, \quad L(t) = -(I + B \operatorname{diag}(t, 1))^{-1}B.$$

For every t in the interval

$$\mathcal{I}_\lambda = [1/(2\lambda), \min\{1, 1/(2\lambda c)\}], \quad (87)$$

this formula is defined, $\|L(t)\| \leq 10\Lambda^2$, and $\|L(t') - L(t)\| \leq 100\Lambda^4|t' - t|$. The true value of t lies in this interval and can be estimated to error u with $O(u^{-2} \log(1/\zeta))$ samples from p_α .

Proof. Since $1 \leq k_i \leq 2\lambda$ and $k_i \geq 2\lambda\alpha_i$, $t \geq 1/(2\lambda)$ and $ct \leq 1/(2\lambda)$, proving the interval claim. In the orthonormal basis obtained by normalizing the two column directions of X , the normalized Hessian is

$$Q(t) = \begin{pmatrix} 1 + 4\lambda(\lambda - 1)ct & -4\lambda\sqrt{ct(\lambda - 1)/2} \\ -4\lambda\sqrt{ct(\lambda - 1)/2} & 2\lambda - 1 \end{pmatrix}.$$

Its trace is at most 4λ and its determinant is $2\lambda - 1 - 4\lambda(\lambda - 1)ct \geq 1$ on (87). Its leading diagonal entry is positive, so both eigenvalues lie in $[1/(4\lambda), 4\lambda]$. On the orthogonal complement the matrix is the identity. This proves (86). For $G = \operatorname{diag}(t, 1)$,

$$L(t) = G^{-1/2}(Q(t)^{-1} - I)G^{-1/2}.$$

This identity proves symmetry and gives $\|L(t)\| \leq 2\lambda(4\lambda + 1) \leq 10\Lambda^2$. Differentiating the inverse formula gives $L'(t) = L(t) \operatorname{diag}(1, 0)L(t)$, hence $\|L'(t)\| \leq 100\Lambda^4$. Finally, t is the expectation of $1/k_I \in [0, 1]$ for $I \sim p_\alpha$. Hoeffding's inequality proves the sample bound. Clipping an estimate to (87) never increases its error. Clipping only to $[1/(2\Lambda), 1]$ would not preserve the determinant bound and is insufficient. $\square$

Corollary 10.4 (Cone Newton systems with supplied block data). *Let A have full row rank and row and column sparsity d , with constant-cost sparse row and column access, a public block partition, and constant-cost conversion between block-local and global indices. For L_c Lorentz blocks, assume $\operatorname{SQ}(z)$ is supplied separately for each block and axial coordinates can be queried; for L_c generalized-power blocks, assume $\operatorname{SQ}(\alpha)$ and $\operatorname{SQ}(z)$ are supplied separately for each block, coordinates of x can be queried, and each block's barrier scalars p and s are supplied. Thus each block's norm is available, not only the norm and sampling law of the concatenated vector. Radial-vector norm and coordinate access are available also when $z = 0$; sampling is requested only for blocks with nonzero radial vector, and a zero block's radial correction is omitted. Supply $\operatorname{SQ}(b)$ and spectral bounds normalizing $S = ADA^T$ to $[\kappa_S^{-1}, 1]$, where $\mathcal{D}$ is $\bigoplus d_x I$ (Lorentz) or $\bigoplus D_0^{-1}$ (power). Then, for $N = AH^{-1}A^T$,*

relative approximation of $N^{-1}b$, capped sampling from it, and relative estimation of $b^T N^{-1}b$ each cost

$$(d+1)^{O(\sqrt{\kappa_S} \log(2L_c \Gamma \kappa_S / \epsilon))} \text{poly}(L_c, \Gamma, \kappa_S, \epsilon^{-1}, \log(1/\zeta), \log(1/\delta)), \quad (88)$$

where $\Gamma = \max \chi$ for Lorentz cones and $\Gamma = \Lambda$ for power cones. The setup and draws have the guarantees of Theorem 10.1; scalar estimation does not use δ .

Proof. The Lorentz case has already been reduced to Theorem 10.1. For power cones take $C = A\mathcal{D}^{1/2}$. The columns in (85) are diagonal contractions of $\alpha/\|\alpha\|$ and embeddings of $z/\|z\|$; their supports are disjoint across and within blocks, which gives $\|X\| \leq 1$. They do not require the exact norm of X_1 . Lemma 10.3 gives $a = 1/(4\Lambda)$, $\beta = 4\Lambda$, $\ell = 10\Lambda^2$ and $r \leq 2L_c$. Only L requires approximation. Estimate each t and clip as in (87), obtaining a fixed block diagonal $\hat{L}$. If $\|\hat{L} - L\| \leq v$, then

$$\|S^{-1/2}(\hat{N} - N)S^{-1/2}\| \leq v, \quad \hat{N} = S + CX\hat{L}X^T C^T.$$

For $v \leq a/2$, therefore, $(a/2)S \preceq \hat{N} \preceq (\beta + a/2)S$. The resolvent identity also gives, for every b ,

$$\|\hat{N}^{-1}b - N^{-1}b\| \leq \|\hat{N}^{-1}\| \|\hat{N} - N\| \|N^{-1}b\| \leq \frac{2\kappa_S v}{a} \|N^{-1}b\|.$$

Take $v \leq \min\{a/2, \epsilon a/(8\kappa_S)\}$, and run the theorem on $\hat{N}$ with tolerance $\epsilon/4$. Since $\epsilon \leq 1/2$, the resulting total relative error is less than ϵ . The estimates of t use accuracy $v/(100\Lambda^4)$ and joint failure budget $\zeta/2$; the remaining setup uses $\zeta/2$. The lemma bounds $\|\hat{L}\| \leq 10\Lambda^2$ even for inaccurate estimates. These are polynomial costs, and the logarithm of every required tolerance has the order asserted in (88). For the scalar estimate, run this vector construction at tolerance $\epsilon a/(4\beta\kappa_S)$ and apply the independent estimator from Corollary 10.2. $\square$

Corollary 10.4 concerns one linear system with given oracles. For fixed d, L_c, Γ, κ_S and accuracy, growing only the cone dimensions does not force a query cost polynomial in them. The corollary does not cover maintaining the supplied access to the iterate or forming b , and does not show that the parameters stay bounded along a path. The cyclic one-cone example in Section 7 has a growing sparse-base condition number and hence does not contradict it. Standard three-dimensional cone blocks give bounded block sparsity directly, so Section 3 already applies when the reduced system has the stated spectral bounds.

10.3 The power-cone scalar oracle cannot be omitted

Corollary 10.4 assumes that each power block's p is supplied. Without it, obtaining p from the vector access can require $\Omega(M)$ queries. Fix uniform weights on $2M$ positive coordinates grouped into pairs. Encode a hidden bit h_j by choosing pair values (a, b) or (c, d) , where $a^2 + b^2 = c^2 + d^2$ and $ab \neq cd$; the norm is then public. A squared-coordinate sample is simulated by choosing a uniform pair and querying its bit; a coordinate query also needs one bit. For the quantum lower bound we fix one quantum version of this access: coherent value queries, and a preparation unitary that prepares a uniform pair index, queries its hidden bit, applies the corresponding public two-coordinate rotation, and uncomputes the bit. Since all pairs have equal norms, this prepares the normalized input vector. Coherent value queries, the preparation unitary and its inverse, and their controlled versions all use $O(1)$ hidden-bit queries. We make no claim about other unitary completions of classical SQ access. However,

$$p = [(ab)^{M-|h|}(cd)^{|h|}]^{1/M}$$

determines $|h|$ exactly and hence its parity. By the classical and quantum bounded-error lower bounds for parity, computing p exactly requires $\Omega(M)$ queries under either access. Even constant relative error is hard when the logarithmic range of the coordinates is unbounded. Take

$$(a, b) = (\sqrt{1 - e^{-2\gamma M}}, e^{-\gamma M}), \quad (c, d) = (1/\sqrt{2}, 1/\sqrt{2}), \quad \gamma = 4.$$

Both squared pair norms equal one, and adjacent possible values of p have ratio

$$[(1/2)/(e^{-4M}\sqrt{1 - e^{-8M}})]^{1/M} > e^4/2 > 3.$$

A relative error less than $1/2$ therefore still determines the integer $|h|$ and its parity, with the same quantum simulation. This proves the same linear lower bound in ideal real arithmetic, using a range of $\log x_i$ that grows with M . If that range is known and bounded, one can instead estimate p through the weighted mean of $\log x_i$. Samples proportional to α_i suffice, but $\text{SQ}(\alpha)$ samples proportional to α_i^2 ; the two coincide for uniform weights. These costs are part of acquiring the input to Corollary 10.4.

The aggregated logarithm cone shows the limits of low-rank algebra alone. In the interior of this cone, where $s = v \sum_i \log(w_i/v) - u > 0$ and $v, w_i > 0$, the Hessian of $-\log s - \log v - \sum_i \log w_i$ is diagonal plus rank at most three. To see this, let $h = (0, 0, (1/w_i)_i)$, and use the positive diagonal with entries 1 , $m/(vs) + 1/v^2$, and $(1 + v/s)/w_i^2$, where m is the number of w coordinates. The remaining terms are $-e_u e_u^T + (\nabla s)(\nabla s)^T/s^2 - (e_v h^T + h e_v^T)/s$; ∇s belongs to $\text{span}\{e_u, e_v, h\}$. By the Woodbury identity, the inverse Hessian differs from the inverse diagonal by rank at most three. This gives no uniform spectral comparison like Lemma 10.3 and no bound on the cost of obtaining the needed scalars, so we make no analogue of Corollary 10.4 for this cone.

10.4 Eccentricity profiles with full output

We now turn to full-output algorithms. The first allows many Lorentz cones, a few with arbitrarily large eccentricity χ_j (defined before (83)). Assume $A = [A_1 \cdots A_{L_c}]$ has full row rank, and each (t_j, z_j) is interior, with $t_j > \|z_j\|$ and $H_j = \nabla^2[-\log(t_j^2 - \|z_j\|^2)]$. Write $N = \sum_j A_j H_j^{-1} A_j^T$ and $S = \sum_j d_j A_j A_j^T$, with the notation of (83). For a threshold $\theta \geq 1$, keep the blocks $B_\theta = \{j : \chi_j > \theta\}$ exact, replace the others by their scalar base, and put

$$P_\theta = \sum_{j \in B_\theta} A_j H_j^{-1} A_j^T + \sum_{j \notin B_\theta} d_j A_j A_j^T, \quad b_\theta = |B_\theta|.$$

Proposition 10.5 (Profile preconditioner and energy certificate). *The matrix P_θ is positive definite and*

$$\theta^{-1} P_\theta \preceq N \preceq \theta P_\theta, \quad \text{rank}(P_\theta - S) \leq 2b_\theta. \quad (89)$$

Preconditioned conjugate gradients reduce the N -energy error by a factor η in $O(\theta \log(2/\eta))$ iterations. If $q = b - N\hat{y}$, then

$$\theta^{-1} q^T P_\theta^{-1} q \leq \|\hat{y} - N^{-1}b\|_N^2 \leq \theta q^T P_\theta^{-1} q. \quad (90)$$

Proof. For each block $j \notin B_\theta$, $\theta^{-1} d_j I \preceq H_j^{-1} \preceq \theta d_j I$. The contributions of blocks in B_θ are identical in the two matrices and positive semidefinite, so summing proves (89). Equation (83) gives the rank bound. The preconditioned spectrum lies in $[\theta^{-1}, \theta]$. The usual Chebyshev residual bound for conjugate gradients is $2((\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1))^k$ in the energy norm; $\kappa \leq \theta^2$ proves the iteration count. If $\theta = 1$, one iteration suffices. Inverting the Loewner bounds and using $\|\hat{y} - N^{-1}b\|_N^2 = q^T N^{-1}q$ proves the certificate. $\square$

The certificate has an exact Newton interpretation. For $H\Delta x + A^T\Delta y = r_d$ and $A\Delta x = r_p$, the reduced right-hand side is $b = AH^{-1}r_d - r_p$. An approximate multiplier induces $\widehat{\Delta x} = H^{-1}(r_d - A^T\widehat{\Delta y})$, and

$$\|\widehat{\Delta x} - \Delta x\|_H^2 = \|\widehat{\Delta y} - \Delta y\|_N^2. \quad (91)$$

Thus (90) bounds the primal error in the barrier norm, without a Euclidean condition-number factor. It does not by itself bound the full KKT residual or a Euclidean residual.

We next bound the arithmetic cost of an explicit implementation. Let A have m rows and n columns, suppose a width- τ elimination order for the row-intersection graph of A is supplied, and write $\bar{\tau} = \tau + 1$. Each column support is a clique of size at most $\bar{\tau}$, so assembling S costs $O(\bar{\tau} \text{nnz } A)$; its positive definite Cholesky factor costs $O(m\bar{\tau}^2)$ and occupies $O(m\bar{\tau})$ words. After deleting zero update columns, write $P_\theta = S + UJU^T$ with at most $2b_\theta$ columns and diagonal J with entries ± 1 . The matrix $J + U^T S^{-1}U$, of order at most $2b_\theta$, is nonsingular: the determinant identity and $P_\theta \succ 0$ prove this even when the update is indefinite. Forming U , solving for $S^{-1}U$, forming its Gram matrix, and factoring $J + U^T S^{-1}U$ give conservative exact-arithmetic costs for setup and for each application of P_θ^{-1} :

$$C_{\text{setup}} = O(n + \bar{\tau} \text{nnz } A + m\bar{\tau}^2 + mb_\theta(\bar{\tau} + b_\theta) + b_\theta^3), \quad (92)$$

$$C_{\text{apply}} = O(m(\bar{\tau} + b_\theta) + b_\theta^2). \quad (93)$$

These counts store the transformed correction columns densely. A matrix-free product by N costs $O(\text{nnz } A + n)$ by applying (83) blockwise. Hence the total solve cost is

$$C_{\text{prof}} = C_{\text{setup}} + O(\theta \log(2/\eta)[\text{nnz } A + n + C_{\text{apply}}]). \quad (94)$$

To choose θ adaptively, sort the eccentricities in decreasing order. Keeping the largest b and choosing $\theta = \chi_{(b+1)}$, with $\chi_{(L_c+1)} = 1$, gives (94) with correction count at most b . Sorting costs $O(L_c \log L_c)$; evaluating the displayed bound for all b then costs $O(L_c)$. Ties do not affect the argument. This trades the number of corrections against the largest remaining eccentricity, in the supplied coordinates.

The rank bound $2b_\theta$ is sharp in the worst case. Take b three-dimensional Lorentz blocks, $A = I$, and choose $t + r_z = \sqrt{2\chi}$, $t - r_z = \sqrt{2/\chi}$. Then $S = I$ and N has three orthogonal b -dimensional eigenspaces E_+, E_0, E_- with eigenvalues $\chi, 1, \chi^{-1}$. If $P = I + R \succ 0$, $R = R^T$, and $\text{rank } R < 2b$, then

$$\kappa(P^{-1/2}NP^{-1/2}) \geq \chi. \quad (95)$$

To prove this, let p_+, p_- be the positive and negative inertia of R . If $p_+ < b$, some $v \in E_+$ has $v^T R v \leq 0$, so its generalized Rayleigh quotient is at least χ . Since $\dim \ker R > b$, $\ker R \cap E_+^\perp$ contains a nonzero vector whose quotient is at most one. Otherwise $p_+ \geq b$ implies $p_- < b$; a vector in E_- has quotient at most χ^{-1} and one in $\ker R \cap E_-^\perp$ has quotient at least one. This proves (95). Thus, among base-plus-update preconditioners, guaranteeing condition number at most θ^2 for every profile with b blocks of eccentricity $\chi > \theta^2$ requires update rank at least $2b$. In the same family with $A = I$ and unequal eccentricities, the stronger Loewner bounds (89) require at least b_θ positive and b_θ negative update directions, by restriction to the high and low eigenspaces of the blocks with $\chi_j > \theta$, so this rank bound is sharp too. These are linear-algebra bounds within a specified class of preconditioners, not oracle lower bounds for optimization.

Signed Woodbury arithmetic can be numerically unstable. An alternative uses $P_\theta = S + UU^T - VV^T$. Blockwise, $d_j I - r_j(t_j - r_j)u_{j,-}u_{j,-}^T \succ 0$, hence $S - VV^T \succ 0$. The augmented matrix

$$\begin{pmatrix} S & V & U \\ V^T & I & 0 \\ U^T & 0 & -I \end{pmatrix}$$

is symmetric quasidefinite: its principal block on the first two groups is positive definite, and its last block is negative definite. Adding the at most $2b_\theta$ vertices for the columns of V and U to every bag of a width- τ tree decomposition for S increases the width by at most $2b_\theta$. Quasidefinite matrices admit pivot-free LDL^T factorization under every symmetric permutation [39]. The supplied width- τ order therefore gives setup cost $O((m + b_\theta)(\bar{\tau} + 2b_\theta)^2)$ and application and storage cost $O((m + b_\theta)(\bar{\tau} + 2b_\theta))$, plus coefficient formation. Either implementation may be used. Neither this count nor (94) is a uniform bit-complexity bound. Product-form factorizations address numerical cancellation in this setting [24]; working precision and residual checks must still meet the accuracy required by the outer solve.

10.5 Latent sparsity and explicit-output comparisons

Dense cone blocks can also have sparse representations with auxiliary variables. For one Lorentz block, let $J_0 = \text{diag}(1, -I)$, $u = J_0(t, z)$ and $s = t^2 - \|z\|^2$. Then

$$H = -2J_0/s + 4uu^T/s^2 = D_x + ww^T, \quad w = 2u/s. \quad (96)$$

The diagonal D_x is indefinite. Collect these diagonals over all blocks into D_x and the embedded columns w into W . The equality-constrained Newton system has the same primal and multiplier solution as

$$\begin{pmatrix} D_x & W & A^T \\ W^T & -I & 0 \\ A & 0 & 0 \end{pmatrix} \begin{pmatrix} \Delta x \\ h \\ \Delta y \end{pmatrix} = \begin{pmatrix} r_d \\ 0 \\ r_p \end{pmatrix}. \quad (97)$$

We call the auxiliary variables h hub variables, one per cone; eliminating them gives H exactly. Full row rank of A and $H \succ 0$ make both systems nonsingular. Define a graph with n coordinate vertices, m equality-row vertices, and two hub vertices per cone (the two-hub factorization below uses both). Join row k to coordinate i whenever $A_{ki} \neq 0$, and join both hubs of a cone to all coordinates of that cone. The matrix in (97) uses a subgraph of this fixed graph.

Proposition 10.6 (Supplied latent width). *If a width- τ_L tree decomposition of this graph, with $O((\tau_L + 1)(n + m + L_c))$ total bag entries, is supplied, then one Newton system can be solved exactly in*

$$O((n + m + L_c)(\tau_L + 1)^2)$$

field operations, after $O(n + \text{nnz } A)$ coefficient formation. This concerns one right-hand side and asserts neither a reusable symmetric factorization nor numerical stability.

Proof. Use (96) to form (97) without square roots. Replace each vertex in each bag by its row and column copies. Every nonzero of the matrix is now covered in its row-column bipartite graph, both copies inherit a connected subtree, and the new width is at most $2\tau_L + 1$. Write $N_{\text{aug}} = n + m + L_c$ for the augmented matrix order and $k = O(\tau_L + 1)$ for the copied decomposition width plus one. The supplied bound on total bag entries implies $|\mathcal{T}| = O(kN_{\text{aug}})$ decomposition nodes. The preprocessing in Section 3 of Fürer et al. [21] converts this compact decomposition to a nice decomposition with

$O(N_{\text{aug}})$ nodes in $O(k(|\mathcal{T}| + N_{\text{aug}})) = O(k^2 N_{\text{aug}})$ time. Their Corollary 3 then solves a consistent rectangular system in $O((\text{rows} + \text{columns})(\text{width} + 1)^2)$ field operations. Our system is square and nonsingular; applying that corollary proves the bound. Its arbitrary pivoting handles exact cancellations, so no zero-free prescribed pivot order is assumed. $\square$

For a reusable symmetric factorization, there is a two-hub alternative. With $r_z > 0$, define

$$D = 2I/s, \quad u_+ = \frac{\sqrt{2r_z(t+r_z)}}{s}(1, -z/r_z), \quad u_- = \frac{\sqrt{2r_z(t-r_z)}}{s}(1, z/r_z).$$

Then $H = D + u_+ u_+^T - u_- u_-^T$ and $u_-^T D^{-1} u_- = 2r_z/(t+r_z) < 1$, so $D - u_- u_-^T \succ 0$. Zero radial vectors give zero hub columns. For the product barrier let U, V collect the positive and negative columns. Replacing the dual zero block by $-\rho I$, $\rho > 0$, gives the expanded matrix

$$\begin{pmatrix} D & V & U & A^T \\ V^T & I & 0 & 0 \\ U^T & 0 & -I & 0 \\ A & 0 & 0 & -\rho I \end{pmatrix}.$$

It is quasidefinite with positive group (x, V) and negative group (U, y) . A width- τ_L order gives factorization cost $O((n+m+L_c)(\tau_L+1)^2)$, and storage and subsequent solve cost $O((n+m+L_c)(\tau_L+1))$. Forming these coefficients uses norms and square roots; these counts are in real arithmetic. If K_0 is the original KKT matrix, $w_0 = K_0^{-1}g \neq 0$, and $\rho \|K_0^{-1}\| < 1$, the Neumann bound gives

$$\frac{\|w_\rho - w_0\|}{\|w_0\|} \leq \frac{\rho \|K_0^{-1}\|}{1 - \rho \|K_0^{-1}\|}.$$

For $g = 0$ both solutions are zero. A public bound on $\|K_0^{-1}\|$ thus controls the regularization bias. Floating-point stability or successful refinement requires additional scaling and backward-error hypotheses.

Two hubs per block are necessary if the base is a positive diagonal and each hub adds a signed rank-one term on a generic block. Suppose all coordinates of $u = J_0(t, z)$ are nonzero and the dimension is at least three. Then a representation $H = D' + \sigma v v^T$ with $D' \succ 0$ diagonal would require $\sigma(v_i/u_i)(v_j/u_j) = 4/s^2$ for every $i \neq j$. Three indices force all ratios to be equal and $\sigma = 1$, leaving $D' = -2J_0/s$, whose time diagonal is negative. This argument does not apply to more general sparse bases.

A concrete graph class has constant latent width. Let the coarse bipartite graph join each cone to the equality rows that use it, and assume that each scalar column of A occurs in at most one row. If the coarse graph has width w , replace each cone vertex in each bag by its two hubs. For a coordinate used by row k , attach a leaf bag $\{u_j, v_j, x_i, y_k\}$ to a bag containing the corresponding coarse edge; unused coordinates need only the first three vertices. Edge coverage and connectedness of vertex bags are preserved. Consequently

$$\tau_L \leq \max\{2w + 1, 3\}.$$

Coarse forests have $\tau_L \leq 3$, independently of block dimensions and number of cones, although the materialized Hessians may contain arbitrarily large cliques. The smaller of (94) and the bound of Proposition 10.6 is a valid exact-arithmetic cost for a classical full output. The profile method tolerates large latent width when few blocks are highly eccentric; the latent-width method tolerates many eccentric blocks.

These comparisons concern the cost of one Newton solve, not whether a different solver produces the same iterates. Replacing the solver in an interior-point method that tolerates inexact directions requires the structural and access assumptions, with uniform parameter bounds, at every iterate the method may visit, and a classical direction that meets the method’s residual requirement. For example, even without cone structure, for a consistent full-column-rank system $Kx = f$ of total row and column dimension D_K , with M nonzeros and condition number κ_K , CGLS started from zero produces iterates with

$$\|Kx_k - f\| \leq 2 \left(\frac{\kappa_K - 1}{\kappa_K + 1} \right)^k \|f\|.$$

This follows by applying the CG energy bound to $K^T K$, whose condition number is κ_K^2 , and identifying that energy norm with the residual norm. It gives the familiar exact-arithmetic cost $O((M + D_K)\kappa_K \log(2/\eta))$ for relative residual η ; this is not a new result. For a positive definite system of order D , sparse direct elimination with a supplied width- w order costs $O(D(w + 1)^2)$. A quantum solver that returns the same explicit direction should be compared with these elementary classical costs.

In particular, suppose $n + m = D$, $\text{nnz } A \leq D^{1+o(1)}$, and either profile parameters $\bar{\tau}, \theta, b_\theta = D^{o(1)}$ or a latent width $\tau_L = D^{o(1)}$ are supplied. Then each Newton solve takes $D^{1+o(1)}$ arithmetic operations, up to logarithmic factors in the accuracy. In the latent-width case the bound on $\text{nnz } A$ is automatic: width τ_L bounds the edges of the structural incidence graph, hence $\text{nnz } A$, by $O(D(\tau_L + 1))$, since $L_c \leq n$. Writing out a full direction already takes $\Omega(D)$ word writes. So for these classes, with the same input access for both solvers and these assumptions holding uniformly over all iterates, replacing only the linear solver in a method that writes out every iterate cannot give a speedup polynomial in the dimension. This does not apply to scalar, sample, or quantum-state outputs. Nor does a dense assembled matrix or a large raw condition number by itself put an instance outside either bound.

11 Conclusion

The complexity of a scalar Newton quantity need not equal that of the Newton direction. Classical local estimation of a positive inverse form to relative error has a variance factor linear in the condition number, while separate sparse families with full SQ access show that the statistical factor and the exponential factor are each necessary. Explicit composition shows when both factors apply to the same scalar, and why the baseline of a hidden signal cannot be ignored. Under plain block access, the known variable-time algorithm is optimal up to logarithmic factors, but the lower bound does not hold when exact sparse entries can be queried.

In our optimization constructions, hardness can sit in an ordinary optimal value, an optimizer coordinate, or a small Newton decrement, with all access to the formulation simulated. It can also add up over a sequence of requested scalars or in one endpoint coordinate. The same examples show the limits: the optimal value without the objective tilt may be public, a normalized state may lose a hard sign or scale, and one hard Newton direction may be reused along a long trajectory. Comparisons of whole optimization methods must therefore account for input acquisition, the outputs requested, and information kept between solves.

For structured systems, a sparse full-rank base with a few controlled corrections allows scalar estimation and sampling without forming a dense inverse or normalizing small correction columns, so cone dimension alone does not determine query complexity. For explicit directions, eccentricity profiles and latent sparsity give different classical solve bounds; their assumptions must hold at the

iterates used, and their residual guarantees must match the outer method. The parameter regimes in which our upper and lower bounds do not match remain open.

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